Quantum Chaos and Information Scrambling

Exercises and Solutions

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Abstract

A worked set of exercises in quantum chaos and information scrambling, grouped by topic and solved in full — including the intermediate algebra that is usually left to the reader. Each topic is followed, where it helps, by flashcards and short derivations for self-review, and every numerical answer is reproduced by an accompanying Python script in `verify_exercises/`, so nothing has to be taken on faith.

Contents

1 Mathematical Foundations

Exercise 1.1: Bloch Sphere Geodesics and the Fubini–Study Metric

Motivation:

Pure quantum states live not in Hilbert space $\mathcal{H}$ but in **projective** Hilbert space $\mathbb{CP}^{D-1}$ (since global phases are unphysical). The natural metric on this space is the **Fubini-Study metric**, which measures the intrinsic distance between quantum states — the same quantity that controls the quantum speed limit and the quantum geometric tensor. For a single qubit, $\mathbb{CP}^1 \cong S^2$: the Fubini-Study metric gives the Bloch sphere its geometry.

Problem Statement:

A general single-qubit pure state is

$$|\psi(\theta, \varphi)\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\varphi}\sin\frac{\theta}{2}|1\rangle.$$

(a) Starting from the Fubini-Study line element $ds^2_{\text{FS}} = \langle d\psi|d\psi\rangle - |\langle\psi|d\psi\rangle|^2$, show that

$$ds^2_{\text{FS}} = \frac{1}{4}\big(d\theta^2 + \sin^2\theta\, d\varphi^2\big).$$

(b) Compute the geodesic distance between $|0\rangle$ (north pole) and $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ (equator).

Pedantic Step-by-Step Solution:

Part (a): Deriving the metric

Step 1: Compute $|d\psi\rangle$.

Differentiating the state:

$$|d\psi\rangle = -\frac{1}{2}\sin\frac{\theta}{2}\, d\theta\, |0\rangle + e^{i\varphi}\left[\frac{1}{2}\cos\frac{\theta}{2}\, d\theta + i\sin\frac{\theta}{2}\, d\varphi\right]|1\rangle.$$

Step 2: Compute $\langle d\psi|d\psi\rangle$ (Term 1).

The squared norm gives three contributions:

$$\langle d\psi|d\psi\rangle = \frac{1}{4}\sin^2\frac{\theta}{2}\, d\theta^2 + \frac{1}{4}\cos^2\frac{\theta}{2}\, d\theta^2 + \sin^2\frac{\theta}{2}\, d\varphi^2.$$

Using $\sin^2 + \cos^2 = 1$:

$$\langle d\psi|d\psi\rangle = \frac{1}{4}d\theta^2 + \sin^2\frac{\theta}{2}\, d\varphi^2.$$

Step 3: Compute $|\langle\psi|d\psi\rangle|^2$ (Term 2).

The inner product $\langle\psi|d\psi\rangle$ has two contributions:

- From $d\theta$: $\cos\frac{\theta}{2}\cdot(-\frac{1}{2}\sin\frac{\theta}{2}) + \sin\frac{\theta}{2}\cdot\frac{1}{2}\cos\frac{\theta}{2} = 0$. (The real part cancels!)
- From $d\varphi$: $\sin\frac{\theta}{2}\cdot i\sin\frac{\theta}{2} = i\sin^2\frac{\theta}{2}\, d\varphi$.

Therefore:

$$|\langle\psi|d\psi\rangle|^2 = \sin^4\frac{\theta}{2}\, d\varphi^2.$$

Step 4: Combine.

$$ds^2_{\text{FS}} = \frac{1}{4}d\theta^2 + \sin^2\frac{\theta}{2}\, d\varphi^2 - \sin^4\frac{\theta}{2}\, d\varphi^2 = \frac{1}{4}d\theta^2 + \sin^2\frac{\theta}{2}\cos^2\frac{\theta}{2}\, d\varphi^2.$$

Using $\sin^2\frac{\theta}{2}\cos^2\frac{\theta}{2} = \frac{1}{4}\sin^2\theta$:

$$\boxed{ds^2_{\mathrm{FS}} = \frac{1}{4}\big(d\theta^2 + \sin^2\theta\, d\varphi^2\big).}$$

This is the metric of a **round sphere of radius** $R = 1/2$. The Bloch sphere has half the radius of the "geographic" unit sphere — the factor 1/2 arises from the half-angle parametrization of SU(2).

Part (b): Geodesic distance $|0\rangle \to |+\rangle$

The north pole is $\theta = 0$ and $|+\rangle$ is at $\theta = \pi/2$, $\varphi = 0$. Along the meridian ($d\varphi = 0$), the geodesic length is:

$$d_{\mathrm{FS}} = \int_0^{\pi/2} \frac{1}{2}\, d\theta = \frac{\pi}{4}.$$

Equivalently, using the closed-form Fubini-Study distance:

$$d_{\mathrm{FS}}(|\psi\rangle, |\phi\rangle) = \arccos |\langle\psi|\phi\rangle|.$$

$$|\langle 0|+\rangle| = \frac{1}{\sqrt{2}}, \qquad d_{\mathrm{FS}} = \arccos\frac{1}{\sqrt{2}} = \frac{\pi}{4}. \quad \checkmark$$

Exercise 1.2: Entanglement of a Parametric Two-Qubit State

Motivation:

Entanglement is the quintessentially quantum resource — absent in classical physics and central to quantum computation, communication, and metrology. The simplest laboratory for studying entanglement is a two-qubit system parametrized by a single angle θ. This exercise develops the core machinery: partial trace, purity, and von Neumann entropy, using a state that interpolates smoothly from separable to maximally entangled.

Problem Statement:

Consider the parametric two-qubit state:

$$|\psi(\theta)\rangle = \cos\theta\,|00\rangle + \sin\theta\,|11\rangle.$$

(a) Compute the reduced density matrix $\rho_A = \mathrm{Tr}_B(|\psi\rangle\langle\psi|)$.

(b) Compute the purity $\gamma = \mathrm{Tr}(\rho_A^2)$ and the von Neumann entropy $S = -\mathrm{Tr}(\rho_A \ln \rho_A)$ as functions of θ.

(c) Show the state is maximally entangled at $\theta = \pi/4$ and separable at $\theta = 0, \pi/2$.

Pedantic Step-by-Step Solution:

Part (a): Reduced density matrix

The full density matrix is:

$$|\psi\rangle\langle\psi| = \cos^2\theta\,|00\rangle\langle 00| + \sin^2\theta\,|11\rangle\langle 11| + \cos\theta\sin\theta\big(|00\rangle\langle 11| + |11\rangle\langle 00|\big).$$

Tracing over subsystem B: the partial trace acts on the second tensor factor. The diagonal terms survive since $\langle 0|0\rangle = \langle 1|1\rangle = 1$. The off-diagonal terms vanish because $\langle 0|1\rangle = 0$:

$$\mathrm{Tr}_B(|00\rangle\langle 11|) = |0\rangle\langle 1| \cdot \underbrace{\langle 1|0\rangle}_{=0} = 0.$$

Therefore:

$$\boxed{\rho_A = \cos^2\theta\,|0\rangle\langle 0| + \sin^2\theta\,|1\rangle\langle 1| = \begin{pmatrix} \cos^2\theta & 0 \\ 0 & \sin^2\theta \end{pmatrix}.}$$

The reduced state is **diagonal** in the computational basis, with eigenvalues $\lambda_0 = \cos^2\theta$ and $\lambda_1 = \sin^2\theta$.

Part (b): Purity and entropy

Purity:

$$\gamma = \mathrm{Tr}(\rho_A^2) = \cos^4\theta + \sin^4\theta.$$

Using the identity $\cos^4\theta + \sin^4\theta = 1 - 2\cos^2\theta\sin^2\theta = 1 - \frac{1}{2}\sin^2(2\theta)$:

$$\gamma = 1 - \frac{1}{2}\sin^2(2\theta).$$

Von Neumann entropy:

$$S = -\cos^2\theta\,\ln(\cos^2\theta) - \sin^2\theta\,\ln(\sin^2\theta).$$

This is the binary entropy function $h(\cos^2\theta)$.

Part (c): Extremal cases

$\theta = \pi/4$ **(maximally entangled):** $\cos^2(\pi/4) = \sin^2(\pi/4) = 1/2$.

$$\rho_A = \frac{I}{2}, \qquad \gamma = \frac{1}{2}, \qquad S = \ln 2.$$

The reduced state is maximally mixed — one bit of entanglement. The state is the Bell state $(|00\rangle + |11\rangle)/\sqrt{2}$.

$\theta = 0$ **or** $\pi/2$ **(separable):** The state is $|00\rangle$ or $|11\rangle$ — a product state.

$$\rho_A = |0\rangle\langle 0| \text{ or } |1\rangle\langle 1|, \qquad \gamma = 1, \qquad S = 0.$$

Zero entanglement. The purity is maximal ($\gamma = 1$), confirming the reduced state is pure — the hallmark of a product state.

Exercise 1.3: The Depolarizing Channel

Motivation:

The **depolarizing channel** is the simplest model of symmetric noise: it replaces a quantum state with the maximally mixed state with some probability p. It arises from the **Pauli twirl** — averaging any single-qubit channel over the Pauli group yields a depolarizing channel. This universality makes it central to error analysis in quantum computing, randomized benchmarking, and the theory of quantum error correction thresholds.

Problem Statement:

The single-qubit depolarizing channel is $\mathcal{E}_p(\rho) = (1-p)\rho + (p/2)I$, with equivalent Kraus form:

$$\mathcal{E}_p(\rho) = \left(1 - \frac{3p}{4}\right)\rho + \frac{p}{4}\big(X\rho X + Y\rho Y + Z\rho Z\big).$$

(a) Show that $\mathcal{E}_p$ is **unital**: $\mathcal{E}_p(I/2) = I/2$.

(b) Verify the Kraus completeness relation $\sum_k K_k^\dagger K_k = I$.

(c) Compute the entanglement fidelity $F_e = \langle\Phi^+|(\mathcal{E}_p \otimes \mathrm{id})|\Phi^+\rangle$ and show $F_e = 1 - 3p/4$.

Pedantic Step-by-Step Solution:

Part (a): Unitality

A channel is **unital** if it preserves the identity: $\mathcal{E}(I) = I$. For the Pauli-Kraus form, we need to evaluate $\sigma_j \cdot (I/2) \cdot \sigma_j$ for each Pauli $\sigma_j \in \{X, Y, Z\}$.

Since each Pauli matrix is unitary ($\sigma_j^\dagger \sigma_j = I$), we have:

$$\sigma_j \frac{I}{2} \sigma_j = \frac{1}{2}\sigma_j\sigma_j = \frac{I}{2}.$$

Substituting into the channel:

$$\mathcal{E}_p\left(\frac{I}{2}\right) = \left(1 - \frac{3p}{4}\right)\frac{I}{2} + \frac{p}{4}\cdot 3 \cdot \frac{I}{2} = \frac{I}{2}\left(1 - \frac{3p}{4} + \frac{3p}{4}\right) = \frac{I}{2}. \quad \checkmark$$

Interpretation: Unitality means the channel preserves the maximally mixed state — it does not "cool" the system (unlike amplitude damping). The depolarizing channel adds noise symmetrically in all directions.

Part (b): Kraus completeness

The Kraus operators are $K_0 = \sqrt{1-3p/4}\, I$ and $K_j = \sqrt{p/4}\, \sigma_j$ for $j \in \{1,2,3\}$.

$$\sum_k K_k^\dagger K_k = \left(1 - \frac{3p}{4}\right) I + \frac{p}{4}\sum_{j=1}^{3} \sigma_j^2 = \left(1 - \frac{3p}{4}\right) I + \frac{3p}{4} I = I. \quad \checkmark$$

Part (c): Entanglement fidelity

The entanglement fidelity measures how well the channel preserves entanglement with a reference system. For a maximally entangled state $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$:

$$F_e = \sum_k |\langle \Phi^+|(K_k \otimes I)|\Phi^+\rangle|^2.$$

Using the identity $\langle \Phi^+|(A \otimes I)|\Phi^+\rangle = \mathrm{Tr}(A)/2$:

- $K_0 = \sqrt{1-3p/4}\, I$: $|\mathrm{Tr}(K_0)/2|^2 = (1 - 3p/4)$.
- $K_j = \sqrt{p/4}\, \sigma_j$: $|\mathrm{Tr}(\sigma_j)/2|^2 = 0$ (Paulis are traceless).

$$\boxed{F_e = 1 - \frac{3p}{4}.}$$

At $p = 0$ (no noise): $F_e = 1$. At $p = 1$ (full depolarization): $F_e = 1/4$ — the minimum for a single-qubit channel, corresponding to the completely depolarizing channel.

Exercise 1.4: The Adjoint Representation and Dynamical Lie Algebra

Motivation:

The **dynamical Lie algebra** (DLA) of a parametrized quantum circuit determines which unitaries the circuit can generate. If the DLA is a proper subalgebra of $\mathfrak{su}(D)$, the circuit is **restricted** to a submanifold of the unitary group — it cannot prepare arbitrary states. Computing the DLA by iterated commutators is the algebraic procedure that identifies these constraints. A smaller DLA means the gradient variance scales as $\Omega(1/\dim(\mathrm{DLA}))$ rather than $\Omega(1/(D^2-1))$, which can **mitigate barren plateaus** via symmetry.

Problem Statement:

Consider the 2-qubit transverse-field Ising generators:

$$H_1 = \sigma_z \otimes I, \qquad H_2 = I \otimes \sigma_z, \qquad H_3 = \sigma_x \otimes \sigma_x.$$

(a) Compute the commutator $[-iH_1, -iH_3]$ and express it in terms of Pauli tensor products.

(b) Starting from $\{-iH_1, -iH_2, -iH_3\}$, iterate all nested commutators until the algebra closes. Show $\dim(\mathrm{DLA}) = 6$.

(c) Verify that $\sigma_z \otimes \sigma_z$ commutes with all three generators. This $\mathbb{Z}_2$ parity symmetry restricts the DLA to a proper subalgebra of $\mathfrak{su}(4)$.

Pedantic Step-by-Step Solution:

Part (a): The first commutator

We compute $[-iH_1, -iH_3]$. Using the identity $[-iA, -iB] = -[A, B]$:

$$[H_1, H_3] = [\sigma_z \otimes I,\ \sigma_x \otimes \sigma_x].$$

For tensor products, $[A \otimes B, C \otimes D] = [A, C] \otimes BD$ when $[B, D] = 0$ and $\{B, D\} = 2BD$, but more generally we use the identity:

$$[A \otimes B, C \otimes D] = AC \otimes BD - CA \otimes DB.$$

Here $B = I$ commutes with everything, so:

$$[H_1, H_3] = [\sigma_z, \sigma_x] \otimes I \cdot \sigma_x = [\sigma_z, \sigma_x] \otimes \sigma_x.$$

Using the Pauli commutation relation $[\sigma_z, \sigma_x] = +2i\sigma_y$:

$$[-iH_1, -iH_3] = -[H_1, H_3] = -2i\, \sigma_y \otimes \sigma_x.$$

This produces a **new generator** $\sigma_y \otimes \sigma_x$ not in the original set.

Part (b): Iterating to closure

We continue taking commutators of existing generators with each other until no new independent elements appear. Starting from $\{-i\sigma_z I,\ -iI\sigma_z,\ -i\sigma_x\sigma_x\}$ and the new $-i\sigma_y\sigma_x$:
The process generates (up to normalization) the six independent Pauli strings:

$$\{\sigma_z I,\ I\sigma_z,\ \sigma_x\sigma_x,\ \sigma_y\sigma_x,\ \sigma_x\sigma_y,\ \sigma_y\sigma_y\}.$$

All further commutators yield linear combinations of these six elements. Therefore:

$$\boxed{\dim(\mathrm{DLA}) = 6.}$$

Since $\dim \mathfrak{su}(4) = 15$, the circuit can only explore a 6-dimensional submanifold of the full 15-dimensional unitary group. Nine directions in $\mathfrak{su}(4)$ are inaccessible — these correspond to operators that break the $\mathbb{Z}_2$ parity symmetry.

Part (c): Parity symmetry

The parity operator $P = \sigma_z \otimes \sigma_z$ satisfies:

$$[P, \sigma_z \otimes I] = 0, \qquad [P, I \otimes \sigma_z] = 0, \qquad [P, \sigma_x \otimes \sigma_x] = 0.$$

The first two are immediate since $[\sigma_z, \sigma_z] = 0$. For the third: $P \cdot H_3 = (\sigma_z\sigma_x) \otimes (\sigma_z\sigma_x) = (i\sigma_y) \otimes (i\sigma_y) = -\sigma_y\sigma_y$, and $H_3 \cdot P = (\sigma_x\sigma_z) \otimes (\sigma_x\sigma_z) = (-i\sigma_y)(-i\sigma_y) = -\sigma_y\sigma_y$, so $[P, H_3] = 0$. Since P commutes with all generators, by Schur's lemma the Hilbert space decomposes into **parity sectors** ($P = +1$ and $P = -1$), and the DLA cannot connect them. This is why $\dim(\mathrm{DLA}) = 6 < 15$.

Exercise 1.5: Schur–Weyl Decomposition for $k = 2$

Motivation:

The **SWAP operator** F is the fundamental building block of Schur-Weyl duality — the decomposition of $(\mathbb{C}^D)^{\otimes k}$ under the joint action of $U(D)$ and the symmetric group S_k. For $k = 2$, the decomposition is elementary: the two-copy space splits into symmetric and antisymmetric subspaces. The resulting **SWAP trick** $\mathrm{Tr}(AB) = \mathrm{Tr}[(A \otimes B)F]$ is the workhorse identity behind purity estimation, shadow tomography, and Weingarten calculus throughout this book.

Problem Statement:

The SWAP operator F on $\mathbb{C}^D \otimes \mathbb{C}^D$ exchanges tensor factors: $F|a,b\rangle = |b,a\rangle$.
(a) Prove $\mathrm{Tr}(F) = D$ and $\mathrm{Tr}(F^2) = D^2$.
(b) Show $\dim(\mathrm{Sym}^2) = D(D+1)/2$ and $\dim(\wedge^2) = D(D-1)/2$, and verify $\dim(\mathrm{Sym}^2)+\dim(\wedge^2) = D^2$.
(c) For $\rho = I/D$ (maximally mixed), verify the SWAP trick: $\mathrm{Tr}(\rho^2) = \mathrm{Tr}[(\rho\otimes\rho)F] = 1/D$.
Pedantic Step-by-Step Solution:

Part (a): Traces of the SWAP operator

In the computational basis, the matrix elements of F are:

$$F_{(a,b),(c,d)} = \delta_{a,d}\,\delta_{b,c}.$$

Trace of F:

$$\mathrm{Tr}(F) = \sum_{a,b} F_{(a,b),(a,b)} = \sum_{a,b} \delta_{a,b}\,\delta_{b,a} = \sum_{a,b} \delta_{a,b} = D.$$

Only diagonal elements ($a = b$) contribute.
Trace of F^2: Since F is an involution ($F^2 = I$, because swapping twice returns the original), $\mathrm{Tr}(F^2) = \mathrm{Tr}(I_{D^2}) = D^2$.

Part (b): Symmetric and antisymmetric subspace dimensions

Since $F^2 = I$, the eigenvalues of F are ± 1. The projectors onto the eigenspaces are:

$$P_+ = \frac{I+F}{2} \quad \text{(symmetric)}, \qquad P_- = \frac{I-F}{2} \quad \text{(antisymmetric)}.$$

Their dimensions (traces of the projectors):

$$\dim(\mathrm{Sym}^2) = \mathrm{Tr}(P_+) = \frac{D^2+D}{2} = \frac{D(D+1)}{2},$$

$$\dim(\wedge^2) = \mathrm{Tr}(P_-) = \frac{D^2-D}{2} = \frac{D(D-1)}{2}.$$

Check: $\frac{D(D+1)}{2} + \frac{D(D-1)}{2} = \frac{D(D+1+D-1)}{2} = D^2.$ ✓
For qubits ($D = 2$): Sym^2 is the 3-dimensional triplet space, $\wedge^2$ is the 1-dimensional singlet.

Part (c): The SWAP trick

The **SWAP trick** is the identity:

$$\mathrm{Tr}(AB) = \mathrm{Tr}\big[(A\otimes B)\,F\big].$$

Proof: In components,

$$\mathrm{Tr}[(A\otimes B)F] = \sum_{a,b} (A\otimes B)_{(a,b),(c,d)}\,F_{(c,d),(a,b)} = \sum_{a,b} A_{a,b}\,B_{b,a} = \mathrm{Tr}(AB).$$

Application: Setting $A = B = \rho = I/D$:

$$\mathrm{Tr}(\rho^2) = \mathrm{Tr}[(\rho\otimes\rho)F] = \frac{1}{D^2}\,\mathrm{Tr}(F) = \frac{D}{D^2} = \frac{1}{D}. \quad ✓$$

This confirms that the purity of the maximally mixed state equals $1/D$ — it becomes arbitrarily small as the Hilbert space dimension grows.

Exercise 1.6: Choi Matrix of the Amplitude-Damping Channel

Motivation:

Every physical quantum channel (CPTP map) has a **Choi matrix** — a positive semidefinite operator on the doubled Hilbert space that fully characterises the channel's action. The Choi-Jamiołkowski isomorphism $\mathcal{E} \leftrightarrow J(\mathcal{E})$ converts the abstract question "is $\mathcal{E}$ completely positive?" into the concrete check "is $J \geq 0$?". The amplitude-damping channel, which models the irreversible decay of a qubit (T_1 process in NMR/superconducting qubits), is the canonical example of a non-unital quantum channel.

Problem Statement:

The amplitude-damping channel has Kraus operators

$$K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}, \qquad K_1 = \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix},$$

where $\gamma \in [0, 1]$ is the damping probability.

(a) Construct the Choi matrix $J(\mathcal{E}) = \sum_{j,k} \mathcal{E}(|j\rangle\langle k|) \otimes |j\rangle\langle k|$.

(b) Show the eigenvalues of J are $\{2-\gamma, \gamma, 0, 0\}$. Verify $J \geq 0$ for $\gamma \in [0, 1]$.

(c) Interpret the limiting cases $\gamma = 0$ (identity) and $\gamma = 1$ (reset to $|0\rangle$).

Pedantic Step-by-Step Solution:

Prerequisite: Trace preservation

Before constructing the Choi matrix, we verify that the Kraus operators satisfy the trace-preservation condition $\sum_\mu K_\mu^\dagger K_\mu = I$:

$$K_0^\dagger K_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1-\gamma \end{pmatrix}, \qquad K_1^\dagger K_1 = \begin{pmatrix} 0 & 0 \\ 0 & \gamma \end{pmatrix}.$$

$$K_0^\dagger K_0 + K_1^\dagger K_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I. \quad \checkmark$$

Part (a): Constructing the Choi matrix

The Choi matrix lives on $\mathbb{C}^2 \otimes \mathbb{C}^2$ and is constructed by applying the channel to one half of a maximally entangled state. Equivalently, we compute $\mathcal{E}(|j\rangle\langle k|)$ for all basis elements and assemble:

$$J = \sum_{j,k=0}^{1} \mathcal{E}(|j\rangle\langle k|) \otimes |j\rangle\langle k|,$$

where $\mathcal{E}(\rho) = K_0 \rho K_0^\dagger + K_1 \rho K_1^\dagger$.

Computing each block:

$$\mathcal{E}(|0\rangle\langle 0|) = K_0|0\rangle\langle 0|K_0^\dagger + K_1|0\rangle\langle 0|K_1^\dagger = |0\rangle\langle 0| + 0 = |0\rangle\langle 0|.$$

$$\mathcal{E}(|0\rangle\langle 1|) = \sqrt{1-\gamma}\,|0\rangle\langle 1|, \qquad \mathcal{E}(|1\rangle\langle 0|) = \sqrt{1-\gamma}\,|1\rangle\langle 0|.$$

$$\mathcal{E}(|1\rangle\langle 1|) = (1-\gamma)|1\rangle\langle 1| + \gamma|0\rangle\langle 0|.$$

The last equation is the key: the K_1 Kraus operator transfers population from $|1\rangle$ to $|0\rangle$ with probability γ.

Assembling in the ordered basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$:

$$\boxed{J = \begin{pmatrix} 1 & 0 & 0 & \sqrt{1-\gamma} \\ 0 & \gamma & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \sqrt{1-\gamma} & 0 & 0 & 1-\gamma \end{pmatrix}}.$$

Part (b): Eigenvalue analysis

The Choi matrix is **block-diagonal** with respect to the subspaces $\{|00\rangle, |11\rangle\}$ and $\{|01\rangle, |10\rangle\}$.
The $|01\rangle$ entry is simply γ, and the $|10\rangle$ entry is 0. These contribute eigenvalues γ and 0.
The 2×2 block in the $\{|00\rangle, |11\rangle\}$ sector is

$$B = \begin{pmatrix} 1 & \sqrt{1-\gamma} \\ \sqrt{1-\gamma} & 1-\gamma \end{pmatrix}.$$

Its trace and determinant:

$$\mathrm{Tr}(B) = 2-\gamma, \qquad \det(B) = 1 \cdot (1-\gamma) - (\sqrt{1-\gamma})^2 = 0.$$

Since $\det(B) = 0$ and $\mathrm{Tr}(B) = 2-\gamma$, the eigenvalues of B are $\{2-\gamma, 0\}$.
Full spectrum:

$$\boxed{\mathrm{spec}(J) = \{2-\gamma,\ \gamma,\ 0,\ 0\}.}$$

For $\gamma \in [0,1]$: both $2-\gamma \geq 1 > 0$ and $\gamma \geq 0$, so $J \geq 0$. This confirms **complete positivity** via the Choi theorem.

Part (c): Limiting cases

$\gamma = 0$ **(identity channel):** Eigenvalues $\{2,0,0,0\}$ — rank 1. The Choi matrix is $J = 2|\Omega\rangle\langle\Omega|$ where $|\Omega\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ is the maximally entangled state. A rank-1 Choi matrix is the hallmark of a unitary channel.
$\gamma = 1$ **(reset channel):** Eigenvalues $\{1,1,0,0\}$ — rank 2. The channel completely erases the input: $\mathcal{E}(\rho) = |0\rangle\langle 0|$ for all ρ. The two nonzero eigenvalues reflect the two Kraus operators needed.
The transition from rank 1 to rank 2 as γ increases from 0 reflects the onset of genuine decoherence — the channel can no longer be undone by a unitary.

Exercise 1.7: Dimension of the Lie Algebras $\mathfrak{u}(D)$ and $\mathfrak{su}(D)$

Motivation:

The dimension of the Lie algebra $\mathfrak{su}(D)$ determines the number of **independent generators** needed to parametrize arbitrary unitary evolutions on D-dimensional Hilbert space. For N qubits ($D = 2^N$), this dimension grows exponentially. It underlies both the power of quantum computation (an exponentially large parameter space) and the barren plateau problem (gradients diluted over an exponentially large manifold).
Problem Statement:

(a) Count the real degrees of freedom in a $D \times D$ skew-Hermitian matrix and show $\dim_{\mathbb{R}} \mathfrak{u}(D) = D^2$.
(b) The tracelessness constraint $\mathrm{Tr}(X) = 0$ defines $\mathfrak{su}(D) \subset \mathfrak{u}(D)$. Show $\dim_{\mathbb{R}} \mathfrak{su}(D) = D^2 - 1$.
(c) For N qubits ($D = 2^N$), how does $\dim \mathfrak{su}(D)$ scale with N? Relate to the barren plateau phenomenon.
Pedantic Step-by-Step Solution:

Part (a): Dimension of $\mathfrak{u}(D)$

The Lie algebra $\mathfrak{u}(D)$ consists of all skew-Hermitian matrices: $X = -X^\dagger$. This constraint relates entries pairwise.
Diagonal entries (X_{jj}): The constraint $X_{jj} = -\overline{X_{jj}}$ forces each diagonal entry to be purely imaginary: $X_{jj} = i\theta_j$ with $\theta_j \in \mathbb{R}$. This gives D **real parameters**.
Off-diagonal entries ($j < k$): The upper triangle has $D(D-1)/2$ complex entries, each with 2 real parameters. The lower triangle is determined by $X_{kj} = -\overline{X_{jk}}$. This gives $D(D-1)$ **real parameters**.
Total:

$$\boxed{\dim_{\mathbb{R}} \mathfrak{u}(D) = D + D(D-1) = D^2.}$$

Part (b): Dimension of $\mathfrak{su}(D)$

The special unitary algebra $\mathfrak{su}(D)$ imposes the additional constraint $\mathrm{Tr}(X) = 0$. Since

$$\mathrm{Tr}(X) = \sum_{j=1}^{D} X_{jj} = i \sum_{j=1}^{D} \theta_j,$$

the trace vanishes if and only if $\sum_j \theta_j = 0$. This is a **single real constraint** on the D imaginary diagonal parameters. The off-diagonal entries are unaffected.

$$\boxed{\dim_{\mathbb{R}} \mathfrak{su}(D) = D^2 - 1.}$$

This matches the familiar count: SU(2) has $4 - 1 = 3$ generators (the Pauli matrices).

Part (c): Exponential scaling and barren plateaus

For N qubits, $D = 2^N$ and

$$\dim \mathfrak{su}(2^N) = 4^N - 1.$$

This exponential growth has a direct physical consequence: the **gradient variance** in a variational quantum circuit that forms a 2-design scales as

$$\mathrm{Var}\left[\frac{\partial C}{\partial \theta}\right] = O\left(\frac{1}{4^N}\right).$$

The gradient is exponentially small because the cost function "lives" on a manifold with $4^N - 1$ dimensions — the gradient signal is spread over an exponentially large space. This is the **barren plateau** phenomenon.

Exercise 1.8: Surjectivity of the Exponential Map on $U(D)$

Motivation:

The exponential map $\exp : \mathfrak{u}(D) \to U(D)$ connects the Lie algebra (infinitesimal generators) to the Lie group (finite unitaries). A crucial question: is every unitary U reachable as e^X for some generator X? For the unitary group the answer is **yes** — the exponential map is surjective. This is a special property of compact groups and fails for non-compact groups (e.g., $\mathrm{GL}(n, \mathbb{R})$ contains matrices with negative determinant, unreachable from exp).

Problem Statement:

Show that every $U \in U(D)$ can be written as $U = \exp(X)$ for some $X \in \mathfrak{u}(D)$.

Pedantic Step-by-Step Solution:

Step 1: Spectral decomposition of U

Since U is unitary, it is **normal** ($UU^\dagger = U^\dagger U$), so the spectral theorem gives:

$$U = \sum_{j=1}^{D} e^{i\theta_j} |v_j\rangle\langle v_j|,$$

where $\{|v_j\rangle\}$ is an orthonormal eigenbasis and $\theta_j \in (-\pi, \pi]$ are the eigenphases.

Step 2: Construct the generator

Define

$$X = \sum_{j=1}^{D} i\theta_j \,|v_j\rangle\langle v_j|.$$

Step 3: Verify X is skew-Hermitian

$$X^\dagger = \sum_j (-i\theta_j)|v_j\rangle\langle v_j| = -X. \quad \checkmark$$

So $X \in \mathfrak{u}(D)$.

Step 4: Verify $\exp(X) = U$

The matrix exponential acts diagonally on the eigenbasis:

$$\exp(X) = \sum_j \exp(i\theta_j)|v_j\rangle\langle v_j| = U. \quad \checkmark$$

Remark on uniqueness

The map is **not injective**: X and $X + 2\pi i\,|v_j\rangle\langle v_j|$ produce the same U. The choice $\theta_j \in (-\pi, \pi]$ selects the principal logarithm, but other branches exist. This multi-valuedness reflects the compactness of $U(D)$.

Exercise 1.9: Haar Invariance on $U(1)$

Motivation:

The **Haar measure** is the unique translation-invariant probability measure on a compact Lie group. Its existence and uniqueness underpin every Haar-random construction in this book — from random matrix theory to unitary designs. The circle group $U(1)$ is the simplest compact Lie group and provides the clearest illustration of these concepts: uniqueness follows from a Fourier analysis argument, and the connection to determinant phases generalizes directly to $U(D)$.

Problem Statement:

The circle group $U(1) = \{e^{i\theta} : \theta \in [0, 2\pi)\}$ has Haar measure $d\mu = d\theta/(2\pi)$.

(a) Verify translation invariance: $\int f(e^{i(\theta+\alpha)})\,\frac{d\theta}{2\pi} = \int f(e^{i\theta})\,\frac{d\theta}{2\pi}$.

(b) Prove uniqueness: any continuous translation-invariant density $p(\theta)$ must equal $1/(2\pi)$.

(c) For Haar-random $U \in U(D)$, show that the phase of $\det(U)$ is uniformly distributed on $[0, 2\pi)$.

Pedantic Step-by-Step Solution:

Part (a): Translation invariance

Under the substitution $\theta' = \theta + \alpha$, $d\theta' = d\theta$, and the domain $[0, 2\pi)$ maps to $[\alpha, 2\pi + \alpha)$. Since the integrand $f(e^{i\theta'})$ is 2π-periodic in θ', the integral over any interval of length 2π is the same:

$$\int_0^{2\pi} f(e^{i(\theta+\alpha)})\frac{d\theta}{2\pi} = \int_\alpha^{2\pi+\alpha} f(e^{i\theta'})\frac{d\theta'}{2\pi} = \int_0^{2\pi} f(e^{i\theta'})\frac{d\theta'}{2\pi}. \quad \checkmark$$

This is the defining property of the Haar measure: it is invariant under group multiplication (here: phase shifts).

Part (b): Uniqueness via Fourier analysis

Suppose $p(\theta)$ is a translation-invariant density. Expand in Fourier modes:

$$p(\theta) = \sum_{n=-\infty}^{\infty} c_n\, e^{in\theta}.$$

Translation invariance requires $\int f(\theta + \alpha)\, p(\theta)\, d\theta = \int f(\theta)\, p(\theta)\, d\theta$ for all α and all test functions f. This implies:

$$e^{in\alpha} c_n = c_n \qquad \forall\, \alpha.$$

For $n \neq 0$, we can choose α such that $e^{in\alpha} \neq 1$, forcing $c_n = 0$. Only c_0 survives.
Normalization $\int_0^{2\pi} p(\theta)\, d\theta = 1$ fixes $c_0 = 1/(2\pi)$. Therefore:

$$\boxed{p(\theta) = \frac{1}{2\pi}.}$$

The Haar measure is the **unique** translation-invariant measure on $U(1)$.

Part (c): Determinant phase is uniform

For $U \in U(D)$, write $\det(U) = e^{i\Phi}$ where $\Phi = \sum_k \varphi_k$ is the sum of eigenphases.
Left multiplication $U \to e^{i\alpha}U$ shifts every eigenphase by α, so $\det(e^{i\alpha}U) = e^{iD\alpha}\det(U)$. This shifts $\Phi \to \Phi + D\alpha$.
Since the Haar measure is left-invariant, the distribution of $\Phi + D\alpha$ equals the distribution of Φ for all α. By the uniqueness result of part (b) (applied to the group $U(1)$ acting on the determinant phase), Φ must be uniformly distributed on $[0, 2\pi)$.

Exercise 1.10: Lebesgue Measure, Counting Measure, and σ-Additivity

1. Physical Motivation

In quantum information theory, we constantly need to "average over random unitaries" — for instance, to compute the Page entropy of a random state, to derive the concentration of measure that causes barren plateaus, or to prove that random circuits form unitary designs. All of these require integrating functions over the unitary group $U(D)$ with respect to its **Haar measure**.
But what *is* a measure, exactly? And why can't we assign a "size" to every subset of $U(D)$, the way we assign probabilities to subsets of a finite set?
This exercise addresses these questions through two concrete examples:

- The **Lebesgue measure** on $\mathbb{R}$ — the prototypical "translation-invariant volume" that the Haar measure generalizes.
- The **counting measure** on the Pauli group — the simplest Haar measure, on a finite group.

The Vitali argument then shows *why* σ-algebras are necessary: the naive hope that every subset can be measured consistently is mathematically impossible on the continuum. This is the conceptual foundation that makes the Borel σ-algebra and the Haar measure well-defined.

2. Mathematical Toolbox

Before diving into the exercise, we set up the vocabulary. Measure theory has three ingredients that build on each other.

Tool 0 — The cast of characters: Ω, $\mathcal{F}$, μ.

- Ω is the **sample space** — the set of all possible outcomes. Think of it as the universe of objects we want to study. Examples: $\Omega = \{1, 2, 3, 4, 5, 6\}$ for a die roll, $\Omega = \mathbb{R}$ for a continuous variable, $\Omega = U(D)$ for a random unitary matrix.
- $\mathcal{F}$ is a **σ-algebra** on Ω — a collection of subsets of Ω that we declare 'measurable'. A subset $A \subseteq \Omega$ is called **measurable** if $A \in \mathcal{F}$; this means that the measure function μ can be evaluated on A, giving a well-defined size $\mu(A) \geq 0$. A subset that is *not* in $\mathcal{F}$ is **non-measurable**: no consistent size can be assigned to it at all (not 'unknown' or 'infinite', but genuinely undefined). $\mathcal{F}$ must satisfy three axioms: (S1) $\Omega \in \mathcal{F}$, (S2) if $A \in \mathcal{F}$ then $\Omega \setminus A \in \mathcal{F}$ (closed under complements), (S3) if $A_1, A_2, \ldots \in \mathcal{F}$ then $\bigcup_k A_k \in \mathcal{F}$ (closed under countable unions). The pair $(\Omega, \mathcal{F})$ is called a **measurable space** — it tells us *what* we can measure, but not *how much*.

- $\mu : \mathcal{F} \to [0, \infty]$ is a **measure** — the function that actually assigns a non-negative size (length, volume, probability) to each measurable set $A \in \mathcal{F}$. The triple $(\Omega, \mathcal{F}, \mu)$ is called a **measure space**. If $\mu(\Omega) = 1$, then μ is a **probability measure** and $\mu(A)$ is the probability that a randomly chosen element of Ω falls in A.

For the Haar measure on the unitary group, the concrete instantiation is:

$$\Omega = U(D), \quad \mathcal{F} = \mathcal{B}(U(D)) \text{ (Borel } \sigma\text{-algebra)}, \quad \mu = \mu_{\text{Haar}}.$$

Why not just use all subsets? For finite sets (like the Pauli group), we *can* — every subset is measurable under $\mathcal{F} = 2^{\Omega}$ (the power set). But for continuous sets like $\mathbb{R}$ or $U(D)$, the power set is 'too large': the Vitali construction (Tool 4) shows that no consistent size assignment exists on *all* subsets. We must restrict to the Borel σ-algebra.

Tool 1 — What a measure must satisfy. A measure μ on $(\Omega, \mathcal{F})$ must obey two rules:

1. $\mu(\emptyset) = 0$ (the empty set has zero size).
2. **Countable additivity** (σ-additivity): for pairwise disjoint measurable sets $A_1, A_2, \ldots \in \mathcal{F}$:

$$\mu\left(\bigsqcup_{k=1}^{\infty} A_k\right) = \sum_{k=1}^{\infty} \mu(A_k).$$

This is the key property: the measure of a union of non-overlapping pieces equals the sum of the measures. It is the continuous generalization of the familiar rule $P(A \cup B) = P(A) + P(B)$ for disjoint events.

Tool 2 — Lebesgue measure on $\mathbb{R}$. The Lebesgue measure λ is the unique (up to normalization) translation-invariant measure on the Borel σ-algebra $\mathcal{B}(\mathbb{R})$. It assigns to every interval its length: $\lambda([a, b]) = b - a$. Key consequence: single points have measure zero ($\lambda(\{x\}) = 0$), and therefore any countable set has measure zero by σ-additivity.

Tool 3 — Counting measure on a finite group. For a finite group $G = \{g_1, \ldots, g_n\}$, define $\mu(A) = |A|/|G|$. This assigns equal weight $1/|G|$ to each element. It is invariant under left-multiplication ($\mu(gA) = \mu(A)$) because $g\cdot$ is a bijection. This is the **Haar measure on a finite group** — the simplest example of what we will generalize to $U(D)$.

Tool 4 — The Vitali construction (1905). Giuseppe Vitali was an Italian mathematician who proved an impossibility result: there is no way to assign a translation-invariant, σ-additive size to *every* subset of $[0, 1)$ while keeping $\mu([0, 1)) = 1$. The proof constructs a specific pathological set — the **Vitali set** V — by choosing one representative from each equivalence class of the relation '$x \sim y$ if and only if $x - y$ is rational' (an equivalence class of x is just the set of all numbers in $[0, 1)$ that differ from x by a rational number). This construction requires the **axiom of choice** (to simultaneously pick one element from each of the uncountably many equivalence classes). The resulting set V cannot be assigned any consistent size: if $\mu(V) = 0$, then the countable union of its translates covers all of $[0, 1)$ but has measure $0 \neq 1$; if $\mu(V) > 0$, the same union has infinite measure $\neq 1$.

The lesson: on continuous spaces, we *must* restrict to a σ-algebra of well-behaved (Borel) sets.

Tool 5 — Pauli group. The single-qubit Pauli group is $\mathcal{P}_1 = \{\pm I, \pm iI, \pm X, \pm iX, \pm Y, \pm iY, \pm Z, \pm iZ\}$ with $|\mathcal{P}_1| = 16$. Being a finite group, it has no Vitali-type obstruction: every subset is measurable under the power-set σ-algebra $\mathcal{F} = 2^{\mathcal{P}_1}$.

3. Exercise Statement

(a) Lebesgue measure of the rationals. Let $A = \mathbb{Q} \cap [0, 1]$. Using the countability of $\mathbb{Q}$, write $A = \{q_1, q_2, \ldots\}$. Show that $\lambda(A) = 0$ via σ-additivity, and conclude $\lambda([0, 1] \setminus \mathbb{Q}) = 1$.

(b) Counting measure on the Pauli group. Define $\mu(A) = |A|/16$ on $\mathcal{P}_1$. Verify:

- Finite additivity: $\mu(A \sqcup B) = \mu(A) + \mu(B)$.
- Left-invariance: $\mu(gA) = \mu(A)$ for all $g \in \mathcal{P}_1$ and $A \subseteq \mathcal{P}_1$.

(c) The Vitali argument. Reproduce the proof that no translation-invariant, σ-additive measure on all subsets of $[0,1)$ can have $\mu([0,1)) = 1$. Identify where the axiom of choice is used.

4. Analytical Derivation

Part (a): $\lambda(\mathbb{Q} \cap [0,1]) = 0$

Enumerate $\mathbb{Q} \cap [0,1] = \{q_1, q_2, \ldots\}$ (possible since $\mathbb{Q}$ is countable). Each singleton $\{q_k\}$ has Lebesgue measure zero because for any $\varepsilon > 0$:

$$\{q_k\} \subseteq (q_k - \varepsilon/2, q_k + \varepsilon/2), \quad \text{so} \quad \lambda(\{q_k\}) \leq \varepsilon.$$

Since ε is arbitrary, $\lambda(\{q_k\}) = 0$ for each k. By σ-additivity (the singletons are pairwise disjoint):

$$\lambda(\mathbb{Q} \cap [0,1]) = \sum_{k=1}^{\infty} \lambda(\{q_k\}) = \sum_{k=1}^{\infty} 0 = 0.$$

Since $[0,1] = (\mathbb{Q} \cap [0,1]) \sqcup ([0,1] \setminus \mathbb{Q})$ and $\lambda([0,1]) = 1$:

$$\lambda([0,1] \setminus \mathbb{Q}) = 1 - 0 = 1.$$

Physical insight: A dense subset ($\mathbb{Q}$ is dense in $\mathbb{R}$) can have measure zero: denseness is a topological property, while measure is a metric/volumetric property. The two notions of "size" are independent.

Part (b): Counting measure on $\mathcal{P}_1$

Here $\Omega = \mathcal{P}_1$ (the 16-element Pauli group), $\mathcal{F} = 2^{\mathcal{P}_1}$ (all subsets are measurable), and $\mu(A) = |A|/16$.

Finite additivity: For disjoint $A, B \subseteq \mathcal{P}_1$: $|A \sqcup B| = |A| + |B|$ (elementary set theory), so $\mu(A \sqcup B) = (|A| + |B|)/16 = \mu(A) + \mu(B)$. For a finite set, finite additivity implies σ-additivity (only finitely many nonempty disjoint subsets can exist).

Left-invariance: For any fixed $g \in \mathcal{P}_1$, the map $L_g : h \mapsto gh$ is a bijection on $\mathcal{P}_1$ (with inverse $L_{g^{-1}}$). Therefore $|gA| = |L_g(A)| = |A|$, and $\mu(gA) = |A|/16 = \mu(A)$.

Physical insight: This is the Haar measure on $\mathcal{P}_1$. When we write $\mathbb{E}_{P \in \mathcal{P}}[f(P)] = \frac{1}{4^N} \sum_P f(P)$ in the twirling formulas of Chapter 3, we are integrating against this counting measure.

Part (c): The Vitali argument

Here $\Omega = [0,1)$ and we ask: can we define a measure μ on $\mathcal{F} = 2^{[0,1)}$ (the power set — i.e., *all* subsets) with translation invariance, σ-additivity, and $\mu(\Omega) = 1$?

Define an equivalence relation $\sim$ on $[0,1)$: we say $x \sim y$ if and only if the difference $x - y$ is a rational number (modulo 1). For example, $0.3 \sim 0.8$ because $0.3 - 0.8 = -0.5 = -1/2 \in \mathbb{Q}$. The equivalence class of a number x is the set of all numbers in $[0,1)$ that differ from x by a rational: $[x] = \{y \in [0,1) : y - x \in \mathbb{Q}\}$.

Step 1 (Axiom of Choice): Select exactly one representative from each equivalence class to form a set $V \subset [0,1)$. This is where the axiom of choice is used — there are uncountably many equivalence classes and we must choose from all of them simultaneously.

Step 2: Consider the translates $\{V + q \pmod 1 : q \in \mathbb{Q} \cap [0,1)\}$, where $V + q = \{v + q \pmod 1 : v \in V\}$. These translates are:

- **Pairwise disjoint** — if $v_1 + q_1 \equiv v_2 + q_2 \pmod 1$ with $v_1, v_2 \in V$, then $v_1 - v_2$ is rational, so $v_1 \sim v_2$, meaning v_1 and v_2 are in the same equivalence class and hence $v_1 = v_2$ (since V has only one representative per class), which forces $q_1 = q_2$.

- **Cover all of** $[0,1)$ — every $x \in [0,1)$ belongs to some equivalence class, so the representative $v \in V$ of that class satisfies $x \equiv v + q \pmod 1$ for some rational q.

Step 3: By σ-additivity and translation invariance:

$$1 = \mu([0,1)) = \sum_{q \in \mathbb{Q} \cap [0,1)} \mu(V + q) = \sum_q \mu(V).$$

Step 4 (Contradiction): The right-hand side is a countable sum of copies of the same number $c = \mu(V) \geq 0$. Such a sum equals 0 if $c = 0$, and ∞ if $c > 0$. Neither equals 1. Contradiction. $\square$
Where the axiom of choice is used: In Step 1, to simultaneously select one element from each of the uncountably many equivalence classes.
Why it fails for finite/countable sets: For a finite group G with $|G| = n$, each element has weight $c = 1/n > 0$, and the sum of n copies of c is $n \cdot (1/n) = 1$ — a *finite* sum of equal terms can equal 1. The paradox requires a *countably infinite* sum.
7. Summary and Connection to the Haar Measure

— Property — Lebesgue on $\mathbb{R}$ — Counting on $\mathcal{P}_1$ — Haar on $U(D)$ — —:————:——————-—:——————————:———— — Ω (sample space) — $\mathbb{R}$ — $\{g_1, \ldots, g_{16}\}$ — $U(D)$ — — $\mathcal{F}$ (σ-algebra) — $\mathcal{B}(\mathbb{R})$ (Borel) — $2^{\mathcal{P}_1}$ (power set) — $\mathcal{B}(U(D))$ (Borel) — — μ (measure) — λ (length) — $|A|/16$ (counting) — μ_{Haar} — — What 'measurable' means — $A \in \mathcal{B}(\mathbb{R})$ — Any $A \subseteq \mathcal{P}_1$ — $A \in \mathcal{B}(U(D))$ — — Invariance — Translation — Left/right multiplication — Left/right multiplication — — $\mu(\Omega)$ — ∞ (σ-finite) — 1 (probability) — 1 (probability) — — Non-measurable sets? — Yes (Vitali) — No (finite $\to$ power set works) — Yes (analogous to Vitali) —

The Haar measure on $U(D)$ is the direct generalization of both: it assigns a "volume" to Borel subsets of the unitary group in a way that is invariant under multiplication by any fixed unitary. The Lebesgue measure provides the translation-invariance intuition; the counting measure provides the group-invariance intuition. The Vitali argument shows why we must restrict to the Borel σ-algebra $\mathcal{B}(U(D))$ rather than attempting to measure all subsets of $U(D)$.

Exercise 1.11: Borel σ-Algebra and Measurability on $U(D)$

1. Physical Motivation

Every time we write an expression like

$$\mathbb{E}_U\left[f(U)\right] = \int_{U(D)} f(U)\, d\mu_{\text{Haar}}(U),$$

we are implicitly assuming that f is a **measurable function** on $U(D)$ — that is, a function whose domain and range are compatible with the σ-algebras in play. Without this property, the integral is undefined.
But what does 'measurable function' actually mean? And for the functions we care about — fidelities, entropies, OTOCs, cost function gradients — why are they measurable?
The short answer is: **continuous functions are automatically measurable**. For the functions in this book, measurability follows because they can be decomposed into a chain of continuous operations (matrix entry extraction $\to$ polynomial combination $\to$ eigenvalue computation $\to$ entropy evaluation). This exercise makes each step explicit.
The final part constructs the **power-set σ-algebra** of the finite Pauli group, showing that for finite groups the measurability question is trivial — every function is measurable — and only becomes nontrivial for continuous groups like $U(D)$.
2. Mathematical Toolbox
We first recall the key definitions (introduced in Section 1.13 and the companion exercise on Lebesgue/counting measures) and then add the tools specific to measurable functions.
Recall: Ω, $\mathcal{F}$, μ, and what 'measurable' means.

- Ω is the **sample space** (e.g., $U(D)$, $\mathbb{R}$, or a finite group).
- $\mathcal{F}$ is a **σ-algebra** on Ω: a collection of subsets of Ω closed under complements and countable unions. A subset $A \subseteq \Omega$ is called **measurable** if $A \in \mathcal{F}$, meaning we can assign it a well-defined size $\mu(A)$. A **non-measurable** subset ($A \notin \mathcal{F}$) cannot be assigned any consistent size at all.

- $\mu : \mathcal{F} \to [0, \infty]$ is a **measure** — the size assignment. The triple $(\Omega, \mathcal{F}, \mu)$ is a **measure space**.

Tool 1 — Borel σ-algebra. When Ω is a topological space (i.e., it has a notion of 'open sets' — for example, $U(D)$ with the matrix-entry topology), the **Borel σ-algebra** $\mathcal{B}(\Omega)$ is the smallest σ-algebra containing all open sets. It contains all open sets, all closed sets, all countable intersections of open sets, and so on. For the unitary group: $\mathcal{F} = \mathcal{B}(U(D))$. This is the σ-algebra on which the Haar measure lives.

Tool 2 — Measurable functions. A function $f : \Omega \to \mathbb{R}$ is called **Borel-measurable** (or simply 'measurable') if it is compatible with the σ-algebras: for every Borel set $B \subseteq \mathbb{R}$, the preimage $f^{-1}(B) = \{\omega \in \Omega : f(\omega) \in B\}$ belongs to $\mathcal{F}$. Equivalently, for every $a \in \mathbb{R}$, the set $\{\omega : f(\omega) \leq a\}$ is in $\mathcal{F}$. This ensures that questions like 'what is the probability that $f(U)$ lies in $[0.3, 0.7]$?' have well-defined answers.

Tool 3 — Continuous $\implies$ measurable. If $f : \Omega \to \mathbb{R}$ is **continuous**, then f is automatically Borel-measurable. The reason: continuous functions pull open sets back to open sets, and open sets are in $\mathcal{B}(\Omega)$ by definition. This is the key fact that makes almost every function in quantum information measurable.

Tool 4 — Composition rule. If f and g are both continuous (or both measurable), then the composition $g \circ f$ is also continuous (or measurable). This lets us build complex measurable functions from simple ones by chaining operations.

Tool 5 — Eigenvalue continuity. For an $n \times n$ Hermitian matrix H with eigenvalues $\lambda_1 \leq \cdots \leq \lambda_n$, each $\lambda_k(H)$ is a continuous function of the matrix entries. This follows from the Weyl perturbation inequality: $|\lambda_k(H) - \lambda_k(H')| \leq \|H - H'\|_{\mathrm{op}}$.

Tool 6 — Partial trace. For a bipartite Hilbert space $\mathcal{H}_A \otimes \mathcal{H}_B$, the partial trace $\mathrm{Tr}_B[\hat{\rho}]$ has matrix entries that are sums of products of entries of $\hat{\rho}$ — i.e., the partial trace is a **linear** (hence continuous) map. Therefore $U \mapsto \mathrm{Tr}_B[U|00\rangle\langle 00|U^\dagger]$ is continuous.

Tool 7 — Power-set σ-algebra. For a finite set Ω with $|\Omega| = n$, the power set 2^Ω (the collection of *all* subsets) is trivially a σ-algebra with $|2^\Omega| = 2^n$ elements. Every subset is measurable, every function is measurable. The nontrivial σ-algebra structure arises only for infinite/continuous spaces.

3. Exercise Statement

(a) Survival probability is Borel-measurable. For $\Omega = U(2)$ and $f(U) = |\langle 0|U|0\rangle|^2$, prove that f is Borel-measurable by showing it is a composition of continuous functions.

(b) Entanglement entropy is Borel-measurable. For $U \in U(4)$, the entanglement entropy $S(U) = -\mathrm{Tr}[\hat{\rho}_A \log \hat{\rho}_A]$ with $\hat{\rho}_A = \mathrm{Tr}_B[U|00\rangle\langle 00|U^\dagger]$ is measurable. Trace through the chain: matrix entries $\to$ partial trace $\to$ eigenvalues $\to$ entropy.

(c) Power-set σ-algebra of $\mathcal{P}_N$. On the Pauli group $\mathcal{P}_N = \{I, X, Y, Z\}^{\otimes N}$ (ignoring phases), verify axioms (S1)–(S3) for $2^{\mathcal{P}_N}$, and compute $|2^{\mathcal{P}_N}|$ for $N = 1, 2$.

4. Analytical Derivation

Part (a): $f(U) = |U_{00}|^2$ is continuous

We decompose f into a chain of two continuous maps:

$$U \xrightarrow{\text{step 1: extract entry}} U_{00} \xrightarrow{\text{step 2: squared modulus}} |U_{00}|^2$$

- **Step 1:** The map $U \mapsto U_{00}$ extracts the $(0, 0)$ entry of the matrix. This is a coordinate projection from $\mathbb{C}^{D\times D}$ to $\mathbb{C}$ — a linear map, hence continuous.

- **Step 2:** The map $z \mapsto |z|^2 = z\bar{z}$ is a polynomial in $\mathrm{Re}(z)$ and $\mathrm{Im}(z)$: $|z|^2 = (\mathrm{Re}\, z)^2 + (\mathrm{Im}\, z)^2$. Polynomials are continuous.

By the composition rule (Tool 4), $f = |\cdot|^2 \circ \pi_{00}$ is continuous, hence Borel-measurable by Tool 3. $\square$

Part (b): $S(U)$ is continuous

We decompose the entropy into a chain of four continuous operations:

$$U \xrightarrow{\text{(i)}} \hat{\rho} = U|00\rangle\langle 00|U^\dagger \xrightarrow{\text{(ii)}} \hat{\rho}_A = \mathrm{Tr}_B[\hat{\rho}] \xrightarrow{\text{(iii)}} (\lambda_1, \lambda_2) \xrightarrow{\text{(iv)}} S = -\sum_i \lambda_i \log \lambda_i$$

Each step is continuous:

- **(i)** $U \mapsto U|00\rangle\langle 00|U^\dagger$: the entries of $\hat{\rho}$ are $\hat{\rho}_{ij} = U_{i0}\overline{U_{j0}}$ — products of matrix entries, hence continuous (polynomials).
- **(ii)** $\hat{\rho} \mapsto \mathrm{Tr}_B[\hat{\rho}]$: partial trace is a linear map — each entry of $\hat{\rho}_A$ is a sum of entries of $\hat{\rho}$ — hence continuous (Tool 6).
- **(iii)** $\hat{\rho}_A \mapsto (\lambda_1, \lambda_2)$: eigenvalues of a Hermitian matrix depend continuously on the matrix entries (Tool 5, Weyl inequality).
- **(iv)** $(\lambda_1, \lambda_2) \mapsto -\sum_i \lambda_i \log \lambda_i$: the function $h(x) = -x \log x$ is continuous on $[0, 1]$ (with $h(0) = 0$ by convention), so $S = h(\lambda_1) + h(\lambda_2)$ is continuous.

By repeated application of the composition rule, $S(U)$ is a continuous function of the matrix entries of U, hence Borel-measurable. □

Part (c): Power-set σ-algebra

For a finite set $\Omega = \mathcal{P}_N$ with $|\Omega| = 4^N$ elements, $\mathcal{F} = 2^\Omega$ satisfies the three σ-algebra axioms:

- **(S1)** $\Omega \in 2^\Omega$: trivially, Ω is a subset of itself.
- **(S2)** If $A \in 2^\Omega$, then $\Omega \setminus A \in 2^\Omega$: every subset of a finite set is in its power set.
- **(S3)** If $A_1, A_2, \ldots \in 2^\Omega$, then $\bigcup_k A_k \subseteq \Omega$, hence $\bigcup_k A_k \in 2^\Omega$.

The number of elements: $|2^{\mathcal{P}_N}| = 2^{4^N}$.

- $N = 1$: $|2^{\mathcal{P}_1}| = 2^4 = 16$.
- $N = 2$: $|2^{\mathcal{P}_2}| = 2^{16} = 65{,}536$.

Physical insight: For finite groups, every subset is measurable (it belongs to the σ-algebra), and every function is measurable. This is why the 'average over Paulis' $\frac{1}{4^N}\sum_P f(P)$ never requires σ-algebra considerations. The nontrivial structure — which subsets are measurable and which are not — only appears for continuous groups like $U(D)$.

7. Summary and Connection to the Book

This exercise established the **measurability** of the most common functions integrated against the Haar measure in this book:

— Function — Where in the book — Why it is measurable — —:————:—————:————— — $f(U) = |\langle\psi|U|\phi\rangle|^2$ (fidelity) — Ch3: overlap distribution, Ch7: kernel — Composition of coordinate projection + $|\cdot|^2$ — — $S(U)$ (entanglement entropy) — Ch3: Page formula, Ch4: scrambling — Chain: partial trace → eigenvalues → $-x\log x$ — — $C(U) = \mathrm{Tr}[\hat{O}\,U\hat{\rho}U^\dagger]$ (cost) — Ch7: barren plateaus — Bilinear in U entries → polynomial → continuous — — OTOC(U) — Ch4: scrambling dynamics — Polynomial in U entries → continuous —

Every function we integrate against μ_{Haar} is a continuous function of the matrix entries of U, and is therefore Borel-measurable: the preimage of any Borel set in $\mathbb{R}$ belongs to $\mathcal{B}(U(D))$. The Haar integral $\int_{U(D)} f(U)\, d\mu$ is therefore well-defined for all such f.

Exercise 1.12: Sectional Curvature of $SU(2)$

Motivation:

Unitary groups form **curved** Riemannian manifolds. The geometry of geodesics on these manifolds governs how physical systems evolve under random or structured Hamiltonian dynamics. A key descriptor of this geometry is the **sectional curvature**, which describes how much geodesics (time-evolution trajectories) tend to deviate from one another.

For Lie groups equipped with a bi-invariant metric (like the Hilbert-Schmidt metric), the sectional curvature is everywhere non-negative. This implies that nearby trajectories generated by slightly different Hamiltonians will slowly drift apart (or oscillate), but they do not exhibit the exponentially fast divergence characteristic of negative curvature. This sets a fundamental continuous-time bound on quantum chaos before many-body effects are considered.

Preliminaries / Toolbox

1. The Lie Algebra $\mathfrak{u}(D)$: The tangent space at the identity consists of skew-Hermitian matrices $X^\dagger = -X$.

2. The Hilbert-Schmidt Metric: A natural bi-invariant inner product on the Lie algebra:

$$g(A, B) = -\operatorname{Tr}(AB).$$

The minus sign ensures positivity because for skew-Hermitian matrices, BA and AB have negative eigenvalues, so $-\operatorname{Tr}(A^2) = \operatorname{Tr}(A^\dagger A) \geq 0$.

3. Sectional Curvature: For a plane spanned by two orthonormal tangent vectors $X, Y \in \mathfrak{u}(D)$ (such that $g(X, X) = 1$, $g(Y, Y) = 1$, and $g(X, Y) = 0$), the sectional curvature of the bi-invariant metric is:

$$K(X, Y) = \frac{1}{4}\|[X, Y]\|_{HS}^2$$

where $\|Z\|_{HS}^2 = g(Z, Z) = -\operatorname{Tr}(Z^2)$.

4. Spherical Geometry: The n-dimensional sphere $S^n(R)$ of radius R has constant sectional curvature $K = 1/R^2$ everywhere.

Problem Statement:

The group SU(2) is diffeomorphic to the 3-sphere S^3.

(a) Verify that $X = -i\sigma_x/\sqrt{2}$ and $Y = -i\sigma_y/\sqrt{2}$ are orthonormal in the Hilbert-Schmidt metric.

(b) Compute the commutator $[X, Y]$, its norm-squared $\|[X, Y]\|_{HS}^2$, and the resulting sectional curvature $K(X, Y)$.

(c) Verify that this matches the expected curvature $K = 1/R^2$ of S^3 with radius $R = \sqrt{2}$.

Pedantic Step-by-Step Solution:

Part (a): Orthonormality

Check the norm of X:

$$\|X\|^2 = -\operatorname{Tr}(X^2) = -\operatorname{Tr}\left(\left(-i\frac{\sigma_x}{\sqrt{2}}\right)^2\right) = -\operatorname{Tr}\left(-\frac{\sigma_x^2}{2}\right) = \frac{\operatorname{Tr}(I_2)}{2} = \frac{2}{2} = 1. \quad \checkmark$$

Check the norm of Y:

$$\|Y\|^2 = -\operatorname{Tr}(Y^2) = -\operatorname{Tr}\left(-\frac{\sigma_y^2}{2}\right) = \frac{\operatorname{Tr}(I_2)}{2} = 1. \quad \checkmark$$

Check orthogonality:

$$g(X, Y) = -\operatorname{Tr}(XY) = -\operatorname{Tr}\left(\left(-i\frac{\sigma_x}{\sqrt{2}}\right)\left(-i\frac{\sigma_y}{\sqrt{2}}\right)\right) = \frac{1}{2}\operatorname{Tr}(\sigma_x\sigma_y) = \frac{1}{2}\operatorname{Tr}(i\sigma_z) = 0. \quad \checkmark$$

Part (b): Evaluating the curvature formula

Compute the commutator:

$$[X, Y] = \left[-i\frac{\sigma_x}{\sqrt{2}}, -i\frac{\sigma_y}{\sqrt{2}}\right] = -\frac{1}{2}[\sigma_x, \sigma_y] = -\frac{1}{2}(2i\sigma_z) = -i\sigma_z.$$

Compute its norm-squared:

$$\|[X, Y]\|^2 = -\operatorname{Tr}\left((-i\sigma_z)^2\right) = -\operatorname{Tr}(-\sigma_z^2) = \operatorname{Tr}(I_2) = 2.$$

The sectional curvature is therefore:

$$\boxed{K(X, Y) = \frac{1}{4}\|[X, Y]\|^2 = \frac{1}{4} \times 2 = \frac{1}{2}.}$$

Part (c): Geometric interpretation

Because the Lie algebra $\mathfrak{su}(2)$ is 3-dimensional and isotropic under the adjoint action, the sectional curvature is constant for all planes. This implies SU(2) with the bi-invariant Hilbert-Schmidt metric is an isometric to a 3-sphere.
For a sphere of radius R, the curvature is $K = 1/R^2$.

$$K = \frac{1}{2} \implies R^2 = 2 \implies R = \sqrt{2}.$$

Why $\sqrt{2}$? The standard parametrization $U = x_0 I + ix_1\sigma_x + ix_2\sigma_y + ix_3\sigma_z$ with $\sum x_i^2 = 1$ naturally puts it on the unit sphere $S^3(R = 1)$ in $\mathbb{R}^4$. However, the metric induced by $-\operatorname{Tr}(XY)$ absorbs a trace over an N-dimensional space. For qubits, $N = 2$, scaling lengths by $\sqrt{2}$ and areas (radii squared) by 2.

Exercise 1.13: Geodesic Length and the Quantum Speed Limit

Motivation:

How fast can a quantum state evolve? The **quantum speed limit** (Mandelstam-Tamm bound) sets a fundamental lower bound on the time needed to evolve between orthogonal states. This exercise derives the bound from the geometry of the unitary group: Hamiltonian evolution traces a **geodesic** on $U(D)$ whose speed is set by the energy spread ΔH. The connection between Riemannian geometry and quantum dynamics is one of the structural insights in quantum information.

Problem Statement:

In the bi-invariant HS metric on $U(D)$, the geodesic $U(t) = \exp(-iHt)$ has constant speed $\|dU/dt \cdot U^\dagger\| = \sqrt{\operatorname{Tr}(H^2)}$.

(a) Compute the geodesic length from I to $U(T) = \exp(-iHT)$.

(b) For $H = (\omega/2)\sigma_x$ on a qubit, compute the speed and the length to $U(\pi/\omega) = -i\sigma_x$.

(c) Mandelstam-Tamm bound: for $|\psi_0\rangle = |0\rangle$, compute ΔH, predict $T_\perp$, and verify the survival probability vanishes at $t = T_\perp$.

Pedantic Step-by-Step Solution:

Part (a): Geodesic length

The velocity along the geodesic is:

$$\frac{dU}{dt} \cdot U^\dagger = (-iH)e^{-iHt} \cdot e^{+iHt} = -iH.$$

The HS norm on $\mathfrak{u}(D)$ is $\|A\| = \sqrt{-\operatorname{Tr}(A^2)}$ for skew-Hermitian A. Since $-iH$ is skew-Hermitian:

$$v = \| - iH\| = \sqrt{-\operatorname{Tr}((-iH)^2)} = \sqrt{\operatorname{Tr}(H^2)}.$$

The speed is **constant** (independent of t), so the geodesic length is simply:

$$\boxed{L = T \cdot \sqrt{\mathrm{Tr}(H^2)}.}$$

Part (b): Single-qubit Rabi oscillation

For $H = (\omega/2)\sigma_x$:

$$\mathrm{Tr}(H^2) = \frac{\omega^2}{4}\,\mathrm{Tr}(\sigma_x^2) = \frac{\omega^2}{4} \cdot 2 = \frac{\omega^2}{2}.$$

$$v = \frac{\omega}{\sqrt{2}}.$$

At $T = \pi/\omega$ (a π-rotation about the x-axis):

$$L = \frac{\pi}{\omega} \cdot \frac{\omega}{\sqrt{2}} = \frac{\pi}{\sqrt{2}}.$$

Part (c): Mandelstam-Tamm bound

The bound states that the minimum time to evolve to an orthogonal state is:

$$T_\perp = \frac{\pi}{2\Delta H}, \qquad \Delta H = \sqrt{\langle H^2 \rangle - \langle H \rangle^2}.$$

Step 1: Compute the expectation values for $|\psi_0\rangle = |0\rangle$:

$$\langle H \rangle = \frac{\omega}{2}\langle 0|\sigma_x|0\rangle = 0, \qquad \langle H^2 \rangle = \frac{\omega^2}{4}\langle 0|\sigma_x^2|0\rangle = \frac{\omega^2}{4}.$$

$$\Delta H = \frac{\omega}{2}.$$

Step 2: Predict $T_\perp$:

$$T_\perp = \frac{\pi}{2 \cdot \omega/2} = \frac{\pi}{\omega}.$$

Step 3: Verify the survival probability vanishes:

$$|\langle 0|e^{-iHt}|0\rangle|^2 = |\langle 0|e^{-i(\omega/2)\sigma_x t}|0\rangle|^2 = \cos^2\left(\frac{\omega t}{2}\right).$$

At $t = T_\perp = \pi/\omega$: $\cos^2(\pi/2) = 0$. ✓

The bound is **saturated** because $|0\rangle$ is an equal superposition of the eigenstates of σ_x, giving maximal energy spread.

Exercise 1.14: Differential Geometry of Quantum State Space

1. Physical Motivation

A quantum state $|\psi\rangle$ lives in the Hilbert space $\mathcal{H} \cong \mathbb{C}^D$, but physical observables are blind to the overall phase: $|\psi\rangle$ and $e^{i\alpha}|\psi\rangle$ represent the same physics. The *true* arena of quantum mechanics is therefore the **projective Hilbert space** $\mathbb{CP}^{D-1}$, obtained by quotienting out the $U(1)$ phase. This quotient endows $\mathbb{CP}^{D-1}$ with a natural Riemannian metric — the **Fubini–Study metric** — that provides a gauge-invariant notion of distance between quantum states. This metric is not merely an abstract mathematical structure: it appears directly as the **Quantum Fisher Information** (QFI) in parameter estimation, as the **Fubini–Study speed** in quantum dynamics, and as the **quantum geometric tensor** controlling Berry phases and adiabatic transport.
Understanding this geometry is essential for:

- **Quantum metrology**: The QFI bounds the precision of parameter estimation via the

Cramér-Rao bound.

- **Variational algorithms**: The Fubini–Study metric determines the natural gradient on the variational manifold.
- **Scrambling diagnostics**: The QFI concentration bounds the trainability of deep circuits (Chapter 7).

2. Mathematical Toolbox

Tool 1 — Quantum Geometric Tensor (QGT): For a family of states $|\psi(\boldsymbol{\theta})\rangle$ parameterized by $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_p)$, the QGT is:

$$Q_{\mu\nu} = \langle \partial_\mu \psi | \partial_\nu \psi \rangle - \langle \partial_\mu \psi | \psi \rangle \langle \psi | \partial_\nu \psi \rangle.$$

Its real part is the **Fubini–Study metric** $g^{\mathrm{FS}}_{\mu\nu} = \operatorname{Re} Q_{\mu\nu}$; its imaginary part is proportional to the **Berry curvature** $F_{\mu\nu} = -2 \operatorname{Im} Q_{\mu\nu}$.
Tool 2 — Fubini–Study distance: For two pure states,

$$d_{\mathrm{FS}}(|\psi\rangle, |\phi\rangle) = \arccos |\langle \psi | \phi \rangle|.$$

Tool 3 — QFI and Cramér–Rao bound: $F_Q(\theta) = 4\, g^{\mathrm{FS}}_{\theta\theta}$. For unitary encoding $|\psi(\theta)\rangle = e^{-i\theta G}|\psi_0\rangle$: $F_Q = 4 \operatorname{Var}(G)$.
Tool 4 — Berry connection: $A_\mu = i\langle \psi | \partial_\mu \psi \rangle$; the Berry curvature $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ governs geometric phases.
Tool 5 — Sectional curvature: For $X, Y \in \mathfrak{su}(D)$ with $\langle X, Y \rangle = -\operatorname{Tr}(XY)$: $K(X,Y) = \frac{1}{4}\|[X,Y]\|^2$.
3. Exercise Statement

(a) Fubini–Study metric for a qubit. Consider a qubit state $|\psi(\theta, \phi)\rangle = \cos(\theta/2)|0\rangle + e^{i\phi}\sin(\theta/2)|1\rangle$. Compute the full QGT and show:

$$ds^2_{\mathrm{FS}} = \frac{1}{4}\left(d\theta^2 + \sin^2\theta \, d\phi^2\right).$$

(b) Berry curvature as monopole. Extract $F_{\theta\phi}$ and show it corresponds to a magnetic monopole. Compute the total Berry flux and Chern number.
(c) QFI for a rotation. For $|\psi(\theta)\rangle = e^{-i\theta\sigma_z/2}|+\rangle$, verify $F_Q = 4\operatorname{Var}(\sigma_z/2) = 1$.
(d) QFI for general SU(2). Show $F_Q = 1 - (\hat{n} \cdot \langle \boldsymbol{\sigma} \rangle_0)^2$. When does F_Q vanish?
(e) Geodesic distances. Compute $d_{\mathrm{FS}}(|0\rangle, |+\rangle)$ and $d_{\mathrm{FS}}(|0\rangle, |1\rangle)$.
4. Analytical Derivation

Part (a): QGT on the Bloch Sphere

Derivatives: $|\partial_\theta \psi\rangle = (-\sin(\theta/2)/2)|0\rangle + (\cos(\theta/2)/2)e^{i\phi}|1\rangle$, $|\partial_\phi \psi\rangle = i\sin(\theta/2)e^{i\phi}|1\rangle$.

- $Q_{\theta\theta} = 1/4$ (real, since $\langle \psi | \partial_\theta \psi \rangle = 0$).
- $Q_{\phi\phi} = \sin^2(\theta/2)\cos^2(\theta/2) = \sin^2\theta/4$ (real).
- $Q_{\theta\phi} = i\cos(\theta/2)\sin(\theta/2)/2 = i\sin\theta/4$ (purely imaginary $\Rightarrow g_{\theta\phi} = 0$).

The FS metric gives $ds^2 = \frac{1}{4}(d\theta^2 + \sin^2\theta \, d\phi^2)$: the round sphere of radius $R = 1/2$. ✓

Part (b): Berry Curvature

$F_{\theta\phi} = -2\operatorname{Im} Q_{\theta\phi} = -2 \cdot \sin\theta/4 = -\sin\theta/2$.
Total flux: $\Phi = \int_0^{2\pi} d\phi \int_0^{\pi} d\theta \, (-\sin\theta/2) = -2\pi$. Chern number $c_1 = \Phi/(2\pi) = -1$ for spin-$\downarrow$. The *magnitude* $|\Phi| = 2\pi$ confirms a monopole of strength 1/2.

Parts (c)-(d): QFI

For $G = \sigma_z/2$ on $|+\rangle$: $\langle G \rangle = 0$, $\langle G^2 \rangle = 1/4$, so $F_Q = 4(1/4 - 0) = 1$.

General: $G = \hat{n} \cdot \boldsymbol{\sigma}/2$, $G^2 = I/4$, $\langle G \rangle = \hat{n} \cdot \langle \boldsymbol{\sigma} \rangle_0 /2$. Therefore $F_Q = 1 - (\hat{n} \cdot \langle \boldsymbol{\sigma} \rangle_0)^2$. Vanishes when $|\psi_0\rangle$ is eigenstate of G ($\hat{n}$ aligned with Bloch vector).

Part (e): Geodesics

$d_{\text{FS}}(|0\rangle, |+\rangle) = \arccos(1/\sqrt{2}) = \pi/4$, $d_{\text{FS}}(|0\rangle, |1\rangle) = \arccos(0) = \pi/2$.

7. Visual Proof: Geometry of Quantum State Space

Flashcards, Active Recall & Derivations

Card 16: Choi–Jamiołkowski Isomorphism

Goal: Prove the Choi–Jamiołkowski isomorphism mapping a quantum channel $\mathcal{E}$ on system A (dimension d) to a bipartite state $\hat{\rho}_{\mathcal{E}}$ on $A \otimes B$:

$$\hat{\rho}_{\mathcal{E}} = (\mathcal{E} \otimes \mathcal{I}) (|\Phi\rangle \langle\Phi|)$$

where $|\Phi\rangle = \frac{1}{\sqrt{d}} \sum_{i=1}^{d} |i,i\rangle$ is the maximally entangled state, and show that $\mathcal{E}$ is trace-preserving if and only if $\mathrm{Tr}_A(\hat{\rho}_{\mathcal{E}}) = \hat{I}_B/d$.

Key Trick: Represent the maximally entangled state explicitly in index notation and contract the Kraus operators of the channel $\mathcal{E}$ with the state vectors.

Step-by-Step Derivation:

1. Let $|\Phi\rangle = \frac{1}{\sqrt{d}} \sum_i |i,i\rangle$. The bipartite projector $|\Phi\rangle \langle\Phi|$ is:

$$|\Phi\rangle \langle\Phi| = \frac{1}{d} \sum_{i,j=1}^{d} |i,i\rangle \langle j,j| = \frac{1}{d} \sum_{i,j=1}^{d} |i\rangle \langle j|_A \otimes |i\rangle \langle j|_B$$

2. Apply the map $\mathcal{E} \otimes \mathcal{I}$ to this projector:

$$\hat{\rho}_{\mathcal{E}} = (\mathcal{E} \otimes \mathcal{I}) (|\Phi\rangle \langle\Phi|) = \frac{1}{d} \sum_{i,j=1}^{d} \mathcal{E} (|i\rangle \langle j|_A) \otimes |i\rangle \langle j|_B$$

   This is the Choi matrix/state of the channel $\mathcal{E}$.

3. Now, let's take the partial trace over system A:

$$\mathrm{Tr}_A(\hat{\rho}_{\mathcal{E}}) = \frac{1}{d} \sum_{i,j=1}^{d} \mathrm{Tr} \left(\mathcal{E}(|i\rangle \langle j|_A) \right) |i\rangle \langle j|_B$$

4. If the channel $\mathcal{E}$ is trace-preserving (TP), then $\mathrm{Tr}(\mathcal{E}(\hat{\sigma})) = \mathrm{Tr}(\hat{\sigma})$ for any operator $\hat{\sigma}$. Thus, for $\hat{\sigma} = |i\rangle \langle j|$:

$$\mathrm{Tr} \left(\mathcal{E}(|i\rangle \langle j|_A) \right) = \mathrm{Tr}(|i\rangle \langle j|_A) = \langle j|i\rangle = \delta_{ij}$$

5. Substitute this back into the partial trace sum:

$$\mathrm{Tr}_A(\hat{\rho}_{\mathcal{E}}) = \frac{1}{d} \sum_{i,j=1}^{d} \delta_{ij} |i\rangle \langle j|_B = \frac{1}{d} \sum_{i=1}^{d} |i\rangle \langle i|_B = \boxed{\frac{\hat{I}_B}{d}}$$

   Thus, $\mathcal{E}$ is TP if and only if the Choi state has a maximally mixed marginal on system B.

Questions

1. **Question:** What is the practical utility of the Choi state in quantum process analysis?

   Answer: The Choi–Jamiołkowski isomorphism maps the dynamical properties of quantum channels (such as complete positivity and trace preservation) directly onto the static geometric properties of bipartite states. This allows experimentalists to perform quantum process tomography by executing standard quantum state tomography directly on the Choi state.

Derivation Blueprint: Choi–Jamiołkowski Isomorphism

Board Layout:

- **Left Board Panel:** Draw a diagram of the mapping: a channel $\mathcal{E}$ acting on one half of a maximally entangled pair.
- **Middle Panel:** Write down the algebraic derivation of the Choi state $\hat{\rho}_{\mathcal{E}}$ in the computational basis.
- **Right Panel:** Compute the partial trace over A. Show how the trace-preserving condition leads to the maximally mixed state on B.

Common Conceptual Pitfalls: The fact that taking the partial trace of the swap operator over A yields the identity operator on B can be non-intuitive. The contraction acts as a Kronecker delta filter that selects only the diagonal terms of the density matrix, which naturally sum to the identity. **Boundary Limit & Sanity Check:** If the channel is completely depolarizing ($\mathcal{E}(\hat{\rho}) = \hat{I}/D$), the Choi state becomes $\hat{\rho}_{\mathcal{E}} = \hat{I}/D^2$. This maximally mixed Choi state verifies that a completely noisy channel maps any input to the identity, leaving the system and environment completely uncorrelated.

Card 20: Schur–Weyl Duality for Arbitrary k

Goal: Prove the general statement of Schur–Weyl Duality relating the diagonal action of the general linear group $GL(D)$ and the symmetric group S_k on the tensor product space $\mathcal{H}^{\otimes k}$.

Key Trick: Show that the representations of $GL(D)$ and S_k act as mutual commutants of each other on $\mathcal{H}^{\otimes k}$, restricting the decomposition of the space to a direct sum of paired irreducible representations.

Step-by-Step Derivation:

1. Consider the tensor product space $\mathcal{H}^{\otimes k}$ where $\mathcal{H} = \mathbb{C}^D$.
2. Define the diagonal representation of the group $GL(D)$ on this space:
$$\rho_{\text{diag}}(g) = g^{\otimes k} = g \otimes g \otimes \cdots \otimes g \quad \forall g \in GL(D)$$
3. Define the representation of the symmetric group S_k which acts by permuting the tensor factors:
$$\sigma(v_1 \otimes v_2 \otimes \cdots \otimes v_k) = v_{\sigma^{-1}(1)} \otimes v_{\sigma^{-1}(2)} \otimes \cdots \otimes v_{\sigma^{-1}(k)} \quad \forall \sigma \in S_k$$
4. The actions of $GL(D)$ and S_k commute on $\mathcal{H}^{\otimes k}$:
$$[\rho_{\text{diag}}(g), \sigma] = 0 \quad \forall g \in GL(D),\ \sigma \in S_k$$
5. Let $\mathcal{A}$ be the algebra spanned by $\{g^{\otimes k}\}$ and $\mathcal{B}$ be the algebra spanned by $\{\sigma\}$. Since they commute, $\mathcal{A} \subseteq \mathcal{B}'$ and $\mathcal{B} \subseteq \mathcal{A}'$ where X' denotes the commutant.
6. Schur's Lemma states that if these two algebras act irreducibly on the component spaces, the global space decomposes as a direct sum:
$$\boxed{\mathcal{H}^{\otimes k} \cong \bigoplus_{\lambda} \mathcal{U}_\lambda \otimes \mathcal{V}_\lambda}$$
where λ labels the partitions of k (Young diagrams), $\mathcal{U}_\lambda$ are irreducible representations of $GL(D)$, and $\mathcal{V}_\lambda$ are irreducible representations of S_k.

Questions

1. **Question:** Why is Schur-Weyl duality a cornerstone of quantum information theory?

   Answer: Schur-Weyl duality bridges the gap between the permutation symmetry of tensor product systems (S_k) and global unitary transformations ($GL(D)$). It serves as the fundamental algebraic backbone for Weingarten calculus, quantum state estimation, and the mathematical characterization of unitary t-designs.

Derivation Blueprint: Schur–Weyl Duality for Arbitrary k

Board Layout:

- **Left Panel:** State the two representations on $\mathcal{H}^{\otimes k}$ and show that they commute.
- **Middle Panel:** State the double commutant theorem. Draw a Young diagram λ to visualize how the partitions label the irreducible blocks.
- **Right Panel:** Write down the direct sum decomposition. Highlight the pairing between $GL(D)$ and S_k representations.

Common Conceptual Pitfalls: The connection between Young diagrams and tensor spaces can be challenging to visualize. For $k = 3$, the Young diagrams correspond to the three possible permutation symmetries: fully symmetric (horizontal row), fully anti-symmetric (vertical column), and mixed-symmetry. **Boundary Limit & Sanity Check:** For the lowest non-trivial case $k = 2$, the partitions decompose $\mathcal{H}^{\otimes 2}$ into the symmetric subspace (dimension $D(D+1)/2$) and the anti-symmetric subspace (dimension $D(D-1)/2$), summing exactly to the full Hilbert space dimension D^2.

2 Random Matrix Theory

Exercise 2.1: The 2×2 GOE and Linear Level Repulsion

Motivation:

The **Wigner surmise** — the nearest-neighbour spacing distribution of an integrable or chaotic quantum system — is one of the most celebrated results in random matrix theory. The exact distribution for 2×2 random matrices provides an excellent approximation even for large matrices. This exercise derives the GOE spacing distribution from first principles, demonstrating the **linear** level repulsion characteristic of systems with time-reversal symmetry ($\beta = 1$).

Problem Statement:

Consider a 2×2 GOE matrix $H = \begin{pmatrix} a & c \\ c & b \end{pmatrix}$ with GOE measure $P(H) \propto \exp\bigl(-\operatorname{Tr}(H^2)/2\bigr)$.

(a) Write down the joint distribution $P(a, b, c)$.
(b) Find the energy spacing $s = \lambda_+ - \lambda_-$ in terms of a, b, c.
(c) Change variables to $t = a + b$, $x = a - b$, $y = 2c$. Integrate out the trace and the angular variable to derive $P(s) \propto s\, e^{-s^2/4}$.

Pedantic Step-by-Step Solution:

Part (a): Joint distribution

The trace of H^2 is:

$$\operatorname{Tr}(H^2) = a^2 + b^2 + 2c^2.$$

(The factor 2 on c^2 arises because c appears both above and below the diagonal.) The GOE measure is:

$$P(a, b, c) \propto \exp\left(-\frac{a^2 + b^2 + 2c^2}{2}\right).$$

This means $a, b \sim \mathcal{N}(0, 1)$ independently, and $c \sim \mathcal{N}(0, 1/2)$ (the off-diagonal has half the variance).

Part (b): Spacing formula

The eigenvalues of the 2×2 symmetric matrix are:

$$\lambda_\pm = \frac{a + b}{2} \pm \sqrt{\frac{(a - b)^2}{4} + c^2}.$$

The spacing is:

$$s = \lambda_+ - \lambda_- = 2\sqrt{\frac{(a - b)^2}{4} + c^2} = \sqrt{(a - b)^2 + 4c^2}.$$

Part (c): Deriving the Wigner surmise

Step 1: Change variables. Let $t = a + b$, $x = a - b$, $y = 2c$. Then $a = (t + x)/2$, $b = (t - x)/2$, $c = y/2$, and:

$$a^2 + b^2 + 2c^2 = \frac{t^2 + x^2}{2} + \frac{y^2}{2} = \frac{t^2 + x^2 + y^2}{2}.$$

The Jacobian is $|\partial(a, b, c)/\partial(t, x, y)| = 1/4$. The distribution becomes:

$$P(t, x, y) \propto \exp\left(-\frac{t^2}{4}\right) \cdot \exp\left(-\frac{x^2 + y^2}{4}\right).$$

Step 2: Integrate out the trace. The variable t (the trace $\operatorname{Tr}(H)$) decouples completely and integrates to a constant $\int e^{-t^2/4}\, dt = 2\sqrt{\pi}$.

Step 3: Switch to polar coordinates. The spacing is $s = \sqrt{x^2 + y^2}$. In polar coordinates (s, θ) with $x = s\cos\theta$, $y = s\sin\theta$:

$$dx\,dy = s\,ds\,d\theta.$$

Integrating over the angle $\theta \in [0, 2\pi)$ gives a factor 2π:

$$\boxed{P(s) = \frac{s}{2}\,e^{-s^2/4}, \qquad s \geq 0.}$$

(Normalization: $\int_0^\infty \frac{s}{2}e^{-s^2/4}\,ds = 1$, verified by substitution $u = s^2/4$.)
Physical interpretation: The factor s in $P(s) \propto s$ as $s \to 0$ is the **linear level repulsion** of the GOE ($\beta = 1$). Compare: the GUE has $P(s) \propto s^2$ (quadratic, as shown in Exercise 2.3), and the Poisson ensemble (no correlations) has $P(s) \propto$ const (no repulsion).

Exercise 2.2: Moments of the Wigner Semicircle

Motivation:
The **Wigner semicircle law** describes the limiting eigenvalue distribution of large random Hermitian matrices from the Gaussian Unitary Ensemble (GUE). Its moments — the Catalan numbers — connect random matrix theory to combinatorics (non-crossing pair partitions) and free probability. Computing these moments is the starting point for understanding spectral universality.
Problem Statement:
The Wigner semicircle density on $[-2, 2]$ is

$$\rho_{\text{sc}}(\lambda) = \frac{1}{2\pi}\sqrt{4-\lambda^2}.$$

Define the moments $m_k = \int_{-2}^{2} \lambda^k\,\rho_{\text{sc}}(\lambda)\,d\lambda$.
(a) Show all odd moments vanish: $m_{2k+1} = 0$.
(b) Compute m_2 and verify $m_2 = 1$.
(c) Compute m_4 and show $m_4 = 2$. Identify the pattern: $m_{2k} = C_k$ (Catalan numbers).
(d) Interpret $m_4/m_2^2 = 2 < 3$ (platykurtic).
Pedantic Step-by-Step Solution:

Part (a): Odd moments vanish

The density $\rho_{\text{sc}}(\lambda)$ is an **even function**: $\rho_{\text{sc}}(-\lambda) = \rho_{\text{sc}}(\lambda)$. For odd k, the integrand $\lambda^k\rho_{\text{sc}}(\lambda)$ is odd on the symmetric interval $[-2, 2]$:

$$m_{2k+1} = \int_{-2}^{2} \lambda^{2k+1}\,\rho_{\text{sc}}(\lambda)\,d\lambda = 0. \quad \checkmark$$

Part (b): Second moment

Use the substitution $\lambda = 2\sin\theta$, $d\lambda = 2\cos\theta\,d\theta$, and $\sqrt{4-\lambda^2} = 2\cos\theta$:

$$m_2 = \frac{1}{2\pi}\int_{-\pi/2}^{\pi/2} (2\sin\theta)^2 \cdot 2\cos\theta \cdot 2\cos\theta\,d\theta = \frac{8}{\pi}\int_{-\pi/2}^{\pi/2} \sin^2\!\theta\,\cos^2\!\theta\,d\theta.$$

Using the identity $\sin^2\!\theta\cos^2\!\theta = \frac{1}{4}\sin^2(2\theta)$ and $\int_{-\pi/2}^{\pi/2}\sin^2(2\theta)\,d\theta = \pi/2$:

$$m_2 = \frac{8}{\pi}\cdot\frac{1}{4}\cdot\frac{\pi}{2} = 1. \quad \checkmark$$

The variance of the semicircle distribution equals 1 (with our normalization).

Part (c): Fourth moment and Catalan numbers

Similarly:

$$m_4 = \frac{32}{\pi}\int_{-\pi/2}^{\pi/2} \sin^4\!\theta\,\cos^2\!\theta\,d\theta.$$

Using the reduction formula $\int_{-\pi/2}^{\pi/2} \sin^4\theta \cos^2\theta \, d\theta = \pi/16$:

$$m_4 = \frac{32}{\pi} \cdot \frac{\pi}{16} = 2. \quad \checkmark$$

The even moments m_{2k} are the **Catalan numbers** $C_k = \frac{1}{k+1}\binom{2k}{k}$:

$$C_0 = 1, \quad C_1 = 1, \quad C_2 = 2, \quad C_3 = 5, \quad C_4 = 14, \quad \ldots$$

Combinatorially, C_k counts the number of **non-crossing pair partitions** of $2k$ elements — the same objects that appear in the Weingarten calculus of Chapter 3.

Part (d): Kurtosis

The excess kurtosis is characterized by the ratio:

$$\frac{m_4}{m_2^2} = \frac{2}{1} = 2 < 3.$$

A Gaussian has $m_4/m_2^2 = 3$ (mesokurtic). The semicircle is **platykurtic**: it has lighter tails because the spectral density has compact support $[-2, 2]$ — eigenvalues cannot fluctuate beyond this hard edge.

Exercise 2.3: Eigenphase Repulsion in the 2×2 CUE

Motivation:

The eigenvalues of a Haar-random unitary matrix are not independent — they **repel** each other. This level repulsion is one of the predictions of random matrix theory and is observed throughout quantum chaos, from nuclear spectra to the Riemann zeta function. The 2×2 CUE is the simplest setting where this phenomenon can be derived exactly: the repulsion is **quadratic** (proportional to the spacing squared), which is the hallmark of unitary symmetry.

Problem Statement:

A 2×2 CUE matrix (Haar-random $U \in U(2)$) has eigenvalues $e^{i\theta_1}, e^{i\theta_2}$ with joint density proportional to the Vandermonde squared:

$$P(\theta_1, \theta_2) \propto |e^{i\theta_1} - e^{i\theta_2}|^2.$$

(a) Show $|e^{i\theta_1} - e^{i\theta_2}|^2 = 2 - 2\cos(\theta_1 - \theta_2)$.
(b) With $\varphi = \theta_1 - \theta_2$, show the spacing density is $P(\varphi) = (1 - \cos\varphi)/(2\pi)$.
(c) Verify $P(0) = 0$ (level repulsion) and find the most probable spacing.

Pedantic Step-by-Step Solution:

Part (a): Algebraic identity

Expand the modulus squared:

$$|e^{i\theta_1} - e^{i\theta_2}|^2 = (e^{i\theta_1} - e^{i\theta_2})(e^{-i\theta_1} - e^{-i\theta_2})$$

$$= 1 - e^{i(\theta_1-\theta_2)} - e^{-i(\theta_1-\theta_2)} + 1 = 2 - 2\cos(\theta_1 - \theta_2). \quad \checkmark$$

Key feature: the expression depends only on the **spacing** $\varphi = \theta_1 - \theta_2$, not on the individual phases.

Part (b): Marginal spacing density

Change variables to the spacing $\varphi = \theta_1 - \theta_2$ and the center of mass $\Theta = (\theta_1 + \theta_2)/2$.
The joint density is $P(\varphi, \Theta) \propto 2 - 2\cos\varphi$. Integrating over the center of mass $\Theta \in [0, 2\pi)$ gives a factor 2π.
The normalization constant is

$$Z = \int_0^{2\pi} (2 - 2\cos\varphi)\, d\varphi = [2\varphi - 2\sin\varphi]_0^{2\pi} = 4\pi.$$

Therefore the normalized spacing density is

$$\boxed{P(\varphi) = \frac{1 - \cos\varphi}{2\pi}.}$$

Part (c): Level repulsion and most probable spacing

Level repulsion: At $\varphi = 0$:

$$P(0) = \frac{1 - 1}{2\pi} = 0.$$

Eigenphases **cannot coincide** — there is a strict vanishing of the density at zero spacing. Near $\varphi = 0$, $P(\varphi) \approx \varphi^2/(4\pi)$: the repulsion is **quadratic** ($\beta = 2$ for the CUE).

Most probable spacing: Maximize $P(\varphi) = (1 - \cos\varphi)/(2\pi)$:

$$\frac{dP}{d\varphi} = \frac{\sin\varphi}{2\pi} = 0 \quad \Rightarrow \quad \varphi = \pi.$$

Eigenphases prefer to sit **diametrically opposite** on the unit circle — the maximally separated configuration. This is the most basic manifestation of spectral rigidity.

Card 7: $\hat{T}^2 = \pm\hat{I}$

Goal: Prove that the square of any antiunitary time-reversal operator $\hat{T}$ must equal $\pm\hat{I}$.

Key Trick: Use the decomposition of an antiunitary operator into a unitary operator and complex conjugation ($\hat{T} = \hat{U}_T\hat{K}$), and apply the associativity of operators alongside the properties of complex conjugation.

Step-by-Step Derivation:

1. Any antiunitary operator $\hat{T}$ can be decomposed as $\hat{T} = \hat{U}_T\hat{K}$, where $\hat{U}_T$ is a unitary operator and $\hat{K}$ is the complex conjugation operator in a chosen basis.

2. The action of $\hat{K}$ on any operator $\hat{A}$ is $\hat{K}\hat{A}\hat{K} = \hat{A}^*$.

3. Compute $\hat{T}^2$:
$$\hat{T}^2 = (\hat{U}_T\hat{K})(\hat{U}_T\hat{K}) = \hat{U}_T(\hat{K}\hat{U}_T\hat{K}) = \hat{U}_T\hat{U}_T^*$$
Since the composition of two antilinear operators is a linear operator, $\hat{T}^2$ is linear. Since it is the product of two unitary operators, it is also unitary: $\left(\hat{T}^2\right)^\dagger\left(\hat{T}^2\right) = \hat{I}$.

4. Now, we use the fact that $\hat{T}^3 = \hat{T}\hat{T}^2 = \hat{T}^2\hat{T}$ to constrain $\hat{T}^2$.

5. Note that $\hat{T}^2 = \hat{U}_T\hat{U}_T^*$ is *linear* (the two antilinear conjugations $\hat{K}$ compose to the identity) and unitary. To pin it down, examine the action on an arbitrary state $|\psi\rangle$: since $\hat{T}$ is antiunitary, $\hat{T}(c\,|\psi\rangle) = c^*\hat{T}\,|\psi\rangle$ for any scalar c, whereas $\hat{T}^2(c\,|\psi\rangle) = c\,\hat{T}^2\,|\psi\rangle$.

6. Apply $\hat{T}$ to both sides of the identity $\hat{T}^2 = \phi\hat{I}$ (by Schur's Lemma, since $\hat{T}^2$ commutes with the time-reversal symmetric Hamiltonian):
$$\hat{T}\hat{T}^2 = \hat{T}(\phi\hat{I}) = \phi^*\hat{T}$$
But we also know that:
$$\hat{T}^2\hat{T} = \phi\hat{I}\hat{T} = \phi\hat{T}$$
Since $\hat{T}\hat{T}^2 = \hat{T}^2\hat{T}$, we must have:
$$\phi^*\hat{T} = \phi\hat{T} \implies \phi^* = \phi$$
Thus, the proportionality constant ϕ must be a real number!

7. Since $\hat{T}^2$ is unitary, the absolute value of its eigenvalues must be 1:
$$|\phi| = 1$$

8. A real number with absolute value 1 can only be $+1$ or -1. Thus:
$$\boxed{\hat{T}^2 = \pm\hat{I}}$$

Questions

1. **Question:** What is the physical meaning of the two cases $\hat{T}^2 = +\hat{I}$ and $\hat{T}^2 = -\hat{I}$?

   Answer: The case $\hat{T}^2 = +\hat{I}$ corresponds to spinless systems or systems with an even number of spin-1/2 particles (governed by the Gaussian Orthogonal Ensemble, GOE). The case $\hat{T}^2 = -\hat{I}$ corresponds to systems with an odd number of spin-1/2 particles, which leads to Kramers' degeneracy and is governed by the Gaussian Symplectic Ensemble (GSE).

Derivation Blueprint: $\hat{T}^2 = \pm\hat{I}$

Board Layout:

- **Left Board Panel:** Write out the definition of antiunitarity: $\langle \hat{T}\phi|\hat{T}\psi\rangle = \langle\psi|\phi\rangle$ and antilinearity. Show the decomposition $\hat{T} = \hat{U}_T\hat{K}$.
- **Middle Panel:** Write down the steps to compute $\hat{T}^2 = \hat{U}_T\hat{U}_T^*$. Show why it commutes with $\hat{T}$.
- **Right Panel:** Apply the commutativity argument to prove that the phase ϕ is real, leading to the binary choice $+1$ or -1.

Common Conceptual Pitfalls: The complex conjugation operator $\hat{K}$ is an anti-unitary operator, which makes it highly basis-dependent. One must keep in mind that despite its basis dependence, it always acts as an involution: applying it twice returns the identity, $\hat{K}^2 = \hat{I}$.
Boundary Limit & Sanity Check: The time-reversal operator $\hat{T}$ is antiunitary and real, so its eigenvalues are restricted to the unit circle. Since $\hat{T}^2$ is a linear, unitary operator that commutes with $\hat{T}$, its own eigenvalues must be real and have absolute value 1, restricting them strictly to ± 1.

Card 8: 2×2 GUE Jacobian

Goal: Prove that the change of variables from GUE matrix elements to its eigenvalues yields the level repulsion factor $|\lambda_+ - \lambda_-|^2$:

$$P(\lambda_+, \lambda_-) \propto |\lambda_+ - \lambda_-|^2\, e^{-(\lambda_+^2 + \lambda_-^2)}$$

Key Trick: Write the 2×2 hermitian matrix in terms of real parameters, compute the eigenvalues, and evaluate the Jacobian determinant of the coordinate transformation.

Step-by-Step Derivation:

1. Let H be a 2×2 GUE matrix:
$$H = \begin{pmatrix} a & x + iy \\ x - iy & b \end{pmatrix}$$
where $a, b, x, y \in \mathbb{R}$ are independent Gaussian variables under the measure:
$$dP(H) \propto da\, db\, dx\, dy \exp\left(-\operatorname{Tr}(H^2)\right)$$
2. Compute $\operatorname{Tr}(H^2)$ in terms of the parameters:
$$\operatorname{Tr}(H^2) = a^2 + b^2 + 2(x^2 + y^2)$$
3. The eigenvalues of H are:
$$\lambda_\pm = \frac{a+b}{2} \pm \frac{1}{2}\sqrt{(a-b)^2 + 4(x^2+y^2)}$$
4. Define new coordinates:
$$\begin{aligned} &s = a + b \implies a = \frac{s+d}{2} \\ &d = a - b \implies b = \frac{s-d}{2} \\ &x = r\cos\phi \\ &y = r\sin\phi \implies x^2 + y^2 = r^2 \end{aligned}$$

The Jacobian of this first-stage transformation is:

$$J_1 = \det \frac{\partial(a,b,x,y)}{\partial(s,d,r,\phi)} = \det \begin{pmatrix} 1/2 & 1/2 & 0 & 0 \\ 1/2 & -1/2 & 0 & 0 \\ 0 & 0 & \cos\phi & -r\sin\phi \\ 0 & 0 & \sin\phi & r\cos\phi \end{pmatrix}$$

Evaluating the determinant by block expansion yields:

$$J_1 = \det \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix} \cdot \det \begin{pmatrix} \cos\phi & -r\sin\phi \\ \sin\phi & r\cos\phi \end{pmatrix} = \left(-\frac{1}{4} - \frac{1}{4}\right)\left(r\cos^2\phi + r\sin^2\phi\right) = -\frac{1}{2}r$$

Taking the absolute value, the volume element is:

$$da\,db\,dx\,dy = |J_1|\,ds\,dd\,dr\,d\phi = \frac{1}{2}r\,ds\,dd\,dr\,d\phi$$

5. In these coordinates, the eigenvalue difference is $w = \lambda_+ - \lambda_- = \sqrt{d^2 + 4r^2}$.
6. Now change variables from (d, r) to the spacing variables (w, θ) where θ parameterizes the eigenvector rotation:

$$d = w\cos\theta, \qquad r = \frac{1}{2}w\sin\theta$$

The Jacobian of this second-stage transformation is:

$$J_2 = \det \frac{\partial(d,r)}{\partial(w,\theta)} = \det \begin{pmatrix} \cos\theta & -w\sin\theta \\ \frac{1}{2}\sin\theta & \frac{1}{2}w\cos\theta \end{pmatrix} = \frac{1}{2}w\cos^2\theta + \frac{1}{2}w\sin^2\theta = \frac{1}{2}w$$

Thus, the differential volume element updates as:

$$dd\,dr = |J_2|\,dw\,d\theta = \frac{1}{2}w\,dw\,d\theta$$

7. Express this in terms of the eigenvalues λ_+ and λ_-: Since $w = \lambda_+ - \lambda_-$ and $s = \lambda_+ + \lambda_-$, the linear transformation $(s, w) \to (\lambda_+, \lambda_-)$ has Jacobian:

$$\det \frac{\partial(s,w)}{\partial(\lambda_+,\lambda_-)} = \det \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = -2 \implies ds\,dw = 2\,d\lambda_+\,d\lambda_-$$

8. Collect the full volume element:

$$da\,db\,dx\,dy \propto r\,dd\,dr\,d\phi = \left(\frac{1}{2}w\sin\theta\right)\left(\frac{1}{2}w\,dw\,d\theta\right)d\phi \propto w^2\sin\theta\,ds\,dw\,d\theta\,d\phi$$

Substituting $w = |\lambda_+ - \lambda_-|$:

$$da\,db\,dx\,dy \propto |\lambda_+ - \lambda_-|^2 \sin\theta\,d\lambda_+\,d\lambda_-\,d\theta\,d\phi$$

9. Integrate out the angular coordinates θ and ϕ (which represent the eigenvectors). The Gaussian exponent is invariant under unitary rotations:

$$\mathrm{Tr}(H^2) = \lambda_+^2 + \lambda_-^2$$

10. The resulting marginal probability density of the eigenvalues is:

$$\boxed{P(\lambda_+,\lambda_-) \propto |\lambda_+ - \lambda_-|^2\, e^{-(\lambda_+^2 + \lambda_-^2)}}$$

Questions

1. **Question:** How does the level repulsion factor generalize to $D \times D$ matrices?

   Answer: For a general $D \times D$ random matrix ensemble, the change of variables from matrix elements to eigenvalues yields the Vandermonde determinant Jacobian:

$$J(\lambda) = \prod_{i<j} |\lambda_i - \lambda_j|^\beta$$

   where $\beta = 1$ for the Orthogonal (GOE), $\beta = 2$ for the Unitary (GUE), and $\beta = 4$ for the Symplectic (GSE) ensembles. The 2×2 GUE derivation explicitly proves how $\beta = 2$ arises from the two independent polar variables of the complex off-diagonal elements.

Derivation Blueprint: 2x2 GUE Jacobian

Board Layout:

- **Left Board Panel:** Write down the GUE matrix and eigenvalues. State the Gaussian measure in terms of matrix entries.
- **Middle Panel:** Show the change of variables $(a, b, x, y) \to (s, d, r, \phi)$. Highlight the polar coordinate transformation for x and y.
- **Right Panel:** Perform the Jacobian derivation step-by-step. Show how the $r\,dr$ polar measure picks up the spacing factor $(\lambda_+ - \lambda_-)^2$ after integration.

Chalk-Talk Visual Guide:

- Draw a 2D plane representing the coordinate variables $(d, 2r)$ where $d = a - b$ is the diagonal difference, and $2r$ is the off-diagonal magnitude. Show that the eigenvalue gap $w = \lambda_+ - \lambda_-$ corresponds exactly to the radial distance from the origin in this 2D plane. Draw a circle of radius w, and write the polar coordinate equations $d = w \cos\theta$ and $2r = w \sin\theta$. Highlight that the area element in this plane is $d(\text{Area}) = w\,dw\,d\theta$, explaining why the integration measure picks up a factor of the gap w, which when multiplied by the off-diagonal polar element $r \propto w \sin\theta$ yields the quadratic spacing factor $w^2 \propto (\lambda_+ - \lambda_-)^2$ characteristic of level repulsion.

Common Conceptual Pitfalls: The transition to polar coordinates (w, θ) can appear unmotivated. It arises naturally because the eigenvalue spacing for the 2×2 matrix is $\sqrt{d^2 + 4r^2}$, which maps geometrically to a circle of radius w in the $(d, 2r)$ plane. **Boundary Limit & Sanity Check:** When eigenvalues approach degeneracy ($\lambda_+ \to \lambda_-$), the joint probability density $P(\lambda_+, \lambda_-)$ vanishes quadratically as $|\lambda_+ - \lambda_-|^2 \to 0$. This quadratic level repulsion at $s = |\lambda_+ - \lambda_-| \to 0$ contrasts with the Poisson level spacing of integrable systems, serving as a robust signature of quantum chaos.

Numerical Sandbox: 2x2 GUE Level Repulsion Verification

```
import numpy as np
import matplotlib.pyplot as plt
N_samples = 100000
gaps = []
for _ in range(N_samples):
    # Generate 2x2 GUE matrix
    a, b = np.random.normal(size=2)
    x, y = np.random.normal(scale=1/np.sqrt(2), size=2)
    H = np.array([[a, x + 1j*y], [x - 1j*y, b]])
    # Compute eigenvalues and gap
    eigenvals = np.linalg.eigvalsh(H)
    gaps.append(eigenvals[1] - eigenvals[0])
# Plot spacing distribution histogram and compare to Wigner Surmise
plt.hist(gaps, bins=100, density=True, alpha=0.6, color='blue', label="Numerical")
s = np.linspace(0, 4, 200)
p_s = (32 / np.pi**2) * (s**2) * np.exp(-4 * s**2 / np.pi)  # Wigner GUE Surmise
print("Level Spacing simulation completed. Run plot to visualize quadratic repulsion s^2 at
```

Card 17: Resolvent Self-Consistent Equation

Goal: Prove that the averaged resolvent $G(z) = \langle \mathrm{Tr}(z-H)^{-1}\rangle/D$ of a Wigner matrix satisfies the self-consistent quadratic equation:

$$G(z) = \frac{1}{z - G(z)}$$

in the planar diagrammatic limit ($D \to \infty$).

Key Trick: Expand the resolvent as a geometric series, average term-by-term using Wick's theorem, and represent the non-crossing planar diagram contractions as a nested self-similar recurrence relation.

Step-by-Step Derivation:

1. Write the resolvent $G(z)$ as a geometric series for large z:

$$G(z) = \frac{1}{D}\,\mathrm{Tr}\left(\frac{1}{z-H}\right) = \frac{1}{z}\sum_{n=0}^{\infty}\frac{1}{z^n}\left(\frac{1}{D}\,\mathrm{Tr}(H^n)\right)$$

2. Take the expectation value over the ensemble:

$$\langle G(z)\rangle = \frac{1}{z}\sum_{n=0}^{\infty}\frac{1}{z^{2n}}\langle\frac{1}{D}\,\mathrm{Tr}(H^{2n})\rangle$$

   (since all odd moments of Wigner matrices vanish by symmetry).

3. By Wick's theorem, the expectation value $\langle\frac{1}{D}\,\mathrm{Tr}(H^{2n})\rangle$ is evaluated as a sum over pairings of the $2n$ matrix elements. In the planar diagrammatic limit ($D \to \infty$), only non-crossing (planar) contractions survive. The number of such planar diagrams is exactly the Catalan number C_n:

$$C_n = \frac{1}{n+1}\binom{2n}{n}$$

   For low-orders, these correspond to the number of non-crossing pairings:

   - For $n = 1$ (2 matrices): $C_1 = 1$, representing the single contraction $\langle\mathrm{Tr}(H^2)\rangle/D = 1$.

- For $n = 2$ (4 matrices): $C_2 = 2$, representing two non-crossing pairings (outer-inner contraction $\langle \mathrm{Tr}(H_1 H_2 H_2 H_1)\rangle$ and sequential contraction $\langle \mathrm{Tr}(H_1 H_1 H_2 H_2)\rangle$).
- For $n = 3$ (6 matrices): $C_3 = 5$ planar diagrams.

4. The Catalan numbers satisfy the fundamental self-similar recurrence relation:

$$C_n = \sum_{k=0}^{n-1} C_k C_{n-1-k}$$

Physically, this recurrence represents a recursive nesting: in any planar pairing of $2n$ elements, the first element must contract with some even element $2k+2$. This contraction partitions the remaining elements into a nested planar diagram containing $2k$ elements inside the loop, and an independent planar diagram containing $2(n-1-k)$ elements outside the loop.

5. Mathematically, this nested recurrence means that if we multiply by the corresponding powers of $1/z$, the full sum for $\langle G(z)\rangle$ recursively folds into itself:

$$\langle G(z)\rangle = \frac{1}{z} + \frac{1}{z}\sum_{n=1}^{\infty} \frac{1}{z^{2n}} \left(\sum_{k=0}^{n-1} C_k C_{n-1-k} \right)$$

The inner Catalan convolution $\sum_{k=0}^{n-1} C_k C_{n-1-k}$ is exactly the Cauchy product that defines $\langle G(z)\rangle^2$: writing $\langle G(z)\rangle = \sum_{n\geq 0} C_n z^{-(2n+1)}$, one finds $\langle G(z)\rangle^2 = \sum_{m\geq 0} \big(\sum_{k=0}^{m} C_k C_{m-k} \big) z^{-(2m+2)}$. Substituting this back collapses the double sum to a single self-consistent term:

$$\langle G(z)\rangle = \frac{1}{z}\Big(1 + \langle G(z)\rangle^2\Big),$$

the standard Catalan generating-function identity $C(x) = 1 + x\, C(x)^2$ in the variable $x = z^{-2}$ (with $C = z\langle G\rangle$). Multiplying through by z gives $z\langle G\rangle = 1 + \langle G\rangle^2$, i.e. the quadratic equation:

$$\boxed{\langle G(z)\rangle = \frac{1}{z - \langle G(z)\rangle}}$$

Questions

1. **Question:** How is the Wigner semicircle law recovered from the averaged resolvent $G(z)$?

   Answer: Solving the quadratic self-consistent equation $\langle G\rangle^2 - z\langle G\rangle + 1 = 0$ yields the physical branch $\langle G(z)\rangle = \frac{z-\sqrt{z^2-4}}{2}$. The spectral density of eigenvalues is recovered via the branch cut discontinuity along the real axis: $\rho(\lambda) = -\frac{1}{\pi}\mathrm{Im}\langle G(\lambda + i0^+)\rangle = \frac{1}{2\pi}\sqrt{4-\lambda^2}$.

Derivation Blueprint: Resolvent Self-Consistent Equation

Board Layout:

- **Left Board Panel:** Define the resolvent operator $G(z)$ and write down the geometric series expansion.
- **Middle Panel:** Draw a diagram of crossing vs. non-crossing contractions for $n = 4$ to show why planar diagrams dominate at $D \to \infty$.
- **Right Panel:** Write out the nested recursive sum and complete the factorization into the quadratic equation.

Chalk-Talk Visual Guide:

- Draw a horizontal line representing the sequence of $2n$ matrix operators $HHH \dots H$. Show a contraction as a semi-circular arc connecting two operators.
- Underneath it, draw two distinct diagrams for the four-operator case $H_1H_2H_3H_4$:
  1. **Planar Contraction (Non-Crossing):** Draw an arc from H_1 to H_4 and an arc from H_2 to H_3. Explain that this represents a nested contraction where the outer pair shields the inner pair, corresponding to the term C_0C_1.
  2. **Non-Planar Contraction (Crossing):** Draw an arc from H_1 to H_3 and an arc from H_2 to H_4. These arcs cross. Show that this contraction contributes a factor of $1/D^2$, which vanishes in the $D \to \infty$ planar limit, proving why only non-crossing diagrams survive.

Common Conceptual Pitfalls: The diagrammatic nesting of the planar contractions in RMT resolvent equations is highly complex. One should visualize this as a self-similar fractal boundary: the large planar contraction encloses a sequence of smaller, independent nested contractions which represent the self-energy correction. **Boundary Limit & Sanity Check:** Turning off the coupling terms by setting $H = 0$ reduces the self-consistent quadratic equation to $G(z) = 1/z$, representing the resolvent of the zero matrix $\hat{O}$ and verifying the expected vanishing interaction limit.

3 Haar Measure & Weingarten Calculus

Exercise 3.1: Overlap of Haar-Random States and Measure Concentration

Motivation:

How much do two randomly chosen quantum states overlap? In low dimensions, there is significant variability. But in high dimensions, a phenomenon occurs: the overlap $|\langle\psi|\phi\rangle|^2$ **concentrates** around $1/D$ — two random states are nearly orthogonal with overwhelming probability. This concentration of measure underlies the typicality arguments used throughout quantum information: random states behave like typical states, and typical states are nearly maximally entangled.

Problem Statement:

The squared overlap $\chi = |\langle\psi|\phi\rangle|^2$ between two independent Haar-random pure states in D dimensions has density:

$$P(\chi) = (D-1)(1-\chi)^{D-2}, \qquad \chi \in [0,1].$$

(This is a $\text{Beta}(1, D-1)$ distribution.)

(a) Compute $\mathbb{E}[\chi]$ and verify $\mathbb{E}[\chi] = 1/D$.

(b) Compute $\text{Var}(\chi)$ and show it scales as $1/D^2$ for large D.

Pedantic Step-by-Step Solution:

Part (a): Mean overlap

Using the substitution $u = 1-\chi$, we have $\chi = 1-u$, $d\chi = -du$:

$$\mathbb{E}[\chi] = (D-1)\int_0^1 \chi(1-\chi)^{D-2}\, d\chi = (D-1)\int_0^1 (1-u)\, u^{D-2}\, du.$$

Expanding:

$$= (D-1)\left[\frac{1}{D-1} - \frac{1}{D}\right] = 1 - \frac{D-1}{D} = \frac{1}{D}.$$

$$\boxed{\mathbb{E}[\chi] = \frac{1}{D}.}$$

Two random states have expected overlap $1/D$ — they are nearly orthogonal in high dimensions.

Part (b): Variance

First compute the second moment:

$$\mathbb{E}[\chi^2] = (D-1)\int_0^1 (1-u)^2\, u^{D-2}\, du = (D-1)\left[\frac{1}{D-1} - \frac{2}{D} + \frac{1}{D+1}\right].$$

Simplifying:

$$\mathbb{E}[\chi^2] = 1 - \frac{2(D-1)}{D} + \frac{D-1}{D+1} = \frac{2}{D(D+1)}.$$

The variance is:

$$\text{Var}(\chi) = \frac{2}{D(D+1)} - \frac{1}{D^2} = \frac{2D-(D+1)}{D^2(D+1)} = \frac{D-1}{D^2(D+1)}.$$

$$\boxed{\text{Var}(\chi) = \frac{D-1}{D^2(D+1)} \sim \frac{1}{D^2} \quad \text{as } D \to \infty.}$$

The standard deviation scales as $1/D$, which is the **same order** as the mean itself. This means the *relative* fluctuation $\sigma/\mu \sim 1/\sqrt{D} \to 0$: the overlap concentrates around $1/D$ with vanishing relative spread.

Exercise 3.2: Operator Averages using Weingarten Calculus

Motivation:

The **first moment formula** (Schur's lemma) is the simplest and most widely used identity in Haar integration:

$$\mathbb{E}_U[UAU^\dagger] = \frac{\mathrm{Tr}(A)}{D}\, I.$$

It says that the Haar average of any operator, conjugated by a random unitary, is proportional to the identity — all directional information is washed out, and only the trace survives. This is the mathematical content of **depolarization**: a fully random rotation destroys all structure except the trace.

Problem Statement:

Let A, B be arbitrary $D \times D$ complex matrices. Using the first moment formula, compute

$$\mathbb{E}_U\left[\mathrm{Tr}(UAU^\dagger B)\right].$$

Pedantic Step-by-Step Solution:

Step 1: Apply the first moment formula

The trace is linear, so we can bring the Haar average inside:

$$\mathbb{E}_U\left[\mathrm{Tr}(UAU^\dagger B)\right] = \mathrm{Tr}\Big(\mathbb{E}_U[UAU^\dagger] \cdot B\Big).$$

By the first moment formula:

$$\mathbb{E}_U[UAU^\dagger] = \frac{\mathrm{Tr}(A)}{D}\, I.$$

Step 2: Evaluate the trace

$$\mathrm{Tr}\left(\frac{\mathrm{Tr}(A)}{D}\, I \cdot B\right) = \frac{\mathrm{Tr}(A)}{D}\, \mathrm{Tr}(B).$$

Result

$$\boxed{\mathbb{E}_U\left[\mathrm{Tr}(UAU^\dagger B)\right] = \frac{\mathrm{Tr}(A)\,\mathrm{Tr}(B)}{D}.}$$

Interpretation: The Haar average factorizes the product into individual traces — all correlations between A and B are erased. Only the "scalar content" (traces) of each operator survives the random rotation. This factorization is the trace-level manifestation of depolarization.

Exercise 3.3: The Fourth Moment of the Trace

1. Physical Motivation

The fourth moment $\mathbb{E}[|\mathrm{Tr}(U)|^4]$ is the simplest non-trivial application of the $k = 2$ Weingarten formula. The result — exactly 2 for **all** $D \geq 2$ — is completely dimension-independent. This universality:

- Serves as the defining test that a random circuit ensemble forms a **unitary 2-design**
- Sets the plateau value of the spectral form factor $\mathrm{SFF}(t = 0) = \mathbb{E}[|\mathrm{Tr}(U)|^4]$ in CUE statistics
- Follows from the CUE moment theorem: $\mathbb{E}[|\mathrm{Tr}(U)|^{2k}] = k!$ for $D \geq k$, giving $2! = 2$ for $k = 2$

2. Mathematical Preliminaries

The symmetric group S_k

The **symmetric group** S_k is the group of all permutations of $\{1, \dots, k\}$, containing $k!$ elements. For $k = 2$: $S_2 = \{\mathrm{id}, (12)\}$, where (12) is the transposition swapping elements 1 and 2.

The Weingarten function

The **Weingarten function** $\mathrm{Wg}(\pi, D)$ assigns a weight to each permutation $\pi \in S_k$, depending on the Hilbert space dimension D and the cycle structure of π. It is the inverse of the Gram matrix of permutation operators under the Haar measure:

$$\sum_{\sigma \in S_k} \mathrm{Wg}(\sigma, D) \cdot D^{\#\mathrm{cycles}(\pi\sigma^{-1})} = \delta_{\pi,\mathrm{id}}.$$

For $k = 2$ there are two values:

$$\mathrm{Wg}(\mathrm{id}, D) = \frac{1}{D^2 - 1}, \qquad \mathrm{Wg}\big((12), D\big) = \frac{-1}{D(D^2 - 1)}.$$

Note: $|\mathrm{Wg}(\mathrm{id})| > |\mathrm{Wg}((12))|$ — the identity (uncrossed) pairing dominates.

The Weingarten formula ($k = 2$)

$$\boxed{\mathbb{E}[U_{i_1 j_1} U_{i_2 j_2} U^*_{\ell_1 m_1} U^*_{\ell_2 m_2}] = \sum_{\pi, \sigma \in S_2} \mathrm{Wg}(\pi^{-1}\sigma, D) \prod_{a=1}^{2} \delta_{i_a \ell_{\pi(a)}} \delta_{j_a m_{\sigma(a)}}}$$

3. Analytical Derivation

Step 1: Index expansion

$|\operatorname{Tr}(U)|^4 = \operatorname{Tr}(U)^2 \overline{\operatorname{Tr}(U)}^2 = \sum_{i_1, i_2, \ell_1, \ell_2} U_{i_1 i_1} U_{i_2 i_2} U^*_{\ell_1 \ell_1} U^*_{\ell_2 \ell_2}$
Setting $j_a = i_a$ and $m_a = \ell_a$ (the trace constraint).

Step 2: Enumerate the four (π, σ) terms

— (π, σ) — $\pi^{-1}\sigma$ — Wg weight — Left deltas — Right deltas — Loop factor — ———————— ———————— — $(\mathrm{id}, \mathrm{id})$ — id — $\frac{1}{D^2-1}$ — $\delta_{i_1\ell_1}\delta_{i_2\ell_2}$ — $\delta_{i_1\ell_1}\delta_{i_2\ell_2}$ — D^2 — — $((12), (12))$ — id — $\frac{1}{D^2-1}$ — $\delta_{i_1\ell_2}\delta_{i_2\ell_1}$ — $\delta_{i_1\ell_2}\delta_{i_2\ell_1}$ — D^2 — — $(\mathrm{id}, (12))$ — (12) — $\frac{-1}{D(D^2-1)}$ — $\delta_{i_1\ell_1}\delta_{i_2\ell_2}$ — $\delta_{i_1\ell_2}\delta_{i_2\ell_1}$ — D — — $((12), \mathrm{id})$ — (12) — $\frac{-1}{D(D^2-1)}$ — $\delta_{i_1\ell_2}\delta_{i_2\ell_1}$ — $\delta_{i_1\ell_1}\delta_{i_2\ell_2}$ — D —

Why D vs D^2? When $\pi = \sigma$, the left and right deltas agree, giving 2 free indices $\to D^2$. When $\pi \neq \sigma$, they conflict, forcing all 4 indices equal $\to$ only 1 free index $\to D$.

Step 3: Combine

$$\mathbb{E}[|\operatorname{Tr}(U)|^4] = \frac{2D^2}{D^2 - 1} - \frac{2}{D^2 - 1} = \frac{2(D^2 - 1)}{D^2 - 1} = \boxed{2}$$

4. Physical Interpretation

Tensor network perspective: Each of the four (π, σ) terms corresponds to a distinct wiring of the tensor diagram:

- **Planar pairings** ($\pi^{-1}\sigma = \mathrm{id}$): uncrossed wires, producing 2 independent loops $\to D^2$ per term
- **Non-planar pairings** ($\pi^{-1}\sigma = (12)$): crossed wires, forcing all indices to collapse to 1 loop $\to D$ per term

The exact cancellation $\frac{2D^2}{D^2-1} - \frac{2}{D^2-1} = 2$ is the algebraic manifestation of a **topological** fact: the number of independent wiring topologies, not the dimension of the group, controls the moments. This universality is why $|\operatorname{Tr}(U)|^4 = 2$ serves as the benchmark test for 2-design convergence.

Exercise 3.4: The Asymptotic Page Formula via Marchenko–Pastur

Motivation:

Page's formula gives the average entanglement entropy of a subsystem in a Haar-random state. It is the single most important result connecting random state geometry to quantum information: it tells us that **typical states are nearly maximally entangled**. The asymptotic form $\mathbb{E}[S] \approx \ln m - m/(2n)$ (valid for $m \leq n$) combines two ideas: (1) the eigenvalues of a random reduced state follow the **Marchenko-Pastur** distribution, and (2) the entropy correction can be computed by a simple Taylor expansion.

Problem Statement:

Derive the asymptotic Page formula

$$\mathbb{E}[S(\rho_A)] \approx \ln m - \frac{m}{2n}$$

by treating the eigenvalues of ρ_A as following the Marchenko-Pastur law with ratio $c = m/n$.

Pedantic Step-by-Step Solution:

Step 1: Setup

The m eigenvalues λ_j of ρ_A satisfy $\sum_j \lambda_j = 1$ and $\lambda_j \geq 0$. Introduce rescaled variables $x_j = m\lambda_j$ so that $\langle x \rangle = 1$. The entropy is:

$$S = -\sum_j \lambda_j \ln \lambda_j = -\sum_j \frac{x_j}{m} \ln \frac{x_j}{m} = \ln m - \frac{1}{m} \sum_j x_j \ln x_j.$$

In the large-m limit, the empirical spectral distribution of the x_j converges to the Marchenko-Pastur law with parameter $c = m/n$.

Step 2: Taylor expansion

Define $f(x) = x \ln x$. Taylor-expand around the mean $x = 1$:

$$f(x) \approx f(1) + f'(1)(x-1) + \frac{1}{2} f''(1)(x-1)^2 = 0 + (x-1) + \frac{1}{2}(x-1)^2.$$

(We used $f(1) = 0$, $f'(x) = 1 + \ln x$ so $f'(1) = 1$, $f''(x) = 1/x$ so $f''(1) = 1$.)

Step 3: Integrate against the spectral distribution

Average over the Marchenko-Pastur distribution:

$$\langle f(x) \rangle \approx \underbrace{\langle x - 1 \rangle}_{=0} + \frac{1}{2} \underbrace{\langle (x-1)^2 \rangle}_{=\mathrm{Var}(x)=c=m/n}.$$

(The variance of the Marchenko-Pastur distribution with mean 1 and ratio c is exactly c.)
Substituting back:

$$\mathbb{E}[S] = \ln m - \frac{1}{m} \cdot m \cdot \frac{m}{2n} = \ln m - \frac{m}{2n}.$$

$$\boxed{\mathbb{E}[S(\rho_A)] \approx \ln m - \frac{m}{2n}.}$$

Physical interpretation: The leading term $\ln m$ is the maximal entanglement entropy — a random state is nearly maximally entangled. The correction $-m/(2n)$ quantifies the small deficit and vanishes when $n \gg m$ (the complement is much larger than the subsystem).

Exercise 3.5: The SWAP Trick in Action

Motivation:

The SWAP trick $\mathrm{Tr}(AB) = \mathrm{Tr}[(A \otimes B)F]$ is the computational engine behind purity estimation, Rényi entropy measurements, and the Weingarten calculus. In experiments, it allows measuring the purity of a quantum state (a nonlinear function of ρ) via **two-copy interference** — no tomography needed. This exercise develops fluency with the SWAP trick and connects it to Haar-averaged entanglement.

Problem Statement:

(a) Prove $\mathrm{Tr}[(A \otimes B)F] = \mathrm{Tr}(AB)$ for the SWAP operator F.
(b) For the Bell state $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$, compute ρ_A and verify $\mathrm{Tr}(\rho_A^2) = \mathrm{Tr}[(\rho_A \otimes \rho_A)F_A] = 1/2$.
(c) Compare with the Haar-averaged purity $\mathbb{E}[\gamma] = (m+n)/(mn+1)$ at $m = n = 2$. Is the Bell state more or less entangled than a typical random state?

Pedantic Step-by-Step Solution:

Part (a): Proof of the SWAP trick

The SWAP operator acts as $F|i,j\rangle = |j,i\rangle$. Therefore:

$$\mathrm{Tr}[(A \otimes B)F] = \sum_{i,j} \langle i,j|(A \otimes B)|j,i\rangle = \sum_{i,j} A_{ij}\, B_{ji} = \mathrm{Tr}(AB). \quad \checkmark$$

The SWAP converts a two-copy tensor product into a single-copy trace — trading one quantum copy for a classical computation.

Part (b): Bell state purity

Step 1: The reduced state of the Bell state is:

$$\rho_A = \mathrm{Tr}_B\, |\Phi^+\rangle\langle\Phi^+| = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{I}{2}.$$

Step 2 (direct): $\mathrm{Tr}(\rho_A^2) = \mathrm{Tr}(I/4) = 2/4 = 1/2$.
Step 3 (SWAP trick):

$$\rho_A \otimes \rho_A = \frac{I}{2} \otimes \frac{I}{2} = \frac{I_4}{4}.$$

$$\mathrm{Tr}[(\rho_A \otimes \rho_A)F] = \frac{1}{4}\,\mathrm{Tr}(F) = \frac{1}{4} \cdot 2 = \frac{1}{2}. \quad \checkmark$$

Both methods agree: $\gamma = 1/2 = 1/D_A$, confirming the Bell state is maximally entangled.

Part (c): Comparison with Haar average

The Haar-averaged purity for a bipartite system with subsystem dimensions m and n is:

$$\mathbb{E}[\gamma] = \frac{m+n}{mn+1}.$$

For $m = n = 2$ (two qubits on a 4-dimensional Hilbert space):

$$\mathbb{E}[\gamma] = \frac{2+2}{2 \cdot 2 + 1} = \frac{4}{5} = 0.8.$$

The Bell state has $\gamma_{\text{Bell}} = 0.5 < 0.8 = \mathbb{E}[\gamma]$. The Bell state is **more entangled** than a typical Haar-random state on 2 qubits.

This makes sense: at $D = 4$, the Page prediction is only approximate, and the typical state is not yet close to maximal entanglement. From $\mathbb{E}[\gamma] = (m+n)/(mn+1)$, the maximally mixed value $1/m$ is reached only when $n \gg m$; for the equal bipartition $m = n$ here, $\mathbb{E}[\gamma] \to 2/m$, twice the minimum purity.

Exercise 3.6: Concentration of Purity

Motivation:

Concentration of measure is the mathematical mechanism behind quantum typicality: for Haar-random states in high-dimensional Hilbert spaces, observables such as entanglement entropy and purity are concentrated around their mean values. Lévy's lemma provides the quantitative bound — the probability of a deviation ϵ from the mean decays exponentially in the Hilbert space dimension D. This exercise applies Lévy's lemma to the purity, showing that in multi-qubit systems, virtually every random state has nearly the same subsystem purity.

Preliminaries / Toolbox

Before diving into the solution, recall the following concepts:

1. Purity: The purity of a state ρ is $\gamma = \mathrm{Tr}(\rho^2)$. For pure states $\gamma = 1$, and for maximally mixed states $\gamma = 1/D$.

2. Haar Measure: The uniform distribution over the unitary group $U(D)$. Integrals over polynomials of U_{ij} and $\bar{U}_{kl}$ can be evaluated using Weingarten calculus.

3. Lévy's Lemma: Given a Lipschitz-continuous function $f : \mathbb{S}^{2D-1} \to \mathbb{R}$ on the unit hypersphere, the probability that $f(|\psi\rangle)$ deviates from its median by more than ϵ decays exponentially with dimension D: $\Pr(|f - \bar{f}| \geq \epsilon) \leq 2\exp\left(-\frac{CD\epsilon^2}{\eta^2}\right)$.

4. Partial Trace: The reduced density matrix of subsystem A is $\rho_A = \mathrm{Tr}_B(|\psi\rangle\langle\psi|)$, where $|\psi\rangle$ is a bipartite state on $\mathcal{H}_A \otimes \mathcal{H}_B$.

Problem Statement:

The purity $\gamma = \mathrm{Tr}(\rho_A^2)$ of a Haar-random state has mean $\mathbb{E}[\gamma] = (m+n)/(mn+1)$. The function $f(\psi) = \mathrm{Tr}(\rho_A^2)$ is Lipschitz with constant $\eta_f = 4$.

(a) Using Lévy's lemma, write the concentration bound $\Pr(|\gamma - \mathbb{E}[\gamma]| > \epsilon)$.

(b) For an equal bipartition ($m = n = 2^{N/2}$), find the minimum N such that $\Pr(\gamma > \mathbb{E}[\gamma] + 0.01) < 10^{-6}$.

Pedantic Step-by-Step Solution:

Part (a): Lévy's lemma bound

Lévy's lemma states: if $f : S^{2D-1} \to \mathbb{R}$ is Lipschitz with constant η_f, then for Haar-random $|\psi\rangle$:

$$\Pr(|f(\psi) - \mathbb{E}[f]| > \epsilon) \leq 2\exp\left(-\frac{c\,D\,\epsilon^2}{\eta_f^2}\right),$$

where $c > 0$ is a universal constant of order unity and $D = mn$ is the total Hilbert space dimension. For the purity with $\eta_f = 4$:

$$\boxed{\Pr(|\gamma - \mathbb{E}[\gamma]| > \epsilon) \leq 2\exp\left(-\frac{c\,D\,\epsilon^2}{16}\right).}$$

The concentration is **exponentially tight** in D: doubling the number of qubits squares the Hilbert space dimension and drives the failure probability to zero exponentially fast.

Part (b): Critical qubit number

For the equal bipartition $m = n = 2^{N/2}$, we have $D = 2^N$. The condition is:

$$2\exp\left(-\frac{c \cdot 2^N \cdot 10^{-4}}{16}\right) < 10^{-6}.$$

Taking logarithms (with $c \approx 1$):

$$\frac{2^N \cdot 10^{-4}}{16} > \ln(2 \times 10^6) \approx 14.5.$$

$$2^N > \frac{16 \times 14.5}{10^{-4}} \approx 2.3 \times 10^6.$$

Since $2^{21} \approx 2.1 \times 10^6$ and $2^{22} \approx 4.2 \times 10^6$:

$$\boxed{N \geq 22 \text{ qubits.}}$$

With just 22 qubits (a 22-qubit Hilbert space of dimension $\sim 4 \times 10^6$), the probability that the purity deviates by more than 0.01 from its mean is less than one in a million. This extreme concentration is why typicality arguments are so powerful in quantum many-body physics.
Visual Pedagogical Proof

Exercise 3.7: Tensor Network Weingarten Calculus

1. Physical Motivation

Many quantities in quantum information — purities, frame potentials, OTOCs, channel fidelities — reduce to polynomial averages over Haar-random unitaries. The **Weingarten calculus** is a systematic tool for computing these averages. It is the unitary-group analogue of Wick's theorem for Gaussian integrals: just as Gaussian expectations reduce to sums over pair contractions, Haar expectations reduce to sums over **permutation pairings**, weighted by the **Weingarten function**.
The tensor network (Penrose) representation makes these sums geometrically transparent. The goal of this notebook is to master the three-step recipe:

1. **Draw** the unevaluated tensor network
2. **Replace** U/U^* pairs with permutation pairings
3. **Count** closed loops (each loop = factor of D)

2. Mathematical Preliminaries (Self-Contained)

2.1 The symmetric group S_k

The **symmetric group** S_k is the group of all permutations of k objects $\{1, 2, \ldots, k\}$. It has $k!$ elements.

- $S_1 = \{\mathrm{id}\}$: one element (the identity).
- $S_2 = \{\mathrm{id}, (12)\}$: two elements — the identity and the swap of objects 1 and 2.
- S_3: six elements — id, three transpositions $(12), (13), (23)$, and two 3-cycles $(123), (132)$.

A permutation $\pi \in S_k$ maps each index $a \in \{1, \ldots, k\}$ to $\pi(a)$. The **cycle structure** of π decomposes it into disjoint cycles. For example, (123) means $1 \to 2 \to 3 \to 1$ — a single cycle of length 3. The number of cycles determines how many independent index sums survive after applying Kronecker deltas.

2.2 The Weingarten function

The **Weingarten function** $\mathrm{Wg}(\pi, D)$ is a real-valued function on S_k (depending on Hilbert space dimension D) that serves as the weight for each permutation pairing in the Haar integral. It is defined implicitly as the inverse of the Gram matrix:

$$\sum_{\sigma \in S_k} \mathrm{Wg}(\sigma, D) \cdot D^{\#\mathrm{cycles}(\pi\sigma^{-1})} = \delta_{\pi,\mathrm{id}}.$$

Key properties:

- $\mathrm{Wg}(\pi, D)$ depends only on the **cycle type** (conjugacy class) of π
- The identity permutation has the **largest** weight — it is the "planar" (uncrossed) pairing

- Non-identity permutations ("crossings") are suppressed by powers of $1/D$

2.3 The Weingarten formula

The k-th moment of the Haar measure over $U(D)$ is:

$$\boxed{\mathbb{E}_{\text{Haar}}[U_{i_1j_1}\cdots U_{i_kj_k}U^*_{\ell_1m_1}\cdots U^*_{\ell_km_k}] = \sum_{\pi,\sigma\in S_k} \text{Wg}(\pi^{-1}\sigma, D)\prod_{a=1}^{k}\delta_{i_a\ell_{\pi(a)}}\delta_{j_am_{\sigma(a)}}}$$

Interpretation: The permutation π controls how the *output* (left) indices of U are paired with those of U^*. The permutation σ does the same for *input* (right) indices. Each pairing is weighted by $\text{Wg}(\pi^{-1}\sigma, D)$.

2.4 Explicit Weingarten values

— k — Permutation τ — Cycle type — $\text{Wg}(\tau, D)$ — ———————— — 1 — id — (1) — $1/D$ — — 2 — id — (1)(2) — $1/(D^2-1)$ — — 2 — (12) — (12) — $-1/[D(D^2-1)]$ — — 3 — id — (1)(2)(3) — $(D^2-2)/[D(D^2-1)(D^2-4)]$ — — 3 — any transposition — (2, 1) — $-1/[(D^2-1)(D^2-4)]$ — — 3 — any 3-cycle — (3) — $2/[D(D^2-1)(D^2-4)]$ —

2.5 Tensor network rules

- **Box** = matrix/tensor; **wire** = index; **connected wire** = contracted (summed) index
- **Closed loop** (wire with no open ends) = contributes a factor of D
- **Cup** (semicircular arc connecting two wires) = Kronecker delta δ_{ij}

3. Part (a): First Moment — $\mathbb{E}[\text{Tr}(UAU^\dagger B)]$

Analytical derivation

Goal: Compute $\mathbb{E}_U[\text{Tr}(\hat{U}\hat{A}\hat{U}^\dagger\hat{B})]$ for fixed matrices $\hat{A}, \hat{B}$.

Step 1 — Index expansion:

$$\text{Tr}(UAU^\dagger B) = \sum_{i,j,k,l} U_{ij}\,A_{jk}\,U^*_{lk}\,B_{li}$$

Step 2 — Apply $k=1$ Weingarten ($|S_1|=1$, only $\pi=\sigma=\text{id}$):

$$\mathbb{E}[U_{ij}U^*_{lk}] = \frac{1}{D}\,\delta_{il}\,\delta_{jk}$$

Step 3 — Contract deltas:

- δ_{jk} closes a loop through $\hat{A}$: $\sum_j A_{jj} = \text{Tr}(\hat{A})$
- δ_{il} closes a loop through $\hat{B}$: $\sum_i B_{ii} = \text{Tr}(\hat{B})$

Result: $\mathbb{E}[\text{Tr}(UAU^\dagger B)] = \frac{\text{Tr}(\hat{A})\,\text{Tr}(\hat{B})}{D}$

4. Part (b): Second Moment — Haar-Averaged Purity

Analytical derivation

Goal: Derive $\mathbb{E}[\text{Tr}(\hat{\rho}_A^2)]$ for a Haar-random pure state $|\psi\rangle = U|0\rangle$ on $\mathcal{H}_A\otimes\mathcal{H}_B$ with $\dim\mathcal{H}_A = D_A$, $\dim\mathcal{H}_B = D_B$, $D = D_AD_B$.

Step 1 — Setup: Using the SWAP trick, $\text{Tr}(\hat{\rho}_A^2) = \text{Tr}(\mathbb{F}_A\cdot\hat{\rho}^{\otimes 2})$. This requires $k=2$ copies of U and U^*.

Step 2 — Index structure: All input indices are fixed to the reference state $|0\rangle$, meaning $j_1 = j_2 = m_1 = m_2 = 0$. This makes $\delta_{j_a,m_{\sigma(a)}} = \delta_{0,0} = 1$ for **every** $\sigma\in S_2$. Therefore the σ sum is **free**: every permutation σ contributes equally.

Step 3 — Sigma sum:

$$\sum_{\sigma\in S_2} \mathrm{Wg}(\pi^{-1}\sigma, D) = \sum_{\tau\in S_2} \mathrm{Wg}(\tau, D) = \frac{1}{D^2-1} - \frac{1}{D(D^2-1)} = \frac{1}{D(D+1)}$$

This is the same for every π (since we're summing $\tau = \pi^{-1}\sigma$ over all of S_2).
Step 4 — Loop counting: The output wiring has SWAP on A and identity on B. The effective permutation for output pairing π is:
— π — Eff. on A: $\pi \circ \mathrm{SWAP}$ — Eff. on B: $\pi \circ \mathrm{id}$ — Loop factor — ——————————— — id — (12): 1 cycle $\to D_A^1$ — id: 2 cycles $\to D_B^2$ — $D_A D_B^2$ — — (12) — id: 2 cycles $\to D_A^2$ — (12): 1 cycle $\to D_B^1$ — $D_A^2 D_B$ —
Step 5 — Assembly:

$$\mathbb{E}[\mathrm{Tr}(\hat{\rho}_A^2)] = \frac{1}{D(D+1)}(D_A D_B^2 + D_A^2 D_B) = \frac{D_A D_B(D_A + D_B)}{D_A D_B(D_A D_B + 1)} = \frac{D_A + D_B}{D_A D_B + 1}$$

5. Part (c): Third Moment — $\mathbb{E}[|\,\mathrm{Tr}(U)|^6]$ via Full S_3 Calculus

Analytical derivation

Goal: Compute $\mathbb{E}[|\,\mathrm{Tr}(U)|^6]$ for Haar-random $U \in U(D)$.
Step 1 — Index expansion: $|\,\mathrm{Tr}(U)|^6 = \sum_{i_1,i_2,i_3,\ell_1,\ell_2,\ell_3} U_{i_1 i_1} U_{i_2 i_2} U_{i_3 i_3} U^*_{\ell_1\ell_1} U^*_{\ell_2\ell_2} U^*_{\ell_3\ell_3}$.
This sets $j_a = i_a$ and $m_a = \ell_a$ (the trace constraint: all indices are "diagonal").
Step 2 — Weingarten sum: For each $(\pi, \sigma) \in S_3 \times S_3$, the delta product $\prod_a \delta_{i_a,\ell_{\pi(a)}} \delta_{i_a,\ell_{\sigma(a)}}$ forces $\ell_{\pi(a)} = \ell_{\sigma(a)}$ for all a. The number of free summation indices equals $D^{\#\mathrm{cycles}(\pi^{-1}\sigma)}$.
Step 3 — Group by cycle type of $\pi^{-1}\sigma$: Since Wg depends only on cycle type, we count how many of the 36 pairs (π, σ) give each cycle type of $\tau \equiv \pi^{-1}\sigma$. For fixed τ there are exactly $|S_3| = 6$ pairs (π, σ) with $\pi^{-1}\sigma = \tau$ (choose π freely, then $\sigma = \pi\tau$). Hence

$$\mathbb{E}[|\,\mathrm{Tr}\,U|^6] = \sum_{\pi,\sigma\in S_3} D^{\#\mathrm{cycles}(\pi^{-1}\sigma)}\,\mathrm{Wg}(\pi^{-1}\sigma, D) = 6\sum_{\tau\in S_3} D^{\#\mathrm{cycles}(\tau)}\,\mathrm{Wg}(\tau, D).$$

S_3 has one identity (3 cycles), three transpositions (2 cycles each), and two 3-cycles (1 cycle each), so

$$\mathbb{E}[|\,\mathrm{Tr}\,U|^6] = 6\Big[D^3\,\mathrm{Wg}(\mathrm{id}) + 3\,D^2\,\mathrm{Wg}((12)) + 2\,D\,\mathrm{Wg}((123))\Big].$$

Step 4 — Weingarten values and the sum identity: For $D \geq 3$ the S_3 Weingarten functions are

$$\mathrm{Wg}(\mathrm{id}) = \frac{D^2-2}{D(D^2-1)(D^2-4)}, \quad \mathrm{Wg}((12)) = \frac{-1}{(D^2-1)(D^2-4)}, \quad \mathrm{Wg}((123)) = \frac{2}{D(D^2-1)(D^2-4)}.$$

Placing all three terms over the common denominator $D(D^2-1)(D^2-4)$, the bracketed combination collapses to 1 identically:

$$D^3\,\mathrm{Wg}(\mathrm{id}) + 3D^2\,\mathrm{Wg}((12)) + 2D\,\mathrm{Wg}((123)) = \frac{D^3(D^2-2) - 3D^3 + 4D}{D(D^2-1)(D^2-4)} = \frac{D^5 - 5D^3 + 4D}{D(D^2-1)(D^2-4)} = \frac{D(D^2-1)(D^2}{D(D^2-1)(D^2}$$

using $D^4 - 5D^2 + 4 = (D^2-1)(D^2-4)$ (one checks numerically at $D = 3$: $\frac{7}{120}\cdot 27 - 3\cdot\frac{1}{40}\cdot 9 + 2\cdot\frac{1}{60}\cdot 3 = 1.575 - 0.675 + 0.1 = 1$). Therefore

$$\boxed{\mathbb{E}[|\,\mathrm{Tr}\,U|^6] = 6 = 3! \quad \text{exactly for all } D \geq 3.}$$

This is the $k = 3$ case of the general identity $\sum_{\tau\in S_k} D^{\#\mathrm{cycles}(\tau)}\,\mathrm{Wg}(\tau, D) = 1$ (the Weingarten functions are the inverse of the Gram matrix $G_{\sigma\tau} = D^{\#\mathrm{cycles}(\sigma^{-1}\tau)}$), which gives $\mathbb{E}[|\,\mathrm{Tr}\,U|^{2k}] = k!$ for all $D \geq k$.

6. Physical Interpretation

Why does $\mathbb{E}[|\operatorname{Tr}(U)|^{2k}] = k!$?

This result connects Haar-random unitaries to **complex Gaussian random variables**. For a single complex Gaussian $z \sim \mathcal{N}_{\mathbb{C}}(0,1)$, the moments are $\mathbb{E}[|z|^{2k}] = k!$ (by Wick's theorem). In the large-D limit, $\operatorname{Tr}(U)$ for a Haar-random $U \in U(D)$ behaves like a standard complex Gaussian — this is a consequence of the central limit theorem applied to the D eigenvalues $e^{i\theta_j}$ of U.

The Weingarten calculus gives $k!$ **exactly** for all $D \geq k$, not just asymptotically. This is because the Weingarten function is the exact inverse of the representation-theoretic Gram matrix, and the exact cancellations between planar and non-planar contributions produce dimension-independent trace moments.

The topological structure

1. **Planar pairings** (identity permutation, uncrossed wires) produce the most loops and dominate at large D
2. **Non-planar pairings** (crossings) produce fewer loops and are suppressed by $1/D$ per crossing
3. The $1/D$ suppression hierarchy mirrors the **genus expansion** of matrix models — a connection between random matrix theory and two-dimensional quantum gravity

Theoretical Guidance: The Intuition of Twirling

To master the mathematics of random quantum states and channels, one must develop a robust physical intuition for the process of **twirling**. In theoretical physics and quantum information, a twirl is the mathematical representation of "complete rotational washing." It acts as a physical filter that systematically strips away directional or basis-dependent features, leaving only the isotropic, coordinate-free skeleton of an operator.

1. The Geometric Picture: Isotropic Spherical Washing

Consider a single-qubit density matrix $\hat{\rho}$ represented as a vector $\vec{r}$ inside the 3D Bloch sphere:

$$\hat{\rho} = \frac{\hat{I} + \vec{r} \cdot \vec{\hat{\sigma}}}{2}$$

If we apply a rotation $\hat{U}$ to the qubit, the Bloch vector $\vec{r}$ undergoes a corresponding 3D rotation to $\vec{r}' = R_U \vec{r}$. If we select a rotation $\hat{U}$ uniformly at random from the full rotation group $SU(2)$ and average over all possible orientations (the Haar twirl), we write:

$$\mathbb{E}_{U\sim\text{Haar}}[\hat{U}\hat{\rho}\hat{U}^\dagger] = \int_{SU(2)} \hat{U}\hat{\rho}\hat{U}^\dagger \, dU$$

Because the Haar measure is perfectly isotropic, it has no preferred direction. The continuous average of the rotated vectors $\vec{r}'$ over the entire sphere cancels out every directional component, collapsing the average vector strictly to the origin ($\vec{r}'_{\text{avg}} = 0$). The resulting state is the maximally mixed state:

$$\mathbb{E}_{U\sim\text{Haar}}[\hat{U}\hat{\rho}\hat{U}^\dagger] = \frac{\hat{I}}{2}$$

For a general operator $\hat{O}$ in a D-dimensional Hilbert space, the Haar twirl $\mathbb{E}_U[\hat{U}\hat{O}\hat{U}^\dagger]$ washes away all off-diagonal elements in any fixed basis. The only operator invariant under all possible rotations is the identity matrix $\hat{I}$. Schur's Lemma enforces that the twirled operator must be a scalar multiple of the identity, where conservation of the trace determines the scaling factor: $c = \text{Tr}(\hat{O})/D$.

2. Higher-Order Twirling: Averaging Correlations

While a first-order twirl $\mathbb{E}_U[\hat{U}\hat{O}\hat{U}^\dagger]$ averages out linear vector-like components, it cannot capture quadratic quantities such as variance, purity, or quantum correlations. To average these, we must twirl the tensor-product operator over two copies of the unitary group (a second-order twirl):

$$\mathbb{E}_{U\sim\text{Haar}}[\hat{U}^{\otimes 2}\hat{O}\hat{U}^{\dagger\otimes 2}]$$

Under this doubled rotation, Schur-Weyl duality guarantees that the only operators commuting with the joint rotation $\hat{U} \otimes \hat{U}$ are permutations. For two copies ($k = 2$), the only possible permutations are the identity $\hat{\mathbb{I}}$ (which does nothing) and the SWAP operator $\mathbb{F}$ (which exchanges the two copies). A second-order twirl therefore projects any bipartite operator strictly onto a linear combination of these two coordinate-free generators:

$$\mathbb{E}_{U\sim\text{Haar}}[\hat{U}^{\otimes 2}\hat{O}\hat{U}^{\dagger\otimes 2}] = c_I\hat{\mathbb{I}} + c_{\mathbb{F}}\mathbb{F}$$

The coefficients c_I and $c_{\mathbb{F}}$ represent the projections of the operator $\hat{O}$ onto the symmetric and anti-symmetric subspaces of the tensor product space $\mathcal{H} \otimes \mathcal{H}$.

3. Physical and Experimental Realization

In actual laboratories, Haar-random unitaries are mathematically continuous and require infinitely fine-tuned control, making them impossible to implement directly. However, we can bypass this limitation using two key insights:

1. **Unitary t-Designs:** If we only need to evaluate t-th order moments (such as average purity, which is $t = 2$), we do not need the full infinite unitary group. We can average over a finite subset of unitaries that reproduces the Haar moments up to order t. The Clifford group is a famous example: it forms a perfect unitary 3-design, meaning Clifford twirling yields results identical to Haar twirling for all calculations involving up to three copies of unitaries.

2. **Chaotic Dynamics as Natural Twirls:** In quantum many-body systems, chaotic time evolution under a generic Hamiltonian H behaves as an effective twirling mechanism. Over long times, the system's own dynamics act as a pseudo-random unitary generator, scrambling local information across the global degrees of freedom and thermalizing subsystems back to the Haar-average maximally mixed state.

Card 1: First Haar Moment

Goal: Prove that the Haar average of a twirled operator $\hat{O}$ is proportional to the identity:

$$\mathbb{E}_{U\sim\text{Haar}}[\hat{U}\hat{O}\hat{U}^\dagger] = \frac{\text{Tr}(\hat{O})}{D}\,\hat{I}$$

Key Trick: Left-invariance of the Haar measure ($dU = d(VU)$) forces the twirled operator to commute with all unitaries, which by Schur's Lemma restricts it to a multiple of the identity.

Step-by-Step Derivation:

1. Let $\hat{M} = \mathbb{E}_{U\sim\text{Haar}}[\hat{U}\hat{O}\hat{U}^\dagger] = \int_{U(D)} \hat{U}\hat{O}\hat{U}^\dagger\, dU$, where dU is the normalized Haar measure satisfying $\int dU = 1$.

2. Multiply $\hat{M}$ by any fixed unitary $\hat{V} \in U(D)$ from the left, and $\hat{V}^\dagger$ from the right:

$$\hat{V}\hat{M}\hat{V}^\dagger = \hat{V}\left(\int_{U(D)} \hat{U}\hat{O}\hat{U}^\dagger\, dU\right)\hat{V}^\dagger = \int_{U(D)} (\hat{V}\hat{U})\hat{O}(\hat{V}\hat{U})^\dagger\, dU$$

3. Define a new integration variable $\hat{U}' = \hat{V}\hat{U}$. Since the Haar measure is left-invariant, the volume element is invariant under translation: $dU' = dU$. The integration domain $U(D)$ is a group and remains unchanged under group multiplication. Thus:

$$\hat{V}\hat{M}\hat{V}^\dagger = \int_{U(D)} \hat{U}'\hat{O}\hat{U}'^\dagger\, dU' = \hat{M}$$

4. Multiply both sides by $\hat{V}$ from the right to get:

$$\hat{V}\hat{M} = \hat{M}\hat{V} \quad \forall \hat{V} \in U(D)$$

5. Since $\hat{M}$ commutes with all unitaries, it commutes with any linear combination of unitaries. Every operator can be written as a linear combination of unitaries. Thus, $\hat{M}$ commutes with all operators on the Hilbert space. By Schur's Lemma, any operator that commutes with all operators must be a scalar multiple of the identity:

$$\hat{M} = c\,\hat{I}$$

6. To determine the scalar c, take the trace of both sides:

$$\text{Tr}(\hat{M}) = \text{Tr}(c\,\hat{I}) = c\,D$$

7. Using the linearity of the trace and the expectation value:

$$\text{Tr}(\hat{M}) = \mathbb{E}_U[\text{Tr}(\hat{U}\hat{O}\hat{U}^\dagger)] = \mathbb{E}_U[\text{Tr}(\hat{O}\hat{U}^\dagger\hat{U})] = \mathbb{E}_U[\text{Tr}(\hat{O})] = \text{Tr}(\hat{O})$$

8. Equating the two trace values yields $c\,D = \text{Tr}(\hat{O}) \implies c = \frac{\text{Tr}(\hat{O})}{D}$. Substituting this back gives the final result:

$$\boxed{\mathbb{E}_U[\hat{U}\hat{O}\hat{U}^\dagger] = \frac{\text{Tr}(\hat{O})}{D}\,\hat{I}}$$

Questions

1. **Question:** Does this hold for any state?

   Answer: Yes, setting $\hat{O} = |\psi_0\rangle\langle\psi_0|$ immediately yields $\mathbb{E}[|\psi\rangle\langle\psi|] = \hat{I}/D$. The average state under Haar-random twirling is always the maximally mixed state, reflecting complete isotropic randomness.

Derivation Blueprint: First Haar Moment

Board Layout:

- **Left Board Panel:** Draw a unit circle representing the unitary group $U(D)$, showing left-translation invariance $U' = VU$. Write the normalization condition $\int dU = 1$ in a prominent box at the top.
- **Middle Panel:** Write the 3-step proof that $VMV^\dagger = M$. Write Schur's Lemma in a side column to explain step 5.
- **Right Panel:** Take the trace of both sides to extract c, highlighting the cyclic property of the trace.

Chalk-Talk Visual Guide:

- Draw a 3D sphere representing the state space. Mark an initial state $|\psi_0\rangle$ on the surface. Draw several arrows showing random unitary rotations $\hat{U}_i$ scattering $|\psi_0\rangle$ over the sphere. Explain that since the Haar measure is perfectly isotropic (has no preferred direction), averaging these rank-1 projectors over the entire sphere cancels all directional components, collapsing the average state strictly to the origin (the maximally mixed state $\hat{I}/D$).

Common Conceptual Pitfalls: A common point of confusion is distinguishing the Haar twirling of a general *operator* $\mathbb{E}_U[\hat{U}\hat{O}\hat{U}^\dagger]$ from the twirling of a *state vector* $\mathbb{E}_U[|\psi\rangle\langle\psi|]$. State vectors are simply rank-1 operators; since any pure state satisfies $\mathrm{Tr}(|\psi\rangle\langle\psi|) = 1$, the general operator formula instantly simplifies to the state vector case. **Boundary Limit & Sanity Check:** If we choose the identity operator $\hat{O} = \hat{I}$, its trace is $\mathrm{Tr}(\hat{I}) = D$, which yields the coefficient $c = \mathrm{Tr}(\hat{I})/D = 1$. The Haar twirl thus maps the identity back to itself ($\mathbb{E}_U[\hat{U}\hat{I}\hat{U}^\dagger] = \hat{I}$), satisfying the expected identity preservation of the channel.

Numerical Sandbox: First Haar Moment Twirl Verification

```python
import numpy as np
D = 4  # Hilbert space dimension
O = np.diag([1.0, 2.0, 3.0, 4.0])  # Input operator O
# Generate a Haar-random unitary using QR decomposition
X = np.random.normal(size=(D, D)) + 1j * np.random.normal(size=(D, D))
Q, R = np.linalg.qr(X)
U = Q @ np.diag(np.diagonal(R) / np.abs(np.diagonal(R)))
# Verify single twirled sample and compare trace expectation
twirled = U @ O @ U.conj().T
print(f"Tr(O)/D * I:\n{np.real(np.trace(O) / D * np.eye(D))}")
print(f"Sample U O U^\dagger (average over many matches this):\n{np.real(twirled)}")
```

Card 2: SWAP Operator and Swap Trick

Goal: Prove that the trace of a tensor product contracted with the SWAP operator $\mathbb{F}$ yields the trace of the product of the operators:

$$\mathrm{Tr}[(\hat{A} \otimes \hat{B})\,\mathbb{F}] = \mathrm{Tr}(\hat{A}\hat{B})$$

Key Trick: Represent the trace as a sum over the computational basis and apply the permutation definition of $\mathbb{F}$ directly to the index structure.

Step-by-Step Derivation:

1. The SWAP operator $\mathbb{F}$ is defined on the tensor product space $\mathcal{H} \otimes \mathcal{H}$ by its action on computational basis states:
$$\mathbb{F}\,|i\rangle\,|j\rangle = |j\rangle\,|i\rangle$$
2. We can write $\mathbb{F}$ in bra-ket notation as:
$$\mathbb{F} = \sum_{i,j=1}^{D} |j,i\rangle\,\langle i,j|$$
3. Now, write the trace of the operator $(\hat{A} \otimes \hat{B})\mathbb{F}$ explicitly in the computational basis:
$$\mathrm{Tr}[(\hat{A} \otimes \hat{B})\,\mathbb{F}] = \sum_{m,n=1}^{D} \langle m,n|\,(\hat{A} \otimes \hat{B})\,\mathbb{F}\,|m,n\rangle$$
4. Apply the SWAP operator to the right ket: $\mathbb{F}\,|m,n\rangle = |n,m\rangle$. Substitute this back:
$$\mathrm{Tr}[(\hat{A} \otimes \hat{B})\,\mathbb{F}] = \sum_{m,n=1}^{D} \langle m,n|\,(\hat{A} \otimes \hat{B})\,|n,m\rangle$$
5. Factor the tensor product matrix elements into separate single-system matrix elements:
$$\langle m,n|\,(\hat{A} \otimes \hat{B})\,|n,m\rangle = \langle m|\,\hat{A}\,|n\rangle\,\langle n|\,\hat{B}\,|m\rangle = A_{mn}B_{nm}$$
6. Summing over n yields the matrix product element $(AB)_{mm}$:
$$\mathrm{Tr}[(\hat{A} \otimes \hat{B})\,\mathbb{F}] = \sum_{m=1}^{D}\left(\sum_{n=1}^{D} A_{mn}B_{nm}\right) = \sum_{m=1}^{D}(\hat{A}\hat{B})_{mm} = \boxed{\mathrm{Tr}(\hat{A}\hat{B})}$$

Questions

1. **Question:** What is the 1-qubit representation of the SWAP operator $\mathbb{F}$?

   Answer: On a 2-dimensional doubled Hilbert space (two qubits), the SWAP operator can be represented as:
$$\mathbb{F} = \frac{1}{2}\left(\hat{I} \otimes \hat{I} + \hat{\sigma}_x \otimes \hat{\sigma}_x + \hat{\sigma}_y \otimes \hat{\sigma}_y + \hat{\sigma}_z \otimes \hat{\sigma}_z\right)$$
   This decomposition holds because the identity and Pauli matrices form an orthonormal basis for 2×2 matrices under the Hilbert-Schmidt inner product.

Derivation Blueprint: SWAP Operator and Swap Trick

Board Layout:

- **Left Board Panel:** Write the tensor state index mappings clearly using arrows: $|i\rangle_1|j\rangle_2 \xrightarrow{\mathbb{F}} |j\rangle_1|i\rangle_2$. Underneath, write the bra-ket representation of $\mathbb{F}$.
- **Middle Panel:** Show the index matrix element expansion $\langle m,n|(\hat{A}\otimes\hat{B})|n,m\rangle = A_{mn}B_{nm}$. Use colored chalk to highlight which indices go to which operator.
- **Right Panel:** Write out the final sum $\sum_m (AB)_{mm} = \mathrm{Tr}(AB)$ to complete the proof.

Chalk-Talk Visual Guide:

- Draw a standard tensor network diagram: draw two boxes representing operators $\hat{A}$ and $\hat{B}$ with incoming wires from the bottom and outgoing wires at the top. Below them, draw the SWAP operator $\mathbb{F}$ as a literal cross (”X”) of the two wires. Show how connecting the wires of $(\hat{A}\otimes\hat{B})\mathbb{F}$ and wrapping them to form a trace topologically loops the output of $\hat{A}$ into the input of $\hat{B}$ and vice-versa, forming a single continuous closed loop which is exactly the trace of the product $\mathrm{Tr}(\hat{A}\hat{B})$.

Common Conceptual Pitfalls: It is easy to misinterpret the action of $\mathbb{F}_A \otimes \hat{I}_B$ on the bipartite doubled space. The identity operator $\hat{I}_B$ leaves the environment indices completely unchanged, whereas the swap operator $\mathbb{F}_A$ acts strictly on the indices of system A. **Boundary Limit & Sanity Check:** If we set $\hat{B} = \hat{I}$, the tensor product reduces to $\hat{A}\otimes\hat{I}$. Substituting this into the swap trick yields $\mathrm{Tr}[(\hat{A}\otimes\hat{I})\mathbb{F}] = \mathrm{Tr}(\hat{A}\hat{I}) = \mathrm{Tr}(\hat{A})$, correctly collapsing the bipartite tensor contraction back to a single subsystem trace.

Numerical Sandbox: Swap Trick Verification

```
import numpy as np
D = 3  # Dimension
A = np.random.normal(size=(D, D)) + 1j * np.random.normal(size=(D, D))
B = np.random.normal(size=(D, D)) + 1j * np.random.normal(size=(D, D))
# Construct SWAP operator F explicitly in the computational basis
F = np.zeros((D**2, D**2))
for i in range(D):
    for j in range(D):
        F[i*D + j, j*D + i] = 1.0
# Compare tensor product contraction to standard matrix product trace
tensor_contraction = np.trace((np.kron(A, B) @ F))
standard_trace = np.trace(A @ B)
print(f"Tr[(A \otimes B) F]: {tensor_contraction:.6f}")
print(f"Tr(A B):             {standard_trace:.6f}")
```

Card 3: Second Haar Moment

Goal: Prove the general formula for the second-order Haar twirl of an operator $\hat{O}$:

$$\mathbb{E}_{U\sim\mathrm{Haar}}[\hat{U}^{\otimes 2}\hat{O}\,\hat{U}^{\dagger\otimes 2}] = c_I\,\hat{\mathbb{I}} + c_{\mathbb{F}}\,\mathbb{F}$$

Key Trick: Schur-Weyl duality states that the commutant of $\hat{U}^{\otimes 2}$ is spanned by the permutations $\hat{\mathbb{I}}$ and $\mathbb{F}$. Contracting with both generators yields a solvable 2×2 system of linear equations.

Step-by-Step Derivation:

1. Let $\hat{M}^{(2)} = \mathbb{E}_U[\hat{U}^{\otimes 2}\hat{O}\,\hat{U}^{\dagger\otimes 2}]$. By the same left-invariance argument as Card 1, $\hat{M}^{(2)}$ commutes with $\hat{V}^{\otimes 2}$ for all $\hat{V} \in U(D)$.

2. By Schur-Weyl duality, the only operators commuting with $\hat{V}^{\otimes 2}$ are linear combinations of permutations. For $k = 2$, the permutation group is $S_2 = \{\mathrm{id}, (12)\}$, corresponding to the identity $\hat{\mathbb{I}}$ and the SWAP operator $\mathbb{F}$. Thus:

$$\hat{M}^{(2)} = c_I\,\hat{\mathbb{I}} + c_{\mathbb{F}}\,\mathbb{F}$$

3. To find the coefficients c_I and $c_{\mathbb{F}}$, take the trace of both sides after multiplying by $\hat{\mathbb{I}}$ and $\mathbb{F}$ respectively.

4. **Contraction with $\hat{\mathbb{I}}$:**

$$\mathrm{Tr}(\hat{M}^{(2)}\hat{\mathbb{I}}) = \mathrm{Tr}(\hat{M}^{(2)}) = \mathbb{E}_U[\mathrm{Tr}(\hat{U}^{\otimes 2}\hat{O}\,\hat{U}^{\dagger\otimes 2})] = \mathrm{Tr}(\hat{O})$$

   On the right-hand side:

$$\mathrm{Tr}[(c_I\,\hat{\mathbb{I}} + c_{\mathbb{F}}\,\mathbb{F})\hat{\mathbb{I}}] = c_I\;\mathrm{Tr}(\hat{\mathbb{I}}) + c_{\mathbb{F}}\;\mathrm{Tr}(\mathbb{F}) = c_I D^2 + c_{\mathbb{F}} D$$

   Thus, our first equation is: $\mathrm{Tr}(\hat{O}) = c_I D^2 + c_{\mathbb{F}} D$.

5. **Contraction with $\mathbb{F}$:** Since $\mathbb{F}$ commutes with $\hat{U}^{\otimes 2}$ (it is a permutation), we have:

$$\mathrm{Tr}(\hat{M}^{(2)}\mathbb{F}) = \mathbb{E}_U[\mathrm{Tr}(\hat{U}^{\otimes 2}\hat{O}\,\hat{U}^{\dagger\otimes 2}\mathbb{F})] = \mathbb{E}_U[\mathrm{Tr}(\hat{U}^{\otimes 2}\hat{O}\mathbb{F}\,\hat{U}^{\dagger\otimes 2})] = \mathrm{Tr}(\hat{O}\mathbb{F})$$

   On the right-hand side:

$$\mathrm{Tr}[(c_I\,\hat{\mathbb{I}} + c_{\mathbb{F}}\,\mathbb{F})\mathbb{F}] = c_I\;\mathrm{Tr}(\mathbb{F}) + c_{\mathbb{F}}\;\mathrm{Tr}(\mathbb{F}^2) = c_I D + c_{\mathbb{F}} D^2 \quad (\text{since } \mathbb{F}^2 = \hat{\mathbb{I}})$$

   Thus, our second equation is: $\mathrm{Tr}(\hat{O}\mathbb{F}) = c_I D + c_{\mathbb{F}} D^2$.

6. **Solve the 2×2 system:** We solve the simultaneous system of equations:

$$\begin{cases} D^2 c_I + D c_{\mathbb{F}} = \mathrm{Tr}(\hat{O}) \\ D c_I + D^2 c_{\mathbb{F}} = \mathrm{Tr}(\hat{O}\mathbb{F}) \end{cases}$$

   Representing this in matrix form:

$$\begin{pmatrix} D^2 & D \\ D & D^2 \end{pmatrix} \begin{pmatrix} c_I \\ c_{\mathbb{F}} \end{pmatrix} = \begin{pmatrix} \mathrm{Tr}(\hat{O}) \\ \mathrm{Tr}(\hat{O}\mathbb{F}) \end{pmatrix}$$

   The determinant of the coefficient matrix is:

$$\Delta = \det \begin{pmatrix} D^2 & D \\ D & D^2 \end{pmatrix} = D^4 - D^2 = D^2(D^2 - 1)$$

   Applying Cramer's Rule, the coefficients are solved using determinants of substituted matrices:

$$c_I = \frac{1}{\Delta} \det \begin{pmatrix} \mathrm{Tr}(\hat{O}) & D \\ \mathrm{Tr}(\hat{O}\mathbb{F}) & D^2 \end{pmatrix} = \frac{D^2\,\mathrm{Tr}(\hat{O}) - D\,\mathrm{Tr}(\hat{O}\mathbb{F})}{D^2(D^2-1)} = \boxed{\frac{D\,\mathrm{Tr}(\hat{O}) - \mathrm{Tr}(\hat{O}\mathbb{F})}{D(D^2-1)}}$$

   and:

$$c_{\mathbb{F}} = \frac{1}{\Delta} \det \begin{pmatrix} D^2 & \mathrm{Tr}(\hat{O}) \\ D & \mathrm{Tr}(\hat{O}\mathbb{F}) \end{pmatrix} = \frac{D^2\,\mathrm{Tr}(\hat{O}\mathbb{F}) - D\,\mathrm{Tr}(\hat{O})}{D^2(D^2-1)} = \boxed{\frac{D\,\mathrm{Tr}(\hat{O}\mathbb{F}) - \mathrm{Tr}(\hat{O})}{D(D^2-1)}}$$

Questions

1. **Question:** What are the Weingarten functions for the second Haar moment?

   Answer: The coefficients of the twirled operator can be written in terms of contractions with the Weingarten matrix:

$$\mathbb{E}_U[U^{\otimes 2}\hat{O}U^{\dagger\otimes 2}] = \sum_{\sigma,\tau\in S_2} \mathrm{Tr}(\hat{O}\tau)\,\mathrm{Wg}(\sigma\tau^{-1})\,\sigma$$

   where the second-order Weingarten values are $\mathrm{Wg}(\mathrm{id}) = \frac{1}{D^2-1}$ and $\mathrm{Wg}((12)) = -\frac{1}{D(D^2-1)}$, demonstrating the direct connection to asymptotic group integration theory.

Derivation Blueprint: Second Haar Moment

Board Layout:

- **Left Board Panel:** State the problem clearly: We are averaging products of four unitary matrix elements. Write out Schur-Weyl duality: the commutant of $\hat{U}^{\otimes 2}$ is spanned by $\{\hat{\mathbb{I}}, \mathbb{F}\}$.
- **Middle Panel:** Write out the two contractions side-by-step. Draw boxes around the two simultaneous linear equations.
- **Right Panel:** Write out the 2×2 matrix system and solve it using Cramer's rule. Highlight the final formulas for c_I and $c_{\mathbb{F}}$ in bright chalk.

Common Conceptual Pitfalls: A frequent source of errors is carrying over $\mathrm{Tr}(\hat{I}_D) = D$ from the single-copy space and writing $\mathrm{Tr}(\hat{\mathbb{I}}) = D$ on the doubled space. The correct value is $\mathrm{Tr}(\hat{\mathbb{I}}) = D^2$: since $\hat{\mathbb{I}} = \hat{I}_D \otimes \hat{I}_D$ acts on $\mathcal{H}\otimes\mathcal{H}$, its trace is $D \times D = D^2$ (and likewise $\mathrm{Tr}(\mathbb{F}) = D$).
Boundary Limit & Sanity Check: Setting the Hilbert space dimension $D = 1$ makes the determinant of the contraction matrix $\Delta = D^2(D^2 - 1) = 0$. This singularity arises because the identity $\hat{\mathbb{I}}$ and the SWAP operator $\mathbb{F}$ act identically on a 1D doubled space (both map $|0,0\rangle \to |0,0\rangle$), which breaks their linear independence and requires $D \geq 2$ for the Weingarten expansion.

Active Recall Concept Test: Linear Independence and Singular Weingarten Systems

Question: Why does the determinant of the contraction matrix $\Delta = D^2(D^2-1)$ become zero when the Hilbert space dimension is $D = 1$?

A) Because the Haar measure is only defined for systems with non-trivial continuous degrees of freedom.

B) Because for $D = 1$, the identity operator $\hat{\mathbb{I}}$ and the SWAP operator $\mathbb{F}$ act identically on the 1D doubled space, making the two generators linearly dependent. **[Correct]**

C) Because Weingarten functions require $D \geq 2$ to satisfy the group theory properties of the symmetric group S_2.

Explanation: When $D = 1$, the doubled tensor product space $\mathcal{H} \otimes \mathcal{H}$ is 1-dimensional. In 1D, there is only a single basis state $|0,0\rangle$, meaning the SWAP operator $\mathbb{F}\,|0,0\rangle = |0,0\rangle$ acts identically to the identity operator $\hat{\mathbb{I}}\,|0,0\rangle = |0,0\rangle$. Since the two operators are identical, they are not linearly independent. Trying to represent the average using both generators creates a redundant, singular system of equations, which manifests as a zero determinant $(\Delta = 1^2(1^2 - 1) = 0)$.

Card 4: Average Purity via Weingarten

Goal: Prove that the average subsystem purity $\mathbb{E}[\mathrm{Tr}(\hat{\rho}_A^2)]$ for a bipartite system $\mathcal{H}_A \otimes \mathcal{H}_B$ of dimensions $m \times n$ ($D = mn$) under Haar-random state vectors is given by:

$$\mathbb{E}[\mathrm{Tr}(\hat{\rho}_A^2)] = \frac{m+n}{mn+1}$$

Key Trick: Use the swap trick on the doubled space, write the average in terms of the second Haar moment of the rank-1 state $|0,0\rangle\langle 0,0|$, and evaluate the trace contractions explicitly.

Step-by-Step Derivation:

1. Let $|\psi\rangle = \hat{U}|0,0\rangle_{AB}$ be a Haar-random state on $\mathcal{H}_A \otimes \mathcal{H}_B$. The reduced density matrix is $\hat{\rho}_A = \mathrm{Tr}_B(|\psi\rangle\langle\psi|)$.
2. Write the subsystem purity using the swap trick (Card 2) on the doubled subsystem space $\mathcal{H}_{A_1} \otimes \mathcal{H}_{A_2}$:
$$\mathrm{Tr}(\hat{\rho}_A^2) = \mathrm{Tr}[(\hat{\rho}_A \otimes \hat{\rho}_A)\,\mathbb{F}_A]$$
3. We can embed this into the full doubled Hilbert space $\mathcal{H}_{AB}^{\otimes 2}$ as:
$$\mathrm{Tr}(\hat{\rho}_A^2) = \mathrm{Tr}\left[(|\psi\rangle\langle\psi|)^{\otimes 2}\,(\mathbb{F}_A \otimes \hat{I}_B)\right] = \mathrm{Tr}\left[\hat{U}^{\otimes 2}\,(|0,0\rangle\langle 0,0|)^{\otimes 2}\,\hat{U}^{\dagger\otimes 2}(\mathbb{F}_A \otimes \hat{I}_B)\right]$$
4. By taking the expectation value and utilizing the cyclicity of the trace, we isolate the second Haar moment:
$$\mathbb{E}[\mathrm{Tr}(\hat{\rho}_A^2)] = \mathrm{Tr}\left[\mathbb{E}_U\left[\hat{U}^{\otimes 2}\,(|0,0\rangle\langle 0,0|)^{\otimes 2}\,\hat{U}^{\dagger\otimes 2}\right](\mathbb{F}_A \otimes \hat{I}_B)\right]$$
5. Let $\hat{O} = (|0,0\rangle\langle 0,0|)^{\otimes 2}$. Since $\hat{O}$ is a rank-1 projector on the doubled space:
$$\mathrm{Tr}(\hat{O}) = 1, \qquad \mathrm{Tr}(\hat{O}\mathbb{F}_{AB}) = 1$$
6. Apply the second moment twirl formula from Card 3 with $D = mn$:
$$\mathbb{E}_U[\hat{U}^{\otimes 2}\hat{O}\,\hat{U}^{\dagger\otimes 2}] = c_I\,\hat{\mathbb{I}}_{AB} + c_{\mathbb{F}}\,\mathbb{F}_{AB}$$
where:
$$c_I = c_{\mathbb{F}} = \frac{D(1)-1}{D(D^2-1)} = \frac{D-1}{D(D-1)(D+1)} = \frac{1}{D(D+1)}$$
7. Now contract the average twirled operator with $\mathbb{F}_A \otimes \hat{I}_B$:
$$\mathbb{E}[\mathrm{Tr}(\hat{\rho}_A^2)] = \frac{1}{D(D+1)}\left[\mathrm{Tr}\left(\hat{\mathbb{I}}_{AB}(\mathbb{F}_A \otimes \hat{I}_B)\right) + \mathrm{Tr}\left(\mathbb{F}_{AB}(\mathbb{F}_A \otimes \hat{I}_B)\right)\right]$$
8. Let us evaluate each trace term systematically with full index/basis contractions to prevent any ambiguity:
   - **Term 1:** The doubled identity is $\hat{\mathbb{I}}_{AB} = \hat{I}_{A_1A_2} \otimes \hat{I}_{B_1B_2}$. Contracting it with $\mathbb{F}_A \otimes \hat{I}_{B_1B_2}$ yields:
   $$\mathrm{Tr}\left(\hat{\mathbb{I}}_{AB}(\mathbb{F}_A \otimes \hat{I}_B)\right) = \mathrm{Tr}(\mathbb{F}_A \otimes \hat{I}_{B_1B_2}) = \mathrm{Tr}(\mathbb{F}_A)\,\mathrm{Tr}(\hat{I}_{B_1B_2})$$
   Let $\{|a\rangle\}$ be an orthonormal basis for subsystem A (dim m), and $\{|b\rangle\}$ for B (dim n). Evaluating the trace of $\mathbb{F}_A$ explicitly:
   $$\mathrm{Tr}(\mathbb{F}_A) = \sum_{a_1,a_2=1}^{m}\langle a_1,a_2|\mathbb{F}_A|a_1,a_2\rangle = \sum_{a_1,a_2=1}^{m}\langle a_1,a_2|a_2,a_1\rangle = \sum_{a_1,a_2=1}^{m}\delta_{a_1,a_2}\delta_{a_2,a_1} = \sum_{a_1=1}^{m}\delta_{a_1,a_1} = m$$

The trace of the identity on the doubled space $B_1 B_2$ is:

$$\mathrm{Tr}(\hat{I}_{B_1 B_2}) = \sum_{b_1, b_2=1}^{n} \langle b_1, b_2 | \hat{I}_{B_1 B_2} | b_1, b_2 \rangle = \sum_{b_1, b_2=1}^{n} 1 = n^2$$

Thus:

$$\mathrm{Tr}\left(\hat{\mathbb{I}}_{AB}(\mathbb{F}_A \otimes \hat{I}_B)\right) = m \cdot n^2 = mn^2$$

- **Term 2:** Express the global SWAP operator as $\mathbb{F}_{AB} = \mathbb{F}_A \otimes \mathbb{F}_B$. Using the algebraic property $\mathbb{F}_A^2 = \hat{I}_{A_1 A_2}$:

$$\mathrm{Tr}\left[\mathbb{F}_{AB}(\mathbb{F}_A \otimes \hat{I}_B)\right] = \mathrm{Tr}\left[(\mathbb{F}_A \otimes \mathbb{F}_B)(\mathbb{F}_A \otimes \hat{I}_B)\right] = \mathrm{Tr}(\mathbb{F}_A^2 \otimes \mathbb{F}_B) = \mathrm{Tr}(\hat{I}_A \otimes \mathbb{F}_B) = \mathrm{Tr}(\hat{I}_A)\,\mathrm{Tr}(\mathbb{F}_B)$$

The trace of the identity on the doubled subsystem $A_1 A_2$ is:

$$\mathrm{Tr}(\hat{I}_{A_1 A_2}) = \sum_{a_1, a_2=1}^{m} \langle a_1, a_2 | a_1, a_2 \rangle = m^2$$

The trace of the SWAP operator on the environment subsystem $\mathbb{F}_B$ is:

$$\mathrm{Tr}(\mathbb{F}_B) = \sum_{b_1, b_2=1}^{n} \langle b_1, b_2 | \mathbb{F}_B | b_1, b_2 \rangle = \sum_{b_1, b_2=1}^{n} \delta_{b_1, b_2} \delta_{b_2, b_1} = n$$

Thus:

$$\mathrm{Tr}\left[\mathbb{F}_{AB}(\mathbb{F}_A \otimes \hat{I}_B)\right] = m^2 \cdot n = m^2 n$$

9. Substitute these values back into the expression:

$$\mathbb{E}[\mathrm{Tr}(\hat{\rho}_A^2)] = \frac{mn^2 + m^2 n}{mn(mn+1)} = \frac{mn(m+n)}{mn(mn+1)} = \boxed{\frac{m+n}{mn+1}}$$

Questions

1. **Question:** What happens to the average subsystem purity in the thermodynamic limit when the environment B is infinitely large ($n \to \infty$)?

   Answer: Under the thermodynamic limit $n \to \infty$, the average purity converges as $\mathbb{E}[\mathrm{Tr}(\hat{\rho}_A^2)] \to 1/m$. The subsystem is thus maximally mixed, which is in exact alignment with Page's theorem stating that subsystem entanglement is extremely close to maximal for generic pure states.

Derivation Blueprint: Average Purity via Weingarten

Board Layout:

- **Left Panel:** State the bipartite setup: system A (m), system B (n). Define the purity trace structure on the doubled space using system labels A_1, A_2, B_1, B_2 to keep it absolutely clean.
- **Middle Panel:** Insert the twirl. Show how the rank-1 input state simplifies the coefficients to $c_I = c_{\mathbb{F}} = 1/(D(D+1))$.
- **Right Panel:** Show the two partial traces. Write out the final step where the mn factors cancel in the fraction, leaving the clean result.

Chalk-Talk Visual Guide:

- Draw four parallel horizontal lines representing the doubled systems A_1, A_2 (on top) and B_1, B_2 (on the bottom). Show the trace contract $\mathbb{F}_A \otimes \hat{I}_B$ as a crossing loop linking the output of A_1 to the input of A_2 (and vice-versa), while the B_1 and B_2 wires are closed directly into themselves as simple parallel loops. Below it, show the second term contraction $\mathbb{F}_{AB}(\mathbb{F}_A \otimes \hat{I}_B)$: here, both A and B wires cross. Show that the double crossing on A ($\mathbb{F}_A^2 = \hat{I}_A$) untangles the wires, leaving two parallel loops of dimension m^2, while B's single crossing leaves a single closed loop of dimension n, proving geometrically why the trace evaluates to $m^2 n$.

Physical & Experimental Anchorage:

- Subsystem purity $\mathrm{Tr}(\hat{\rho}_A^2)$ is the foundational metric for measuring entanglement entropy in many-body simulators (e.g. trapped ions, superconducting qubits). In experiments studying quantum thermalization or phase transitions, full state tomography is blocked by the exponential measurement cost. By using the Weingarten twirling protocols (or local Cliffords), experimentalists extract this exact purity directly from random measurement basis outcome collisions, making it a vital diagnostic in quantum supremacy and benchmarking research.

Common Conceptual Pitfalls: Evaluating the double SWAP trace contraction $\mathrm{Tr}[\mathbb{F}_{AB}(\mathbb{F}_A \otimes \hat{I}_B)]$ is highly prone to algebraic slips. Visualizing the wire connections as a tensor network reveals that the double crossing on system A untangles to form a trivial parallel identity connection ($\mathbb{F}_A^2 = \hat{I}_A$), whereas system B's single crossing remains a swap, collapsing the expression cleanly. **Boundary Limit & Sanity Check:** Evaluating the extreme dimensions provides crucial limits: for an environment dimension $n = 1$ (no environment), the average purity is $(m+1)/(m+1) = 1$, corresponding to a pure state. In the limit of an infinitely large environment $n \to \infty$, the purity $\mathbb{E}[\mathrm{Tr}(\hat{\rho}_A^2)] \to 1/m$, which is the maximally mixed state limit.

Numerical Sandbox: Page Subsystem Purity Verification

```
import numpy as np
m, n = 2, 4  # System dimensions
D = m * n
# Generate a random Haar pure state vector
psi = np.random.normal(size=D) + 1j * np.random.normal(size=D)
psi /= np.linalg.norm(psi)
# Reshape to bipartite matrix and compute partial trace over environment B
rho_A = np.reshape(psi, (m, n)) @ np.reshape(psi, (m, n)).conj().T
purity_sample = np.real(np.trace(rho_A @ rho_A))
purity_page = (m + n) / (m * n + 1)
print(f"Bipartite dimensions: A={m} (dim), B={n} (dim)")
print(f"Sample Purity: {purity_sample:.6f} | Page Expectation: {purity_page:.6f}")
```

Card 14: Page Entropy (Asymptotic)

Goal: Prove that the average entanglement entropy $\mathbb{E}[S]$ of a subsystem A of dimension m coupled to an environment B of dimension n ($m \leq n$) asymptotically approaches $\log m - \frac{m}{2n}$.

Key Trick: Evaluate the exact Page formula using the integral approximation of the harmonic sum.

Step-by-Step Derivation:

1. Page's exact formula for the average entanglement entropy is:

$$\mathbb{E}[S] = \sum_{k=n+1}^{mn} \frac{1}{k} - \frac{m-1}{2n}$$

2. Express the sum using the harmonic numbers $H_x = \sum_{k=1}^{x} \frac{1}{k}$:

$$\sum_{k=n+1}^{mn} \frac{1}{k} = H_{mn} - H_n$$

3. Use the asymptotic expansion of the harmonic numbers $H_x \approx \ln x + \gamma + \mathcal{O}(1/x)$, where γ is the Euler-Mascheroni constant:

$$H_{mn} - H_n \approx (\ln(mn) + \gamma) - (\ln(n) + \gamma) = \ln(mn) - \ln(n)$$

4. Simplify the logarithms:

$$\ln(mn) - \ln(n) = \ln(m)$$

5. Substitute this back into the Page formula:

$$\mathbb{E}[S] \approx \ln m - \frac{m-1}{2n}$$

6. For large m, $m - 1 \approx m$. Thus, we obtain the standard asymptotic scaling:

$$\boxed{\mathbb{E}[S] \simeq \log m - \frac{m}{2n}}$$

Questions

1. **Question:** What does the asymptotic Page entropy imply about black hole information?

   Answer: This is the physical core of the Page curve. It proves that during black hole evaporation, the emitted radiation remains highly entangled with the remaining black hole until the "Page time" (when the black hole has evaporated to half its size, $m \approx n$). After this point, the entanglement entropy must decrease, forcing information to escape and preserving quantum unitarity.

Derivation Blueprint: Page Entropy (Asymptotic)

Board Layout:

- **Left Board Panel:** Write down the exact Page formula. Explain what m (subsystem) and n (environment) represent.
- **Middle Panel:** Write down the harmonic numbers H_x and show their logarithmic asymptotic expansion.
- **Right Panel:** Complete the algebraic steps to extract the clean $\log m - \frac{m}{2n}$ formula.

Common Conceptual Pitfalls: The requirement $m \leq n$ (where A is the smaller subsystem) is physically critical. If system A were larger than environment B, its maximum possible entanglement entropy would be bounded by the smaller dimension n, meaning the labels m and n must be swapped to maintain physical validity. **Boundary Limit & Sanity Check:** For a symmetric half-cut bipartition ($m = n = 2^{N/2}$), the Page correction reduces to $m/2n = 1/2$. This yields an average entanglement entropy of $S \approx \log m - 0.5$ nats, which is extremely close to the absolute maximal thermodynamic limit.

4 Quantum Dynamics & Chaos

Exercise 4.1: The C–F Identity

Motivation:

The **out-of-time-order correlator** (OTOC) $F(t) = \langle W^\dagger(t)V^\dagger W(t)V\rangle$ is the primary diagnostic of quantum information scrambling. The C-F identity $C(t) = 2 - 2\,\mathrm{Re}\,F(t)$ connects it to the physically intuitive squared commutator. This relation is the starting point for every scrambling analysis in this book — from Lyapunov growth to design verification.

Preliminaries / Toolbox

Before diving into the solution, recall the following concepts:

1. Out-of-Time-Order Correlators (OTOC): A metric for scrambling that tracks how an initially local operator $W_t = e^{iHt}We^{-iHt}$ grows in time to fail to commute with another local operator V. $F(t) = \langle W_t^\dagger V^\dagger W_t V\rangle$.

2. Squared Commutator: $C(t) = \langle [W_t, V]^\dagger [W_t, V]\rangle$. For unitary W, V, it relates exactly to $F(t)$.

3. Infinite Temperature Average: $\langle O\rangle = \mathrm{Tr}(O)/D$, representing the infinite-temperature equilibrium state.

4. Unitarity: $W^\dagger W = I, V^\dagger V = I$.

Problem Statement:

For unitary operators W, V and the infinite-temperature average $\langle\cdot\rangle = \mathrm{Tr}(\cdot)/D$:

$$F(t) = \langle W^\dagger(t)V^\dagger W(t)V\rangle, \qquad C(t) = \langle [W(t), V]^\dagger [W(t), V]\rangle.$$

(a) Show $C(t) = 2 - 2\,\mathrm{Re}\,F(t)$.

(b) At $t = 0$ with $[W, V] = 0$: verify $C(0) = 0$, $F(0) = 1$. What are the values when $F \to -1$ and $F \to 0$?

(c) Explain why $F \to 0$ (not -1) at late times for Haar-random $W(t)$.

Pedantic Step-by-Step Solution:

Part (a): Expanding the squared commutator

Drop the time argument for brevity. Expand:

$$[W, V]^\dagger [W, V] = (WV - VW)^\dagger (WV - VW).$$

$$= V^\dagger W^\dagger WV - V^\dagger W^\dagger VW - W^\dagger V^\dagger WV + W^\dagger V^\dagger VW.$$

Using unitarity $W^\dagger W = I$ and $V^\dagger V = I$:

$$= \underbrace{V^\dagger V}_{I} - V^\dagger W^\dagger VW - W^\dagger V^\dagger WV + \underbrace{W^\dagger W}_{I} = 2I - F^\dagger - F,$$

where we identified $F = V^\dagger W^\dagger VW = W^\dagger(t)V^\dagger W(t)V$ (the OTOC kernel) and $F^\dagger = W^\dagger V^\dagger WV$. Taking the trace average:

$$\boxed{C(t) = 2 - \langle F\rangle - \langle F^\dagger\rangle = 2 - 2\,\mathrm{Re}\,F(t).}$$

Part (b): Boundary values

At $t = 0$, if $[W, V] = 0$:

$$F(0) = \langle W^\dagger V^\dagger WV\rangle = \langle I\rangle = 1, \qquad C(0) = 0.$$

The two extreme cases:

- $F \to -1$: $C = 2 - 2(-1) = 4$. Maximum non-commutativity (anticommutation).

- $F \to 0$: $C = 2$. This is the **Haar scrambled** value.

Part (c): Why $F \to 0$ for Haar-random unitaries

When $W(t) = U^\dagger W U$ with U drawn from the Haar measure, the OTOC becomes a 4th-order polynomial in U. By the Weingarten calculus (specifically the $k = 2$ formula), the average over U of such a product factorizes into traces. Since W and V are traceless Paulis, the leading contribution vanishes:

$$\mathbb{E}_U[F] = \mathbb{E}_U[\langle U^\dagger W^\dagger U V^\dagger U^\dagger W U V\rangle] \to 0.$$

There is no preferred phase alignment between $W(t)$ and V in a fully scrambled system. The OTOC $F(t)$ decays to zero, not -1, because scrambling **randomizes** the relative phase rather than flipping it.

Exercise 4.2: Krylov Complexity from the Geometric Distribution

Motivation:

Krylov complexity $C_K(t)$ measures how far an operator has spread in the Krylov basis — an orthonormal basis generated by repeated commutation with the Hamiltonian (the Lanczos algorithm). When the Lanczos coefficients grow linearly ($b_n = \alpha n$), the Krylov amplitudes take a closed form: a geometric distribution on the Krylov chain. The resulting complexity $C_K(t) = \sinh^2(\alpha t)$ grows exponentially — the operator-growth analogue of a Lyapunov exponent.

Preliminaries / Toolbox

Krylov basis: Given an initial operator $\mathcal{O}_0$ and Hamiltonian H, the Lanczos algorithm constructs an orthonormal basis $\{\mathcal{O}_n\}$ for the operator Hilbert space such that the Liouvillian $\mathcal{L} = [H, \cdot]$ acts tridiagonally:

$$\mathcal{L}|\mathcal{O}_n) = b_{n+1}|\mathcal{O}_{n+1}) + b_n|\mathcal{O}_{n-1}).$$

The **Lanczos coefficients** $\{b_n\}$ encode the growth rate. The operator at time t is $\mathcal{O}(t) = \sum_n \varphi_n(t)\mathcal{O}_n$.

Krylov complexity: $C_K(t) = \sum_n n\,|\varphi_n(t)|^2$ — the average position on the Krylov chain.

Geometric series: For $|r| < 1$: $\sum_{n=0}^\infty r^n = \frac{1}{1-r}$, $\sum_{n=0}^\infty n r^n = \frac{r}{(1-r)^2}$.

Hyperbolic identities: $\tanh^2 x + \operatorname{sech}^2 x = 1$, $\sinh^2 x = \tanh^2 x \cdot \cosh^2 x$.

Problem Statement:

When $b_n = \alpha n$, the Krylov amplitudes are $\varphi_n(t) = \tau^n/c$ with $\tau(t) = \tanh(\alpha t)$, $c(t) = \cosh(\alpha t)$.

(a) Verify $\sum_n |\varphi_n|^2 = 1$.

(b) Compute $C_K(t) = \sum_n n|\varphi_n|^2$ and show $C_K(t) = \sinh^2(\alpha t)$.

(c) Compute $\mathrm{Var}(n)$.

Pedantic Step-by-Step Solution:

Part (a): Normalization

The probability distribution on the Krylov chain is:

$$p_n = |\varphi_n|^2 = \frac{\tau^{2n}}{c^2} = (1-\tau^2)\,\tau^{2n},$$

where we used $1/c^2 = 1/\cosh^2 = 1 - \tanh^2 = 1 - \tau^2$. This is a **geometric distribution** with ratio $r = \tau^2 < 1$:

$$\sum_{n=0}^\infty p_n = (1-\tau^2)\sum_{n=0}^\infty \tau^{2n} = (1-\tau^2)\cdot\frac{1}{1-\tau^2} = 1. \quad \checkmark$$

Part (b): Krylov complexity

$$C_K = \sum_n n\, p_n = (1-\tau^2)\sum_n n\,\tau^{2n} = (1-\tau^2)\cdot\frac{\tau^2}{(1-\tau^2)^2} = \frac{\tau^2}{1-\tau^2}.$$

Substituting $\tau = \tanh(\alpha t)$ and $1-\tanh^2 = \operatorname{sech}^2 = 1/\cosh^2$:

$$C_K = \tanh^2(\alpha t)\cdot\cosh^2(\alpha t) = \sinh^2(\alpha t).$$

$$\boxed{C_K(t) = \sinh^2(\alpha t) \approx \tfrac{1}{4}e^{2\alpha t} \quad (t \gg 1/\alpha).}$$

This exponential growth at rate 2α is the operator-growth Lyapunov exponent.

Part (c): Variance

$$\langle n^2\rangle = (1-\tau^2)\sum_n n^2\tau^{2n} = (1-\tau^2)\cdot\frac{\tau^2(1+\tau^2)}{(1-\tau^2)^3} = \frac{\tau^2(1+\tau^2)}{(1-\tau^2)^2}.$$

$$\mathrm{Var}(n) = \langle n^2\rangle - \langle n\rangle^2 = \frac{\tau^2(1+\tau^2)}{(1-\tau^2)^2} - \frac{\tau^4}{(1-\tau^2)^2} = \frac{\tau^2}{(1-\tau^2)^2}.$$

In hyperbolic functions: $\mathrm{Var} = \sinh^2\cdot\cosh^2 = \frac{1}{4}\sinh^2(2\alpha t)$.
The standard deviation $\sigma \sim e^{2\alpha t}$ grows at the same rate as the mean — the operator wavepacket spreads ballistically on the Krylov chain.

Visual Pedagogical Proof

Exercise 4.3: SFF Plateau Value for the GUE

Motivation:

The **spectral form factor** (SFF) $K(t) = |Z(it)|^2 = |\operatorname{Tr}(e^{-iHt})|^2$ is the spectral diagnostic of quantum chaos. It probes pair correlations of energy levels. The SFF exhibits three regimes: a **slope** (dip below D^2), a **ramp** (linear in t, reflecting level repulsion), and a **plateau** at $K = D$. This exercise derives the plateau value from time-averaging and relates it to the dephasing of off-diagonal terms.

Preliminaries / Toolbox

Before diving into the solution, recall the following concepts:

1. Spectral Form Factor (SFF): Provides a proxy for the analytic continuation of the partition function to study late-time chaos geometry. $K(t) = |Z(it)|^2 = |\operatorname{Tr}(e^{-iHt})|^2 = \sum_{m,n} e^{-i(E_m-E_n)t}$.

2. Level Repulsion: In chaotic models (like GUE), eigenvalues repel each other ($P(\Delta E) \sim \Delta E^\beta \to 0$). They never perfectly cross.

3. Time Averaging: Off-diagonal terms $e^{-i(E_m-E_n)t}$ strongly oscillate and vanish under time average, isolating diagonal contributions.

4. Coherence vs Dephasing: $K(0) = D^2$ shows full coherence, while $K \to D$ indicates full dephasing into mutually orthogonal eigenvectors.

Problem Statement:

For a $D \times D$ GUE Hamiltonian with eigenvalues $\{E_n\}$: $K(t) = \sum_{m,n} e^{-it(E_m-E_n)}$.

(a) Show the long-time average $\overline{K} = D$.

(b) Explain why $K_{\text{plateau}} = D$ (not D^2) and interpret the ratio $K(0)/\overline{K}$.

Pedantic Step-by-Step Solution:

Part (a): Time-averaging the SFF

Time-average each term separately:

$$\overline{K} = \lim_{T\to\infty} \frac{1}{T}\int_0^T K(t)\,dt = \sum_{m,n} \lim_{T\to\infty} \frac{1}{T}\int_0^T e^{-i(E_m-E_n)t}\,dt.$$

Diagonal terms ($m = n$): The integrand is 1. Each contributes 1. There are D such terms.
Off-diagonal terms ($m \neq n$): Let $\omega_{mn} = E_m - E_n \neq 0$. Then:

$$\frac{1}{T}\int_0^T e^{-i\omega_{mn}t}\,dt = \frac{1 - e^{-i\omega_{mn}T}}{i\omega_{mn}T} \to 0 \quad \text{as } T \to \infty.$$

GUE level repulsion ensures $\omega_{mn} \neq 0$ with probability 1 (eigenvalues don't coincide).

$$\boxed{\overline{K} = D.}$$

Part (b): Interpretation

At $t = 0$: all D^2 terms contribute coherently — $K(0) = D^2$.
At late times: the $D(D-1)$ off-diagonal terms dephase (oscillate with incommensurate frequencies and cancel upon averaging), leaving only the D diagonal terms.
The ratio:

$$\frac{K(0)}{\overline{K}} = \frac{D^2}{D} = D$$

quantifies the **coherent enhancement** at $t = 0$. This dephasing from D^2 to D is the spectral signature of quantum chaos: the transition from coherent to incoherent behavior as the system explores its energy landscape.
Visual Pedagogical Proof

Flashcards, Active Recall & Derivations

Card 5: $C(t) = 2(1 - \operatorname{Re} F(t))$

Goal: Prove that the squared commutator $C(t) = \langle [\hat{W}(t), \hat{V}]^\dagger [\hat{W}(t), \hat{V}] \rangle$ is related to the out-of-time-ordered correlator $F(t) = \langle \hat{W}^\dagger(t) \hat{V}^\dagger \hat{W}(t) \hat{V} \rangle$ for unitary operators $\hat{V}$ and $\hat{W}$.

Key Trick: Use the properties of unitary operators ($\hat{U}^\dagger \hat{U} = \hat{I}$) to systematically expand the squared commutator into four distinct terms and simplify.

Step-by-Step Derivation:

1. Write out the squared commutator explicitly:
$$C(t) = \langle [\hat{W}(t), \hat{V}]^\dagger [\hat{W}(t), \hat{V}] \rangle$$
2. Expand the commutator $[\hat{W}(t), \hat{V}] = \hat{W}(t)\hat{V} - \hat{V}\hat{W}(t)$.
3. Find its adjoint:
$$[\hat{W}(t), \hat{V}]^\dagger = \left(\hat{W}(t)\hat{V} - \hat{V}\hat{W}(t)\right)^\dagger = \hat{V}^\dagger \hat{W}^\dagger(t) - \hat{W}^\dagger(t)\hat{V}^\dagger$$
4. Multiply the two expanded terms:
$$C(t) = \langle \left(\hat{V}^\dagger \hat{W}^\dagger(t) - \hat{W}^\dagger(t)\hat{V}^\dagger\right) \left(\hat{W}(t)\hat{V} - \hat{V}\hat{W}(t)\right) \rangle$$
5. Distribute the terms using linearity of the expectation value:
$$\begin{aligned} C(t) = &\langle \hat{V}^\dagger \hat{W}^\dagger(t) \hat{W}(t) \hat{V} \rangle \\ &- \langle \hat{V}^\dagger \hat{W}^\dagger(t) \hat{V} \hat{W}(t) \rangle \\ &- \langle \hat{W}^\dagger(t) \hat{V}^\dagger \hat{W}(t) \hat{V} \rangle \\ &+ \langle \hat{W}^\dagger(t) \hat{V}^\dagger \hat{V} \hat{W}(t) \rangle \end{aligned}$$
6. Simplify Term 1 and Term 4 using the unitarity of $\hat{V}$ and $\hat{W}$ ($\hat{V}^\dagger\hat{V} = \hat{I}$ and $\hat{W}^\dagger(t)\hat{W}(t) = \hat{I}$):
   - **Term 1:** $\langle \hat{V}^\dagger (\hat{W}^\dagger(t)\hat{W}(t)) \hat{V} \rangle = \langle \hat{V}^\dagger \hat{I} \hat{V} \rangle = \langle \hat{V}^\dagger \hat{V} \rangle = \langle \hat{I} \rangle = 1$.
   - **Term 4:** $\langle \hat{W}^\dagger(t) (\hat{V}^\dagger \hat{V}) \hat{W}(t) \rangle = \langle \hat{W}^\dagger(t) \hat{I} \hat{W}(t) \rangle = \langle \hat{W}^\dagger(t) \hat{W}(t) \rangle = \langle \hat{I} \rangle = 1$.
7. Recognize Term 3 as the out-of-time-ordered correlator:
$$\langle \hat{W}^\dagger(t) \hat{V}^\dagger \hat{W}(t) \hat{V} \rangle = F(t)$$
8. Recognize Term 2 as the complex conjugate of $F(t)$ (using $\langle \hat{A}^\dagger \rangle = \langle \hat{A} \rangle^*$ for state expectations):
$$\langle \hat{V}^\dagger \hat{W}^\dagger(t) \hat{V} \hat{W}(t) \rangle = \langle \left(\hat{W}^\dagger(t) \hat{V}^\dagger \hat{W}(t) \hat{V}\right)^\dagger \rangle = F^*(t)$$
9. Combine the four terms:
$$C(t) = 1 - F^*(t) - F(t) + 1 = 2 - (F(t) + F^*(t))$$
10. Since $z + z^* = 2\operatorname{Re}(z)$ for any complex number, we get the final result:
$$\boxed{C(t) = 2\left(1 - \operatorname{Re} F(t)\right)}$$

Questions

1. **Question:** Does the relation $C(t) = 2(1 - \operatorname{Re} F(t))$ hold if the operators $\hat{V}$ and $\hat{W}$ are Hermitian but not unitary?

   Answer: If the operators are both Hermitian and unitary (such as Pauli operators), the relation holds exactly. If they are Hermitian but not unitary, the expansions of the first and fourth terms do not collapse to 1. In this case, we must normalize the commutator by the operators' norms, and additional terms containing the variances of the operators will appear.

Derivation Blueprint: C(t) = 2(1 - Re F(t))

Board Layout:

- **Left Panel:** Define the commutator $[W(t), V]$ and state the target identity. Write the unitarity conditions on a side panel.
- **Middle Panel:** Write the expansion of the squared commutator term-by-term. Use vertical arrows to show how Term 1 and Term 4 collapse to 1.
- **Right Panel:** Write out the definition of $F(t)$, show the complex conjugate relation for Term 2, and write the final boxed result.

Common Conceptual Pitfalls: The physical origin of the out-of-time-ordered sequence in $F(t) = \langle W^\dagger(t) V^\dagger W(t) V \rangle$ can be difficult to grasp. Unlike a normal correlation function $\langle W(t) V \rangle$ which tracks chronological evolution, the alternating times ($t \to 0 \to t \to 0$) in the OTOC describe a forward-backward-forward-backward time loop that is sensitive to quantum chaos. **Boundary Limit & Sanity Check:** At $t = 0$, local spatially separated operators $\hat{W}$ and $\hat{V}$ commute, giving $F(0) = \langle \hat{W}^\dagger \hat{V}^\dagger \hat{W} \hat{V} \rangle = \langle \hat{W}^\dagger \hat{W} \hat{V}^\dagger \hat{V} \rangle = 1$. The identity yields $C(0) = 2(1 - \mathrm{Re}(1)) = 0$, verifying that the commutator is initially zero before causal spreading.

Active Recall Concept Test: OTOC Early-Time Commutativity

Question: Spatially separated local operators $\hat{W}$ and $\hat{V}$ are evolved under a local lattice Hamiltonian. Why does the OTOC $F(t)$ remain close to 1 at early times?

A) Because of the Lieb-Robinson bound, which restricts the spatial spreading of $\hat{W}(t)$ to a causal lightcone, keeping the commutator $[\hat{W}(t), \hat{V}]$ exponentially close to zero until the wavefront reaches $\hat{V}$'s site. **[Correct]**

B) Because in chaotic systems, dephasing only occurs at late times beyond the Thouless time.

C) Because finite-temperature regularization dampens the early-time fluctuations of the canonical Gibbs density matrix.

Explanation: Spatially separated local operators initially commute ($[\hat{W}, \hat{V}] = 0$). Under local Hamiltonian evolution, the Heisenberg operator $\hat{W}(t)$ grows in size. The Lieb-Robinson bound guarantees that the spatial support of $\hat{W}(t)$ is strictly bounded within a linear lightcone of radius $v_{\mathrm{LR}} t$. Until this lightcone reaches the site where $\hat{V}$ is located, the commutator $[\hat{W}(t), \hat{V}]$ is exponentially close to zero, meaning $C(t) \approx 0$. By our identity $C(t) = 2(1 - \operatorname{Re} F(t))$, this forces the OTOC $F(t)$ to remain close to 1 during this transient causal delay.

Card 6: Scrambling Time

Goal: Prove that the scrambling time t_* at which the squared commutator becomes $C(t_*) \sim \mathcal{O}(1)$ scales logarithmically with the system size N:

$$t_* \sim \frac{1}{2\lambda_L} \log N$$

Key Trick: The OTOC in a chaotic system with N degrees of freedom behaves as $F(t) \approx 1 - \frac{1}{2N} e^{2\lambda_L t}$. Setting the exponential term to balance the $1/N$ prefactor gives the scrambling time $t_* \sim \frac{1}{2\lambda_L} \ln N$.

Step-by-Step Derivation:

1. In a chaotic quantum system (such as a holographic system or all-to-all interacting spin chain), local operators scramble across all degrees of freedom.

2. During the early-time chaotic regime, the commutator grows exponentially, governed by the Lyapunov exponent λ_L:
$$C(t) \approx \frac{1}{N} e^{2\lambda_L t}$$

3. Correspondingly, the OTOC decays from its initial value of 1 as:
$$F(t) \approx 1 - \frac{1}{2N} e^{2\lambda_L t}$$

4. The system is considered fully scrambled when the commutator grows to be order of unity:
$$C(t_*) \sim 1$$

5. Set up the equation at $t = t_*$:
$$\frac{1}{N} e^{2\lambda_L t_*} \sim 1$$

6. Multiply both sides by N:
$$e^{2\lambda_L t_*} \sim N$$

7. Take the natural logarithm of both sides:
$$2\lambda_L t_* \sim \log N$$

8. Solve for t_*:
$$\boxed{t_* \sim \frac{1}{2\lambda_L} \log N}$$

Questions

1. **Question:** Is the logarithmic scrambling time $t_* \sim \frac{1}{2\lambda_L} \log N$ related to the fast scrambling conjecture?

   Answer: Yes, Sekino and Susskind conjectured that no physical system can scramble information faster than $\mathcal{O}(\log N)$. This bound is saturated by maximally chaotic quantum systems (such as black holes and the Sachdev-Ye-Kitaev model) where the Lyapunov exponent saturates the Maldacena-Shenker-Stanford (MSS) temperature bound $\lambda_L \leq 2\pi k_B T/\hbar$.

Derivation Blueprint: Scrambling Time

Board Layout:

- **Left Board Panel:** Draw a cartoon of a local operator spreading out over a spin chain as a function of time (an operator light-cone).
- **Middle Panel:** Write the early-time growth equations for $C(t)$ and $F(t)$.
- **Right Panel:** Solve the algebraic steps to extract t_*. Highlight the $\log N$ dependency.

Physical & Experimental Anchorage:

- The Fast Scrambling Conjecture (conjectured by Sekino and Susskind) posits that black holes are the most rapid information scramblers in the universe, saturating the logarithmic scaling bound $t_* \sim \beta \log N$. This logarithmic slow-down is what makes quantum black holes highly non-trivial dephasers and enables the Hayden-Preskill protocol, showing that a black hole behaves as a highly efficient scrambling mirror. In laboratory quantum simulators, fast scrambling is directly probed in all-to-all interacting networks, such as trapped-ion platforms, Rydberg atom arrays, and the Sachdev-Ye-Kitaev (SYK) model implementations.

Common Conceptual Pitfalls: The $1/N$ scaling factor in the early-time growth of the commutator can be counter-intuitive. Because local operators are initially localized on individual sites, their support overlaps with only a $1/N$ fraction of the total system degrees of freedom, keeping early-time commutators highly suppressed. **Boundary Limit & Sanity Check:** Doubling the system size from N to $2N$ shifts the scrambling time by a constant increment: $t_*(2N) - t_*(N) \approx \frac{\log 2}{2\lambda_L}$. This logarithmic growth ensures that even highly complex systems scramble information over massive numbers of degrees of freedom with minimal temporal delay.

Card 18: Lanczos Algorithm & Krylov Complexity

Goal: Derive the Lanczos orthonormalization recursion relation to construct the Krylov basis for operator growth:

$$|O_n\rangle = \frac{1}{b_n}\left(\mathcal{L}|O_{n-1}\rangle - b_{n-1}|O_{n-2}\rangle\right)$$

where $\mathcal{L}(\hat{X}) = [\hat{H}, \hat{X}]$ is the Liouvillian superoperator.

Key Trick: Define a Hilbert-Schmidt inner product on the operator space, and perform Gram-Schmidt orthonormalization on the Krylov sequence $\mathcal{L}^n|O_0\rangle$.

Step-by-Step Derivation:

1. Define the operator inner product (e.g., infinite-temperature inner product):

   $$\langle A|B\rangle = \frac{1}{D}\,\mathrm{Tr}(\hat{A}^\dagger \hat{B})$$

   The Liouvillian is hermitian under this inner product: $\langle A|\mathcal{L}(B)\rangle = \langle \mathcal{L}(A)|B\rangle$.

2. We start with a normalized local seed operator $|O_0\rangle$ (so $\langle O_0|O_0\rangle = 1$).

3. Define the first Krylov operator $|O_1\rangle$ by applying $\mathcal{L}$ and normalizing:

   $$|\tilde{O}_1\rangle = \mathcal{L}|O_0\rangle - a_0|O_0\rangle \quad \text{where } a_0 = \langle O_0|\mathcal{L}|O_0\rangle = 0 \text{ (for traceless systems)}$$

   Thus $|\tilde{O}_1\rangle = \mathcal{L}|O_0\rangle$. Define $b_1 = \sqrt{\langle \tilde{O}_1|\tilde{O}_1\rangle}$, giving the normalized operator:

   $$|O_1\rangle = \frac{1}{b_1}\mathcal{L}|O_0\rangle$$

4. For the next step, construct $|\tilde{O}_2\rangle$ by applying $\mathcal{L}$ and orthogonalizing against all previous states $|O_1\rangle$ and $|O_0\rangle$:
$$|\tilde{O}_2\rangle = \mathcal{L}|O_1\rangle - \langle O_1|\mathcal{L}|O_1\rangle|O_1\rangle - \langle O_0|\mathcal{L}|O_1\rangle|O_0\rangle$$

5. Since the Liouvillian is self-adjoint, the inner product $\langle O_0|\mathcal{L}|O_1\rangle$ is:
$$\langle O_0|\mathcal{L}|O_1\rangle = \langle \mathcal{L}(O_0)|O_1\rangle = \langle b_1 O_1|O_1\rangle = b_1$$
Assuming the diagonal coefficients $a_n = \langle O_n|\mathcal{L}|O_n\rangle$ vanish by symmetry:
$$|\tilde{O}_2\rangle = \mathcal{L}|O_1\rangle - b_1|O_0\rangle$$
Normalize by $b_2 = \sqrt{\langle \tilde{O}_2|\tilde{O}_2\rangle}$:
$$|O_2\rangle = \frac{1}{b_2}\left(\mathcal{L}|O_1\rangle - b_1|O_0\rangle\right)$$

6. Generalizing this to the n-th step, we prove by mathematical induction that the Gram-Schmidt orthonormalization only couples the state $\mathcal{L}|O_{n-1}\rangle$ to the two immediately preceding states $|O_{n-1}\rangle$ and $|O_{n-2}\rangle$, meaning the Liouvillian is strictly tridiagonal in this basis:
$$\langle O_k|\mathcal{L}|O_{n-1}\rangle = 0 \quad \text{for all } k < n-2$$
To prove this, define the Krylov subspace of order m as:
$$\mathcal{K}_m = \text{span}\{|O_0\rangle, |O_1\rangle, \dots, |O_m\rangle\}$$
By definition of the Lanczos process, applying the Liouvillian superoperator to any state in $\mathcal{K}_m$ can at most increase the power by one, meaning $\mathcal{L}(\mathcal{K}_m) \subseteq \mathcal{K}_{m+1}$. Since the Gram-Schmidt process constructs each $|O_k\rangle$ to be strictly orthogonal to the entire lower Krylov subspace $\mathcal{K}_{k-1}$, we have:
$$\langle O_k|\phi\rangle = 0 \quad \text{for all } |\phi\rangle \in \mathcal{K}_{k-1}$$
Now let $k < n-2$, which implies $k+1 < n-1$. Since $\mathcal{L}$ is a self-adjoint superoperator:
$$\langle O_k|\mathcal{L}|O_{n-1}\rangle = \langle \mathcal{L}(O_k)|O_{n-1}\rangle$$
Since $|O_k\rangle \in \mathcal{K}_k$, we have $\mathcal{L}|O_k\rangle \in \mathcal{K}_{k+1}$. Because $k+1 < n-1$, the state $\mathcal{L}|O_k\rangle$ belongs to $\mathcal{K}_i$ for some index $i \le n-2$. By construction, the state $|O_{n-1}\rangle$ is orthogonal to the entire subspace $\mathcal{K}_{n-2}$. Therefore, the inner product must vanish:
$$\langle \mathcal{L}(O_k)|O_{n-1}\rangle = 0 \quad \forall k < n-2$$
By Hermiticity, this immediately implies the tridiagonal structure where only the terms $b_n = \langle O_n|\mathcal{L}|O_{n-1}\rangle$, $b_{n-1} = \langle O_{n-1}|\mathcal{L}|O_{n-2}\rangle$, and $a_{n-1} = \langle O_{n-1}|\mathcal{L}|O_{n-1}\rangle = 0$ can be non-zero. Thus, the recursion collapses exactly to:
$$\boxed{|O_n\rangle = \frac{1}{b_n}\left(\mathcal{L}|O_{n-1}\rangle - b_{n-1}|O_{n-2}\rangle\right)}$$

Questions

1. **Question:** What is the physical significance of the Lanczos coefficient hypothesis?

   Answer: For chaotic quantum many-body systems, the Lanczos coefficients b_n are hypothesized to exhibit maximal linear growth with n: $b_n \sim \alpha n$. In contrast, integrable systems exhibit sublinear growth (e.g., $b_n \sim \sqrt{n(N-n)}$). This linear growth rate sets the physical bound for the exponential growth of Krylov operator complexity.

Derivation Blueprint: Lanczos Algorithm & Krylov Complexity

Board Layout:

- **Left Board Panel:** Define the operator Hilbert space inner product. Write down the tridiagonal matrix form of the Liouvillian in the Krylov basis.
- **Middle Panel:** Show the Gram-Schmidt steps for $n = 1$ and $n = 2$ explicitly, explaining why the diagonal term a_n vanishes by trace symmetry.
- **Right Panel:** Write out the general tridiagonal recursion formula and discuss the difference in coefficient growth between chaotic and integrable systems.

Chalk-Talk Visual Guide:

- Draw a tridiagonal matrix representing the Liouvillian $\mathcal{L}$ in the Krylov basis $\{|O_n\rangle\}$. Mark the diagonal entries $a_n = 0$, and show the non-zero off-diagonals $b_1, b_2, \ldots$ as hopping couplings.
- Below the matrix, draw a 1D chain of sites representing Krylov states $|O_0\rangle \leftrightarrow |O_1\rangle \leftrightarrow |O_2\rangle \leftrightarrow |O_3\rangle$. Draw arrows showing that the action of $\mathcal{L}$ only allows forward hopping to $|O_{n+1}\rangle$ (with rate b_{n+1}) or backward hopping to $|O_{n-1}\rangle$ (with rate b_n). This maps operator complexity growth to a tight-binding particle hopping along a semi-infinite 1D lattice!

Common Conceptual Pitfalls: The vanishing of the Krylov matrix elements $\langle O_k|\mathcal{L}|O_{n-1}\rangle = 0$ for $k < n-2$ is a direct consequence of the Gram-Schmidt construction. Since the operator $\mathcal{L}$ increases the Krylov index by at most one, and $|O_{n-1}\rangle$ is orthogonal to all operators with lower indices, the inner product must vanish. **Boundary Limit & Sanity Check:** For a non-interacting system where the Hamiltonian is set to zero ($H = 0$), the Liouvillian vanishes ($\mathcal{L} = 0$). Gram-Schmidt orthogonalization terminates immediately at step 1 since $b_1 = 0$, reflecting that operators cannot grow without active couplings.

Numerical Sandbox: Operator Lanczos Coefficient Calculator

```
import numpy as np
# 3-site spin chain Hamiltonian: H = Z_0 Z_1 + Z_1 Z_2 + h (X_0 + X_1 + X_2)
h = 1.05
sz = np.array([[1, 0], [0, -1]])
sx = np.array([[0, 1], [1, 0]])
si = np.eye(2)
# Construct operators on 3 sites
Z0 = np.kron(sz, np.kron(si, si))
Z1 = np.kron(si, np.kron(sz, si))
Z2 = np.kron(si, np.kron(si, sz))
X0 = np.kron(sx, np.kron(si, si))
X1 = np.kron(si, np.kron(sx, si))
X2 = np.kron(si, np.kron(si, sx))
H = np.kron(sz, np.kron(sz, si)) + np.kron(si, np.kron(sz, sz)) + h * (X0 + X1 + X2)
# Initial local seed operator: O_0 = X_0 (normalized)
O0 = X0 / np.sqrt(np.trace(X0 @ X0) / 8)
# Liouvillian superoperator: L(O) = [H, O]
def liouvillian(O, H):
    return H @ O - O @ H
# Gram-Schmidt steps
tilde_O1 = liouvillian(O0, H)
b1 = np.sqrt(np.trace(tilde_O1.conj().T @ tilde_O1) / 8)
O1 = tilde_O1 / b1
tilde_O2 = liouvillian(O1, H) - b1 * O0
b2 = np.sqrt(np.trace(tilde_O2.conj().T @ tilde_O2) / 8)
print(f"Lanczos Coefficient b_1: {b1:.6f}")
print(f"Lanczos Coefficient b_2: {b2:.6f}")
```

Card 21: Regularized OTOC at Finite Temperature

Goal: Prove the definition of the regularized out-of-time-ordered correlator (OTOC) $F_\beta(t) = \mathrm{Tr}\left(\hat{y}\hat{W}(t)\hat{y}\hat{V}\hat{y}\hat{W}(t)\hat{y}\hat{V}\right)$ at finite temperature $T = 1/\beta$, where $\hat{y} = \hat{\rho}_\beta^{1/4}$, and explain why it is mathematically preferred over the unregularized version for chaos bounds.

Key Trick: Distribute the thermal density matrix $\hat{\rho}_\beta = e^{-\beta\hat{H}}/Z$ symmetrically along the imaginary time axis to regularize high-frequency UV divergences.

Step-by-Step Derivation:

1. The unregularized thermal expectation value of the OTOC at finite temperature is:

$$F_{\mathrm{unreg}}(t) = \mathrm{Tr}\left(\hat{\rho}_\beta\hat{W}^\dagger(t)\hat{V}^\dagger\hat{W}(t)\hat{V}\right)$$

   where $\hat{\rho}_\beta = e^{-\beta\hat{H}}/Z$ is the canonical Gibbs state.

2. When analyzing quantum field theories or many-body systems, operator products like $\hat{W}^\dagger(t)\hat{V}^\dagger$ evaluated at the same spatial point can produce unphysical ultraviolet (UV) divergences due to short-distance singularities.

3. To regularize these singularities, we analytically continue the operators into complex time by distributing the thermal density matrix symmetrically along the thermal circle.

4. Define $\hat{y} = \hat{\rho}_\beta^{1/4} = \frac{e^{-\beta\hat{H}/4}}{Z^{1/4}}$. Since $\hat{y}^4 = \hat{\rho}_\beta$, this splits the thermal state into four equal parts.

5. We interleave these thermal factors between the four operators in the OTOC product:

$$\boxed{F_\beta(t) = \mathrm{Tr}\left(\hat{y}\hat{W}(t)\hat{y}\hat{V}\hat{y}\hat{W}(t)\hat{y}\hat{V}\right)}$$

   (assuming the operators $\hat{V}$ and $\hat{W}$ are hermitian for simplicity).

6. This corresponds to placing the operators at complex times:

$$t_1 = t, \quad t_2 = i\beta/4, \quad t_3 = t + i\beta/2, \quad t_4 = 3i\beta/4$$

   which maximizes their separation along the imaginary thermal circle, smoothing out short-distance UV singularities while preserving the chaotic real-time dynamics.

Questions

1. **Question:** Does the regularized OTOC yield the same Lyapunov exponent as the unregularized version?

   Answer: Yes, in the late-time chaotic regime ($t \gg \beta$), both regularized and unregularized correlators grow exponentially with the identical Lyapunov exponent λ_L. The regularization only smooths out the early-time ultraviolet (UV) transients, which is required to mathematically prove the Maldacena-Shenker-Stanford (MSS) chaos bound $\lambda_L \leq 2\pi k_B T/\hbar$.

Derivation Blueprint: Regularized OTOC at Finite Temperature

Board Layout:

- **Left Board Panel:** Draw the thermal circle of circumference β in the complex time plane. Mark the four operator locations at complex coordinates.
- **Middle Panel:** Write down the unregularized and regularized OTOC formulas side-by-step.
- **Right Panel:** Explain the physical source of UV singularities and show how the analytical continuation regularizes them.

Chalk-Talk Visual Guide:

- Draw a large vertical circle representing the imaginary time axis (thermal circle) of circumference $\beta = 1/T$. The origin is at the top (0).
- Mark four points separated by angles of 90° ($\pi/2$ radians):
    1. Top site (0): Place the first operator $\hat{V}$.
    2. Right site ($i\beta/4$): Place the first time-evolved operator $\hat{W}(t)$.
    3. Bottom site ($i\beta/2$): Place the second operator $\hat{V}$.
    4. Left site ($3i\beta/4$): Place the second time-evolved operator $\hat{W}(t)$.
- Draw curved arrows along the perimeter showing that the thermal shields $\hat{y} = \hat{\rho}^{1/4}$ correspond to the $i\beta/4$ imaginary-time propagation intervals between operators, illustrating how this symmetric spacing spreads the operator insertion singularities uniformly along the thermal circle.

Common Conceptual Pitfalls: The choice of $y = \rho^{1/4}$ instead of $\rho^{1/2}$ in the regularized OTOC can seem arbitrary. It is mathematically required because the out-of-time-ordered correlator contains four operators, necessitating four symmetric thermal shields to isolate each adjacent operator in the product. **Boundary Limit & Sanity Check:** Under the infinite-temperature limit ($\beta \to 0$), the regularization factors $\hat{y} = \hat{\rho}_\beta^{1/4} \to \hat{I}/D^{1/4}$. The regularized OTOC reduces to $F_{\beta\to 0}(t) = \mathrm{Tr}(\hat{W}(t)\hat{V}\hat{W}(t)\hat{V})/D$, matching the standard infinite-temperature OTOC and confirming the thermal origin of the regularization.

5 Quantum Designs & Clifford Ensembles

Exercise 5.1: OTOCs and the Failure of Clifford Chaos

Motivation:

Clifford circuits form exact unitary 3-designs — they reproduce the first three moments of the Haar measure. This makes them extremely useful for benchmarking and error correction. Yet they are **efficiently classically simulable** (Gottesman-Knill theorem) and therefore cannot exhibit true quantum chaos.

The OTOC $C(t)$ quantifies scrambling: in chaotic systems, it grows exponentially as $\sim e^{2\lambda t}$ before saturating. This exercise shows that Clifford evolution can only produce $C(t) \in \{0, 4\}$ — a discrete jump, never a smooth exponential. This is the fundamental obstruction to defining a Lyapunov exponent for Clifford dynamics.

Preliminaries / Toolbox

Before diving into the solution, recall the following concepts:

1. Pauli Group: The group generated by I, X, Y, Z with phase factors. Elements either commute $[A, B] = 0$ or anticommute $\{A, B\} = 0$.

2. Clifford Group: The normalizer of the Pauli group. A unitary U is Clifford if $UPU^\dagger$ is another Pauli operator P', for any Pauli P.

3. Traces of Paulis: All non-identity Pauli operators are traceless, $\mathrm{Tr}(P) = 0$, and square to identity $P^2 = I$.

4. OTOC bounds: Because continuous scrambling $\sim e^{\lambda t}$ implies smooth interpolation, a strictly binary-valued squared commutator means true (Lyapunov) chaos cannot occur.

Problem Statement:

Let U be a Clifford unitary on N qubits. For single-qubit Paulis W and V, define the squared commutator:

$$C(t) = \frac{1}{D}\mathrm{Tr}\big([W(t), V]^\dagger [W(t), V]\big), \qquad W(t) = U^\dagger W U.$$

Prove $C(t) \in \{0, 4\}$ and conclude that Clifford circuits have no Lyapunov exponent.

Pedantic Step-by-Step Solution:

Step 1: Clifford conjugation maps Paulis to Paulis

By definition, a Clifford unitary U normalizes the Pauli group: $U^\dagger P U \in \mathcal{P}_N$ (up to a sign $\pm 1, \pm i$) for any Pauli P. In particular, $W(t) = U^\dagger W U$ is itself a Pauli operator (possibly on many qubits).

Step 2: Paulis either commute or anticommute

Any two elements of the Pauli group on N qubits either **commute** or **anticommute** — there is no intermediate possibility. This is because Pauli operators are tensor products of $\{I, X, Y, Z\}$, and each pair of single-qubit Paulis either commutes (if they are equal or one is I) or anticommutes (if they are different non-identity Paulis). The full commutation is the product of the per-site factors.

Step 3: The two cases

Case 1: $[W(t), V] = 0$ (they commute).

$$C(t) = 0.$$

Case 2: $\{W(t), V\} = 0$ (they anticommute). Then $[W(t), V] = 2W(t)V$.

$$C(t) = \frac{1}{D}\mathrm{Tr}\big(4V^\dagger W(t)^\dagger W(t) V\big) = \frac{4}{D}\mathrm{Tr}(I) = 4.$$

(We used $W(t)^\dagger W(t) = I$ and $V^\dagger V = I$ since Paulis are unitary.)

Step 4: No Lyapunov exponent

$$\boxed{C(t) \in \{0, 4\} \quad \text{for all Clifford } U.}$$

Continuous exponential growth $C(t) \sim \epsilon\, e^{2\lambda t}$ requires the OTOC to pass through intermediate values — impossible with the discrete set $\{0, 4\}$. Clifford circuits scramble (the Pauli W can spread to a high-weight operator), but the scrambling is **instantaneous** once the light cone reaches V, not gradual. This is why Cliffords are a 3-design (matching moments up to $k = 3$) but **not** a good model of quantum chaos.

Exercise 5.2: Stabilizer Rényi Entropy of Haar-Random States

Motivation:

Clifford circuits acting on computational basis states produce **stabilizer states**. By the Gottesman-Knill theorem, these states are efficiently classically simulable and thus contain no "quantum magic". To achieve universal quantum computation (and true quantum chaos), non-Clifford resources (like T gates) are required.

The **Stabilizer Rényi Entropy (SRE)** quantifies this magic. It measures how "spread out" a state is in the Pauli basis. This exercise computes the second SRE M_2 for Haar-random states, showing they are near-maximally magical, taking $M_2 \approx N - 2$ bits of magic.

Preliminaries / Toolbox

1. Pauli decomposition: Any state $|\psi\rangle$ can be written as $|\psi\rangle\langle\psi| = \frac{1}{D}\sum_{P\in\mathcal{P}_N}\langle P\rangle P$, where $\langle P\rangle = \langle\psi|P|\psi\rangle$ and $\mathcal{P}_N$ is the set of $D^2 = 4^N$ Pauli strings.

2. Stabilizer states: For a stabilizer state, exactly D Pauli strings have $\langle P\rangle = \pm 1$, and the remaining $D^2 - D$ have $\langle P\rangle = 0$.

3. Stabilizer Rényi Entropy: $M_2(\psi) = -\log_2 \Xi_P(\psi)$, where the Pauli purity is:

$$\Xi_P(\psi) = \frac{1}{D}\sum_{P\in\mathcal{P}_N}\langle P\rangle^4.$$

4. The $k = 2$ fourth-moment Weingarten bound: Over the Haar measure, $\mathbb{E}[\langle\psi|A|\psi\rangle^4]$ projects onto the fully symmetric subspace of 4 copies.

Problem Statement:

(a) Show $\mathbb{E}_{\text{Haar}}[\langle A\rangle^4] = \frac{3}{(D+1)(D+3)}$ for any traceless operator with $A^2 = I$.

(b) Show $\mathbb{E}[\Xi_P] = \frac{4}{D+3}$.

(c) Show $-\log_2(\mathbb{E}[\Xi_P]) = N - 2 + \mathcal{O}(2^{-N})$.

(d) Verify that for a stabilizer state, $\Xi_P = 1$ and $M_2 = 0$.

Pedantic Step-by-Step Solution:

Part (a): Fourth moment of a Pauli string

The 4th moment over the Haar measure projects $A^{\otimes 4}$ onto the fully symmetric subspace of $\mathcal{H}^{\otimes 4}$:

$$\mathbb{E}[\langle\psi|A|\psi\rangle^4] = \mathrm{Tr}\left(A^{\otimes 4}\Pi^{(4)}_{\text{sym}}\right)/\dim(\mathrm{Sym}^4).$$

The dimension is $\binom{D+3}{4} = D(D+1)(D+2)(D+3)/24$. The projector is $\Pi^{(4)}_{\text{sym}} = \frac{1}{24}\sum_{\pi\in S_4}P_\pi$.

$$\mathrm{Tr}(A^{\otimes 4}P_\pi)$$

depends on the cycle structure of π. Since A is traceless, any permutation with a 1-cycle gives $\mathrm{Tr}(A) = 0$. Only two conjugacy classes survive:

- 3 products of two disjoint transpositions (e.g., (12)(34)): $\mathrm{Tr}(A^2)^2 = \mathrm{Tr}(I)^2 = D^2$.
- 6 four-cycles (e.g., (1234)): $\mathrm{Tr}(A^4) = \mathrm{Tr}(I) = D$.

$$\mathrm{Tr}\left(A^{\otimes 4}\sum_\pi P_\pi\right) = 3D^2 + 6D = 3D(D+2).$$

$$\mathbb{E}[\langle A\rangle^4] = \frac{3D(D+2)/24}{D(D+1)(D+2)(D+3)/24} = \frac{3}{(D+1)(D+3)}. \quad \checkmark$$

Part (b): Average Pauli purity

Separate the identity $P = I$ from the $D^2 - 1$ traceless Paulis:

$$\Xi_P = \frac{1}{D}\sum_P \langle P\rangle^4 = \frac{1}{D}\left[\langle I\rangle^4 + \sum_{P\neq I}\langle P\rangle^4\right].$$

Since $\langle I\rangle = \langle\psi|\psi\rangle = 1$:

$$\mathbb{E}[\Xi_P] = \frac{1}{D}\left[1 + (D^2-1)\mathbb{E}[\langle P\rangle^4]\right] = \frac{1}{D}\left[1 + (D-1)(D+1)\frac{3}{(D+1)(D+3)}\right].$$

$$\mathbb{E}[\Xi_P] = \frac{1}{D}\left[1 + \frac{3(D-1)}{D+3}\right] = \frac{1}{D}\left[\frac{D+3+3D-3}{D+3}\right] = \frac{1}{D}\frac{4D}{D+3} = \frac{4}{D+3}. \quad \checkmark$$

Part (c): The magic of typical states

$$M_2 \approx -\log_2 \mathbb{E}[\Xi_P] = -\log_2\left(\frac{4}{D+3}\right) = \log_2\left(\frac{2^N+3}{4}\right).$$

For $N \gg 1$, $2^N + 3 \approx 2^N$:

$$M_2 \approx \log_2(2^{N-2}) = N - 2.$$

A typical state has nearly the maximal possible magic (N bits).

Part (d): Stabilizer states

If $|\psi\rangle$ is a stabilizer state, it lies purely in a stabilizer group $\mathcal{S}$. It has expectation $+1$ or -1 for the $D = 2^N$ Paulis in $\mathcal{S}$, and 0 for all others.

$$\Xi_P = \frac{1}{D}\left(\underbrace{1^4 + \cdots + 1^4}_{D \text{ terms}}\right) = \frac{D}{D} = 1.$$

$$M_2 = -\log_2(1) = 0.$$

Stabilizer states have exactly zero magic.

Visual Pedagogical Proof

Card 9: Clifford Twirl → Depolarizing Channel

Goal: Prove that twirling any quantum channel $\mathcal{E}$ over the Clifford group yields a depolarizing channel.

Key Trick: The Clifford group forms a 2-design, which means its second moment is identical to the Haar measure. The twirling operation projects the channel onto the commutant of the representation, restricting the output to a 1-parameter family.

Step-by-Step Derivation:

1. The twirling of a channel $\mathcal{E}$ over the Clifford group $\mathcal{C}$ is defined as:

$$\mathcal{E}_{\text{twirl}}(\hat{\rho}) = \frac{1}{|\mathcal{C}|} \sum_{C \in \mathcal{C}} \hat{C} \mathcal{E}\left(\hat{C}^\dagger \hat{\rho} \hat{C}\right) \hat{C}^\dagger$$

2. Since the Clifford group forms a 2-design, we can replace the discrete sum over Cliffords with an integration over the Haar measure on the unitary group:

$$\mathcal{E}_{\text{twirl}}(\hat{\rho}) = \int_{U(D)} \hat{U} \mathcal{E}\left(\hat{U}^\dagger \hat{\rho} \hat{U}\right) \hat{U}^\dagger \, dU$$

3. Express the channel $\mathcal{E}$ using its Kraus operators $\mathcal{E}(\hat{\rho}) = \sum_k \hat{A}_k \hat{\rho} \hat{A}_k^\dagger$:

$$\mathcal{E}_{\text{twirl}}(\hat{\rho}) = \sum_k \int_{U(D)} \hat{U} \hat{A}_k \hat{U}^\dagger \hat{\rho} \hat{U} \hat{A}_k^\dagger \hat{U}^\dagger \, dU$$

4. Introduce the doubled space notation to isolate the unitary operators:

$$\mathcal{E}_{\text{twirl}}(\hat{\rho}) = \sum_k \mathrm{Tr}_2 \left[\mathbb{E}_U \left[\hat{U}^{\otimes 2} \left(\hat{A}_k \otimes \hat{\rho}^T \right) \hat{U}^{\dagger \otimes 2} \right] \left(\hat{I} \otimes |\Phi\rangle \langle\Phi| \right) \right]$$

   Alternatively, using the commutant of $\hat{U} \otimes \hat{U}^*$: The twirled map must map the identity to the identity (if trace-preserving) and must commute with all unitary transformations: $\mathcal{E}_{\text{twirl}}(\hat{U}\hat{\rho}\hat{U}^\dagger) = \hat{U}\mathcal{E}_{\text{twirl}}(\hat{\rho})\hat{U}^\dagger$.

5. The only linear maps that are covariant under all unitary operations are of the form:

$$\mathcal{E}_{\text{twirl}}(\hat{\rho}) = p\,\hat{\rho} + q\,\mathrm{Tr}(\hat{\rho}) \frac{\hat{I}}{D}$$

6. Assuming the original channel is trace-preserving, the twirled channel must also be trace-preserving:

$$\mathrm{Tr}\left[\mathcal{E}_{\text{twirl}}(\hat{\rho})\right] = p\,\mathrm{Tr}(\hat{\rho}) + q\,\mathrm{Tr}(\hat{\rho}) = (p+q)\,\mathrm{Tr}(\hat{\rho}) \implies p + q = 1 \implies q = 1 - p$$

7. Thus, the twirled channel is a depolarizing channel:

$$\boxed{\mathcal{E}_{\text{twirl}}(\hat{\rho}) = p\,\hat{\rho} + (1-p) \frac{\hat{I}}{D}}$$

8. To find the parameter p, we evaluate the entanglement fidelity $F_{\text{ent}} = \langle\Phi| \left(\mathcal{I} \otimes \mathcal{E}_{\text{twirl}}\right) |\Phi\rangle \langle\Phi|$ where $|\Phi\rangle$ is the maximally entangled state:

$$F_{\text{ent}} = p + (1-p)\frac{1}{D^2} \implies p = \frac{D^2 F_{\text{ent}} - 1}{D^2 - 1}$$

Questions

1. **Question:** Why can we use the Clifford group instead of Haar-random unitaries for twirling?

   Answer: The twirling operation of a quantum channel only requires matching the first two moments of the Haar measure. Since the Clifford group is a unitary 2-design, it reproduces these moments exactly, allowing experimentalists to perform twirling with simple Clifford circuits rather than complex Haar unitaries.

Derivation Blueprint: Clifford Twirl to Depolarizing Channel

Board Layout:

- **Left Board Panel:** Write down the definition of group twirling. Explain what a unitary 2-design is.
- **Middle Panel:** Write down the covariance condition: twirling commutes with all unitaries. Write down the most general form of a covariant map using Schur's Lemma.
- **Right Panel:** Apply the trace-preserving condition to solve for q, and show the derivation of p from the entanglement fidelity.

Common Conceptual Pitfalls: The restriction of the covariant map to linear combinations of the identity and the input state $\hat{\rho}$ is a direct consequence of Schur's Lemma. Any linear map that commutes with all unitary transformations must act as a simple scalar on the irreducible components of the representation, which decomposes strictly into the trace and the traceless parts of the operator. **Physical & Experimental Anchorage:**

- Twirling a channel over the Clifford group is the foundational basis for **Randomized Benchmarking (RB)**. In real quantum processors (like superconducting qubits or neutral atoms), physical gates have small systematic calibration errors and environmental noise. By interleaved application of random Clifford gates, experimentalists effectively depolarize the noise channel into a single parameter p, which can be extracted by measuring the exponential decay of state fidelity as a function of circuit depth. This provides a platform-independent estimate of the average gate error.

Boundary Limit & Sanity Check: If the channel is completely noiseless ($\mathcal{E} = \mathcal{I}$), the entanglement fidelity is $F_{\text{ent}} = 1$, which yields the twirling parameter $p = (D^2-1)/(D^2-1) = 1$. The depolarizing channel thus reduces back to the identity map ($\mathcal{E}_{\text{twirl}} = \mathcal{I}$), satisfying noiseless boundary preservation.

6 Applications of Scrambling

Exercise 6.1: Tripartite Mutual Information of Scrambled States

Motivation:

The **tripartite mutual information** $I_3(A : B : C)$ is a sharp diagnostic of information scrambling. For unscrambled states, $I_3 \geq 0$ (information is localized). For scrambled states, $I_3 < 0$ — meaning that information about A is **delocalized** across B and C and can only be recovered by accessing both jointly. This exercise computes I_3 for Haar-random states using the Page approximation and verifies numerically.

Preliminaries / Toolbox

Before diving into the solution, recall the following concepts:

1. Von Neumann Entropy: $S(\rho) = -\operatorname{Tr}(\rho \ln \rho)$. Base e (nats) is typical in mathematical physics, though base 2 (bits) is common.

2. Page Approximation: For a Haar-random pure state on $\mathcal{H}_A \otimes \mathcal{H}_B$, if $d_A \leq d_B$, the average entropy of A is $S(A) \approx \ln d_A - d_A/(2d_B)$. To leading order, $S(A) \approx \ln \min(d_A, d_B)$.

3. Mutual Information: $I(A : B) = S(A) + S(B) - S(AB)$. Measures total classical and quantum correlations.

4. Tripartite Mutual Information: $I_3(A : B : C) = I(A : B) + I(A : C) - I(A : BC)$. A negative value indicates topological entanglement or information scrambling, as information about A is hidden in the joint state BC but invisible in B and C locally.

Problem Statement:

For a Haar-random pure state on four equal subsystems A, B, C, D with $d_A = d_B = d_C = d_D = d \gg 1$:

Compute $I(A : B)$, $I(A : C)$, $I(A : BC)$, and show $I_3(A : B : C) \approx -2 \ln d$.

Pedantic Step-by-Step Solution:

Step 1: Subsystem entropies via the Page approximation

For a Haar-random state, the Page approximation gives $S(X) \approx \ln(\min(d_X, d_{\bar{X}}))$ for subsystem X with complement $\bar{X}$:

$$S(A) = \ln d \quad (d_A = d,\ d_{\bar{A}} = d^3)$$

$$S(AB) = 2 \ln d \quad (d_{AB} = d^2 = d_{CD})$$

$$S(ABC) = \ln d \quad (d_{ABC} = d^3,\ d_D = d)$$

Step 2: Mutual informations

$$I(A : B) = S(A) + S(B) - S(AB) = \ln d + \ln d - 2 \ln d = 0.$$

$$I(A : C) = S(A) + S(C) - S(AC) = 0.$$

$$I(A : BC) = S(A) + S(BC) - S(ABC) = \ln d + 2 \ln d - \ln d = 2 \ln d.$$

Step 3: Tripartite mutual information

$$\boxed{I_3(A : B : C) = I(A : B) + I(A : C) - I(A : BC) = 0 + 0 - 2 \ln d = -2 \ln d.}$$

Physical interpretation: To leading order in d, neither B nor C individually shares appreciable mutual information with A, while jointly BC has full mutual information $I(A : BC) = 2 \ln d$. This is the hallmark of **scrambling**: information about A is non-locally encoded across B and C. The

leading magnitude $|I_3| \approx 2\ln d$ grows with the subsystem dimension — more scrambling in larger systems.
Finite-d correction (caution): The values $I(A:B) = I(A:C) = 0$ hold only in the $d \to \infty$ Page limit; the $O(1)$ Page correction $-(d_A - 1)/(2d_B)$ makes the pairwise mutual informations small but *nonzero* at finite d ($I(A:B) = I(A:C) \to +\frac{1}{2}$ nat, consistent with the companion script's numerics). This does not affect the leading $-2\ln d$ scaling.

Exercise 6.2: Fragility of the GHZ Metrological Advantage

Motivation:
The GHZ state achieves the **Heisenberg limit** $F_Q = N^2$ — a quadratic improvement over the classical shot-noise limit $F_Q = N$. But this advantage is catastrophically fragile: losing a *single qubit* to the environment destroys **all** metrological information ($F_Q \to 0$). This exercise demonstrates why: the QFI is encoded entirely in the off-diagonal coherence $|0\cdots0\rangle\langle1\cdots1|$, which vanishes under partial trace. This fragility motivates the search for decoherence-robust metrological states.
Preliminaries / Toolbox
Before diving into the solution, recall the following concepts:
1. Quantum Fisher Information (QFI): Quantifies metrological usefulness. For pure states $|\psi\rangle$ with generator G, $F_Q = 4\mathrm{Var}(G) = 4(\langle G^2\rangle - \langle G\rangle^2)$.
2. QFI for Mixed States: For $\rho = \sum_k \lambda_k |k\rangle\langle k|$, $F_Q = 2\sum_{k,l} \frac{(\lambda_k-\lambda_l)^2}{\lambda_k+\lambda_l} |\langle k|G|l\rangle|^2$.
3. Heisenberg Limit vs Shot Noise Limit: Using unentangled states, measuring phase achieves precision $\Delta\theta \geq 1/\sqrt{N}$ (Shot-noise limit, $F_Q = N$). Using highly entangled states (e.g. GHZ), $\Delta\theta \geq 1/N$ (Heisenberg limit, $F_Q = N^2$).
4. Partial Trace: Losing a single qubit means taking the partial trace $\rho_{N-1} = \mathrm{Tr}_N(|\psi\rangle\langle\psi|)$, which destroys global coherences.
Problem Statement:
The N-qubit GHZ state $|\mathrm{GHZ}\rangle = (|0\cdots0\rangle + |1\cdots1\rangle)/\sqrt{2}$ achieves $F_Q = N^2$ for $G = \frac{1}{2}\sum_j \sigma_z^{(j)}$.
(a) Trace out one qubit and find the reduced state ρ_{N-1}.
(b) Compute the QFI of ρ_{N-1} for the reduced generator $G' = \frac{1}{2}\sum_{j=1}^{N-1} \sigma_z^{(j)}$.
Pedantic Step-by-Step Solution:

Part (a): Partial trace destroys coherence

The full density matrix is:

$$\rho = \frac{1}{2}\big(|0\cdots0\rangle\langle0\cdots0| + |0\cdots0\rangle\langle1\cdots1| + |1\cdots1\rangle\langle0\cdots0| + |1\cdots1\rangle\langle1\cdots1|\big).$$

Tracing out the last qubit: the off-diagonal terms vanish because $\langle 0_N|1_N\rangle = 0$:

$$\rho_{N-1} = \frac{1}{2}\big(|0\cdots0\rangle\langle0\cdots0| + |1\cdots1\rangle\langle1\cdots1|\big).$$

This is a **classical mixture** of two orthogonal product states — all quantum coherence has been destroyed.

Part (b): QFI of the reduced state

The QFI for a mixed state $\rho = \sum_k \lambda_k |k\rangle\langle k|$ is:

$$F_Q = 2\sum_{k,l} \frac{(\lambda_k - \lambda_l)^2}{\lambda_k + \lambda_l} |\langle k|G'|l\rangle|^2.$$

For ρ_{N-1}: $\lambda_0 = \lambda_1 = 1/2$ for the two populated eigenstates, and $\lambda_k = 0$ for all others.
Between the two populated states: $(\lambda_0 - \lambda_1)^2 = 0$. This contribution vanishes.
Between populated and unpopulated: $\lambda_l = 0$ means the denominator vanishes, so these terms are excluded.

Moreover, G' is diagonal in the computational basis (it's a sum of Z operators), so $\langle 0\cdots 0|G'|1\cdots 1\rangle = 0$.

$$\boxed{F_Q(\rho_{N-1}) = 0.}$$

Physical interpretation: The GHZ state stores all metrological information in the global coherence $|0\cdots 0\rangle\langle 1\cdots 1|$. This coherence oscillates under the phase generator G at frequency N (Heisenberg scaling), but is maximally non-local — it requires *all* qubits to be preserved. Losing a single qubit irreversibly destroys this coherence, reducing the QFI from N^2 to exactly 0. This is the "Schrödinger cat" fragility.

Exercise 6.3: Randomized Benchmarking via Clifford Twirling

Motivation:

Randomized benchmarking (RB) is the standard method for characterizing gate fidelity in quantum hardware. It exploits a structural result: averaging any CPTP error channel over the Clifford group (a unitary 2-design) **twirls** it into a depolarizing channel. This twirling theorem is a direct consequence of Schur's lemma and the Weingarten calculus — the same machinery that produces barren plateaus also produces the clean exponential decay of RB.
The protocol measures only a single scalar quantity (survival probability) as a function of circuit depth, yet extracts the average gate fidelity with no need for full process tomography.
Preliminaries / Toolbox

Before diving into the solution, recall the following concepts:
1. Standard Randomized Benchmarking (RB): A protocol to estimate the average gate fidelity. Sequences of length m random Clifford gates C_i are followed by an exact inverse $C_m^{-1}\cdots C_1^{-1}$ to yield an identity operation. Survival probability decays as $Ap^m + B$.
2. Twirling over the Clifford Group: By Schur's lemma, averaging a channel $\mathcal{E}$ over the Clifford group produces a depolarizing channel: $\frac{1}{|\mathcal{C}|}\sum_{C\in\mathcal{C}} C^\dagger \mathcal{E}(C\rho C^\dagger)C = \mathcal{E}_{\text{depol}}(\rho) = p\rho + (1-p)\frac{I}{d}$.
3. Depolarizing Parameter p vs Fidelity: The average gate fidelity is related to p (also written as α) by $\bar{F} = p + \frac{1-p}{d}$.
4. Sequence fidelity: The expected return probability is $\text{Tr}(E\cdot\mathcal{E}_{\text{depol}}^m(\rho_0)) = Ap^m + B$ where A, B absorb state preparation and measurement (SPAM) errors.
Problem Statement:

(a) Prove the **twirling theorem**: for any CPTP map Λ, the Clifford twirl

$$\widetilde{\Lambda}(\rho) = \frac{1}{|\mathcal{C}|}\sum_{C\in\mathcal{C}} C^\dagger \Lambda(C\rho C^\dagger)C$$

is a depolarizing channel: $\widetilde{\Lambda}(\rho) = \alpha\,\rho + (1-\alpha)\frac{I}{D}$.
(b) Show the depolarizing parameter is $\alpha = \frac{D\,F_{\text{avg}}-1}{D-1}$, where F_{avg} is the average gate fidelity.
(c) Derive the RB decay: $\bar{p}(m) = A\,\alpha^m + B$.
(d) Implement and run the full RB protocol numerically; fit α and extract F_{avg}.
(e) Verify numerically that the Clifford twirl of the amplitude-damping channel (a non-depolarizing, non-unital channel) produces a depolarizing channel.
Pedantic Step-by-Step Solution:

Part (a): The twirling theorem

Claim: If $\mathcal{C}$ forms a unitary 2-design, then $\widetilde{\Lambda}(\rho) = \alpha\,\rho + (1-\alpha)\frac{I}{D}$ for some α.
Proof: The twirled channel $\widetilde{\Lambda}$ commutes with all Clifford unitaries: $C\widetilde{\Lambda}(\rho)C^\dagger = \widetilde{\Lambda}(C\rho C^\dagger)$ for all $C \in \mathcal{C}$.
By **Schur's lemma** (applied to the irreducible decomposition of the adjoint representation), any CPTP map that commutes with all elements of a 2-design must act as a scalar on each irreducible

sector. For qubits, the space of $D \times D$ Hermitian matrices decomposes into:

- The **trivial** sector: $\{c \cdot I \mid c \in \mathbb{R}\}$ (dimension 1).
- The **traceless** sector: $\{A \mid \mathrm{Tr}(A) = 0\}$ (dimension $D^2 - 1$).

The twirled channel must act as:

$$\widetilde{\Lambda}(I) = I \quad \text{(trace preservation)}, \qquad \widetilde{\Lambda}(A) = \alpha\, A \quad \text{(traceless sector)}.$$

For general $\rho = \frac{I}{D} + (\rho - \frac{I}{D})$, this gives:

$$\widetilde{\Lambda}(\rho) = \frac{I}{D} + \alpha \left(\rho - \frac{I}{D}\right) = \alpha\, \rho + (1-\alpha)\frac{I}{D}. \quad \checkmark$$

Part (b): Connecting α to average gate fidelity

The **average gate fidelity** of a channel Λ (relative to the identity) is:

$$F_{\text{avg}} = \int d\psi\, \langle\psi|\Lambda(|\psi\rangle\langle\psi|)|\psi\rangle.$$

For the depolarizing channel $\widetilde{\Lambda}(\rho) = \alpha\rho + (1-\alpha)I/D$:

$$F_{\text{avg}} = \int d\psi \left[\alpha\, \langle\psi|\psi\rangle^2 + (1-\alpha)\frac{\langle\psi|\psi\rangle}{D}\right] = \alpha + \frac{1-\alpha}{D}.$$

Solving for α:

$$\boxed{\alpha = \frac{D\, F_{\text{avg}} - 1}{D-1}.}$$

At $F_{\text{avg}} = 1$: $\alpha = 1$ (identity). At $F_{\text{avg}} = 1/D$: $\alpha = 0$ (fully depolarizing).

Part (c): The exponential decay

In the RB protocol, a sequence of m random Clifford gates is applied, each followed by noise Λ. The effective channel for the full sequence is m concatenated twirled channels:

$$\widetilde{\Lambda}^m(\rho) = \alpha^m \rho + (1-\alpha^m)\frac{I}{D}.$$

(Composing depolarizing channels: each application multiplies the traceless component by α, giving α^m after m steps.)

The survival probability after the recovery gate is:

$$\bar{p}(m) = \langle 0|\widetilde{\Lambda}^{m+1}(|0\rangle\langle 0|)|0\rangle = \alpha^{m+1} \underbrace{\langle 0|0\rangle\langle 0|0\rangle}_{=1} + (1-\alpha^{m+1}) \underbrace{\frac{1}{D}}$$

$$= \alpha^{m+1} + \frac{1-\alpha^{m+1}}{D} = \left(1 - \frac{1}{D}\right)\alpha^{m+1} + \frac{1}{D}.$$

This is the characteristic RB form $\bar{p}(m) = A\,\alpha^m + B$ with $A = (1-1/D)\alpha$ and $B = 1/D$. A single exponential fit extracts α.

Parts (d) and (e): Numerical Implementation

Visual Pedagogical Proof

Exercise 6.4: Classical Shadows: Channel Inversion and Tomography

Motivation:

Classical shadows are the most sample-efficient method for estimating local observables of an unknown quantum state. The key insight is that random single-qubit Clifford measurements induce a **depolarizing measurement channel**, which can be analytically inverted. This inversion is a direct application of the Haar/design averaging framework:

1. Averaging $U^\dagger|b\rangle\langle b|U$ over all single-qubit Cliffords U and outcomes b yields a depolarizing channel $\mathcal{M}$.
2. The inverse $\mathcal{M}^{-1}$ gives an unbiased estimator of ρ from each measurement.
3. The sample complexity for k-local observables is $O(3^k/\epsilon^2)$ — **independent of system size** N.

Problem Statement:

(a) Show that the measurement channel over the single-qubit Clifford group is $\mathcal{M}(\sigma) = \frac{\sigma+\mathrm{Tr}(\sigma)I}{3}$.
(b) Prove the inverse is $\mathcal{M}^{-1}(\tau) = 3\tau - \mathrm{Tr}(\tau)I$.
(c) Derive the shadow snapshot: for a measurement outcome $|b\rangle$ in the Clifford basis U, the unbiased estimator is $\hat{\rho} = 3U^\dagger|b\rangle\langle b|U - I$.
(d) For N qubits with independent single-qubit measurements, write the multi-qubit shadow snapshot.
(e) Implement the full protocol and verify on a Bell state.

Pedantic Step-by-Step Solution:

Part (a): Deriving the measurement channel $\mathcal{M}$

When we measure qubit j in the basis defined by Clifford U, the possible outcomes are $|b\rangle \in \{|0\rangle, |1\rangle\}$ with Born probabilities $p(b) = \langle b|U\sigma U^\dagger|b\rangle$. The post-measurement state in the original frame is $\tau_b = U^\dagger|b\rangle\langle b|U$.
Averaging over bases U (drawn uniformly from the 3 Pauli bases, forming a 1-design on the Bloch sphere) and outcomes b:

$$\mathcal{M}(\sigma) = \mathbb{E}_U \sum_b p(b)\, U^\dagger|b\rangle\langle b|U = \mathbb{E}_U \left[U^\dagger \left(\sum_b |b\rangle\langle b|U\sigma U^\dagger|b\rangle\langle b| \right) U \right].$$

The inner sum is the **quantum-to-classical-to-quantum** operation: measure in the U-basis and prepare the outcome. For a single qubit, this equals $\mathcal{M}(\sigma) = \frac{\sigma+\mathrm{Tr}(\sigma)I}{3}$.
Proof via Pauli decomposition: Write $\sigma = \frac{1}{2}(\mathrm{Tr}(\sigma)I + \vec{r}\cdot\vec{\sigma})$. For each Pauli basis $P \in \{X, Y, Z\}$:

$$\sum_{b=0,1} |b_P\rangle\langle b_P|\sigma|b_P\rangle\langle b_P| = \frac{1}{2}\left(\mathrm{Tr}(\sigma)I + r_P \cdot P\right),$$

where $r_P = \mathrm{Tr}(\sigma P)$. Averaging over $P \in \{X, Y, Z\}$ with equal weight:

$$\mathcal{M}(\sigma) = \frac{1}{3}\sum_P \frac{1}{2}\left(\mathrm{Tr}(\sigma)I + r_P \cdot P\right) = \frac{\mathrm{Tr}(\sigma)I}{2} \cdot \frac{1}{1} + \frac{\vec{r}\cdot\vec{\sigma}}{6}.$$

Since $\sigma = \frac{1}{2}(\mathrm{Tr}(\sigma)I + \vec{r}\cdot\vec{\sigma})$, we get $\mathcal{M}(\sigma) = \frac{\sigma+\mathrm{Tr}(\sigma)I}{3}$. ✓

Part (b): Channel inversion

We need $\mathcal{M}^{-1}$ such that $\mathcal{M}^{-1}(\mathcal{M}(\sigma)) = \sigma$. Let $\mathcal{M}^{-1}(\tau) = 3\tau - \mathrm{Tr}(\tau)I$. Check:

$$\mathcal{M}^{-1}\left(\frac{\sigma+\mathrm{Tr}(\sigma)I}{3}\right) = 3\cdot\frac{\sigma+\mathrm{Tr}(\sigma)I}{3} - \mathrm{Tr}\left(\frac{\sigma+\mathrm{Tr}(\sigma)I}{3}\right)I.$$

$$= \sigma + \mathrm{Tr}(\sigma)I - \frac{\mathrm{Tr}(\sigma)+2\,\mathrm{Tr}(\sigma)}{3}I = \sigma + \mathrm{Tr}(\sigma)I - \mathrm{Tr}(\sigma)I = \sigma. \quad ✓$$

(In the second line we used $\mathrm{Tr}(\sigma+\mathrm{Tr}(\sigma)I) = \mathrm{Tr}(\sigma)+2\,\mathrm{Tr}(\sigma) = 3\,\mathrm{Tr}(\sigma)$, so the trace of the argument is $\mathrm{Tr}(\sigma)$.)

Part (c): The shadow snapshot

For a single measurement with Clifford U yielding outcome $|b\rangle$, the post-measurement estimator is $\tau = U^\dagger|b\rangle\langle b|U$ with $\mathrm{Tr}(\tau) = 1$. Applying the channel inverse:

$$\boxed{\hat{\rho} = \mathcal{M}^{-1}(\tau) = 3U^\dagger|b\rangle\langle b|U - I.}$$

By construction, $\mathbb{E}[\hat{\rho}] = \mathcal{M}^{-1}(\mathcal{M}(\rho)) = \rho$: the shadow is an **unbiased estimator**.
Note: $\hat{\rho}$ is *not* a valid density matrix (it can have negative eigenvalues and trace $\neq 1$). That's fine — it is a classical data structure that gives unbiased estimates when averaged.

Part (d): Multi-qubit shadows

For N qubits with independent single-qubit measurements using Cliffords $U_1, \ldots, U_N$ and outcomes $b_1, \ldots, b_N$:

$$\hat{\rho} = \bigotimes_{j=1}^{N} \left(3\,U_j^\dagger|b_j\rangle\langle b_j|U_j - I_j\right).$$

Sample complexity: To estimate a k-local observable O acting on k qubits:

$$\mathrm{Var}[\mathrm{Tr}(O\hat{\rho})] \leq \frac{3^k\|O\|^2_{\text{shadow}}}{T},$$

where T is the number of shadow snapshots. The factor 3^k (not 3^N!) is the crucial advantage: the sample complexity depends only on the **locality** of O, not on the system size.

Part (e): Numerical Implementation

Motivation:

The shadow channel $\mathcal{M}(\sigma) = \frac{\sigma+\mathrm{Tr}(\sigma)I}{D+1}$ arises from averaging the measure-and-prepare map over a unitary 2-design. Its invertibility is the mathematical engine of classical shadows: by applying $\mathcal{M}^{-1}$ to each measurement outcome, we obtain an unbiased estimator of ρ. This exercise derives $\mathcal{M}$ using Weingarten calculus and verifies the inversion algebraically.

Preliminaries / Toolbox

Before diving into the solution, recall the following concepts:

1. Measurement Channel: Averaging over a 2-design, the preparation-measurement channel map is $\mathcal{M}(\rho) = \mathbb{E}_U \sum_b \mathrm{Tr}(U\rho U^\dagger|b\rangle\langle b|)U^\dagger|b\rangle\langle b|U$.
2. Weingarten Integral ($k = 2$): A 2-design average over polynomial degree two. Over D-dimensional space, this yields $\mathcal{M}(\rho) = \frac{\rho+\mathrm{Tr}(\rho)I}{D+1}$.
3. Trace Properties: $\mathrm{Tr}(A + B) = \mathrm{Tr}(A) + \mathrm{Tr}(B)$, $\mathrm{Tr}(cI) = cD$.
4. Single Qubits: For $D = 2$, $\mathcal{M}(\rho) = \frac{\rho+\mathrm{Tr}(\rho)I}{3}$.

Problem Statement:

(a) For a single qubit measured in a random Clifford basis, show $\mathcal{M}(\sigma) = \frac{\sigma+\mathrm{Tr}(\sigma)I}{3}$.
(b) Verify that $\mathcal{M}^{-1}(\tau) = 3\tau - \mathrm{Tr}(\tau)I$ is the inverse.
(c) State the general D-dimensional formula $\mathcal{M}(\sigma) = \frac{\sigma+\mathrm{Tr}(\sigma)I}{D+1}$ and its inverse.

Pedantic Step-by-Step Solution:

Part (a): Deriving $\mathcal{M}$ via the $k = 1$ Weingarten formula

The shadow channel is $\mathcal{M}(\sigma) = \mathbb{E}_U \sum_b \mathrm{Tr}(U\sigma U^\dagger|b\rangle\langle b|)\,U^\dagger|b\rangle\langle b|U$.
In index notation:

$$\mathcal{M}(\sigma)_{ij} = \sum_b \mathbb{E}_U\left[\overline{U}_{bi}\,U_{bj}\,\sum_{k,l} U_{bk}\,\sigma_{kl}\,\overline{U}_{bl}\right].$$

This requires the $k = 2$ Weingarten integral. For a 2-design (single-qubit Cliffords with $D = 2$):

$$\mathbb{E}[U_{bk}U_{bj}\overline{U}_{bl}\overline{U}_{bi}] = \sum_{\pi,\sigma\in S_2} \mathrm{Wg}(\pi^{-1}\sigma)\, \delta^{(\pi)}_{b,b}\, \delta^{(\sigma)}_{\dots}.$$

After contracting, the result for any D is:

$$\mathcal{M}(\sigma) = \frac{\sigma + \mathrm{Tr}(\sigma)I}{D+1}.$$

For $D = 2$: $\mathcal{M}(\sigma) = \frac{\sigma+\mathrm{Tr}(\sigma)I}{3}$. ✓

Part (b): Channel inversion

Claim: $\mathcal{M}^{-1}(\tau) = (D+1)\tau - \mathrm{Tr}(\tau)I$.
Proof: Apply $\mathcal{M}^{-1}$ to $\mathcal{M}(\sigma)$:

$$\mathcal{M}^{-1}\left(\frac{\sigma + \mathrm{Tr}(\sigma)I}{D+1}\right) = (D+1)\frac{\sigma + \mathrm{Tr}(\sigma)I}{D+1} - \mathrm{Tr}\left(\frac{\sigma + \mathrm{Tr}(\sigma)I}{D+1}\right) I.$$

$$= \sigma + \mathrm{Tr}(\sigma)I - \frac{\mathrm{Tr}(\sigma) + D\ \mathrm{Tr}(\sigma)}{D+1} I = \sigma + \mathrm{Tr}(\sigma)I - \mathrm{Tr}(\sigma)I = \sigma. \quad ✓$$

For $D = 2$: $\mathcal{M}^{-1}(\tau) = 3\tau - \mathrm{Tr}(\tau)I$.

Part (c): General D-dimensional formula

$$\boxed{\mathcal{M}(\sigma) = \frac{\sigma + \mathrm{Tr}(\sigma)I}{D+1}, \qquad \mathcal{M}^{-1}(\tau) = (D+1)\tau - \mathrm{Tr}(\tau)I.}$$

This is equivalent to saying that $\mathcal{M}$ acts as $1/(D+1)$ on the traceless sector and maps $I \to I$ (trace preservation). The factor $(D+1)$ arises from the dimension of the symmetric subspace $\mathrm{Sym}^2(\mathbb{C}^D)$ in the $k = 2$ Weingarten formula.

Card 10: Purity from Randomized Measurements

Goal: Prove that the average collision probability $\mathbb{E}_U\left[\sum_x p_U(x)^2\right]$ under Haar-random unitaries is related to the purity of the state $\hat{\rho}$:

$$\mathbb{E}_U\left[\sum_x p_U(x)^2\right] = \frac{\mathrm{Tr}(\hat{\rho}^2)+1}{D+1}$$

Key Trick: Represent the squared probability as a trace on the doubled space, apply the second Haar moment, and contract the resulting identity and swap terms.

Step-by-Step Derivation:

1. The probability of measuring state $\hat{\rho}$ in the computational state $|x\rangle$ after applying $\hat{U}$ is:

$$p_U(x) = \langle x|\,\hat{U}\hat{\rho}\hat{U}^\dagger\,|x\rangle = \mathrm{Tr}\left(|x\rangle\langle x|\,\hat{U}\hat{\rho}\hat{U}^\dagger\right)$$

2. Write the sum of squared probabilities using the doubled space:

$$\sum_x p_U(x)^2 = \sum_x \mathrm{Tr}\left[(|x\rangle\langle x|)^{\otimes 2}\left(\hat{U}\hat{\rho}\hat{U}^\dagger\right)^{\otimes 2}\right]$$

3. Take the expectation value over $\hat{U}$:

$$\mathbb{E}_U\left[\sum_x p_U(x)^2\right] = \sum_x \mathrm{Tr}\left[(|x\rangle\langle x|)^{\otimes 2}\,\mathbb{E}_U\left[\hat{U}^{\otimes 2}\hat{\rho}^{\otimes 2}\hat{U}^{\dagger\otimes 2}\right]\right]$$

4. Apply the second Haar moment formula (Card 3) with $\hat{O} = \hat{\rho}^{\otimes 2}$:

$$\mathbb{E}_U\left[\hat{U}^{\otimes 2}\hat{\rho}^{\otimes 2}\hat{U}^{\dagger\otimes 2}\right] = c_I\,\hat{\mathbb{I}} + c_{\mathbb{F}}\,\mathbb{F}$$

   where:

$$\begin{aligned}\mathrm{Tr}(\hat{O}) &= \mathrm{Tr}(\hat{\rho}^{\otimes 2}) = \mathrm{Tr}(\hat{\rho})^2 = 1\\ \mathrm{Tr}(\hat{O}\mathbb{F}) &= \mathrm{Tr}(\hat{\rho}^2)\end{aligned}$$

   Substitute these into the coefficients:

$$c_I = \frac{D-\mathrm{Tr}(\hat{\rho}^2)}{D(D^2-1)}, \qquad c_{\mathbb{F}} = \frac{D\,\mathrm{Tr}(\hat{\rho}^2)-1}{D(D^2-1)}$$

5. Now substitute the twirled expression back into the trace with $\sum_x (|x\rangle\langle x|)^{\otimes 2}$:

$$\mathbb{E}_U\left[\sum_x p_U(x)^2\right] = c_I\sum_x \mathrm{Tr}\left[(|x\rangle\langle x|)^{\otimes 2}\,\hat{\mathbb{I}}\right] + c_{\mathbb{F}}\sum_x \mathrm{Tr}\left[(|x\rangle\langle x|)^{\otimes 2}\,\mathbb{F}\right]$$

6. Evaluate the two sum-of-trace terms:
   - **Term 1:** $\sum_x \mathrm{Tr}\left[(|x\rangle\langle x|)^{\otimes 2}\,\hat{\mathbb{I}}\right] = \sum_x \mathrm{Tr}(|x\rangle\langle x|)^2 = \sum_x 1 = D.$
   - **Term 2:** $\sum_x \mathrm{Tr}\left[(|x\rangle\langle x|)^{\otimes 2}\,\mathbb{F}\right] = \sum_x \mathrm{Tr}(|x\rangle\langle x|\,|x\rangle\langle x|) = \sum_x 1 = D.$

7. Combine the terms:

$$\mathbb{E}_U\left[\sum_x p_U(x)^2\right] = D(c_I + c_{\mathbb{F}})$$

8. Add the two coefficients:

$$c_I + c_{\mathbb{F}} = \frac{D - \mathrm{Tr}(\hat{\rho}^2) + D\,\mathrm{Tr}(\hat{\rho}^2) - 1}{D(D^2-1)} = \frac{\mathrm{Tr}(\hat{\rho}^2) + 1}{D(D+1)}$$

9. Multiply by D:

$$\boxed{\mathbb{E}_U\left[\sum_x p_U(x)^2\right] = \frac{\mathrm{Tr}(\hat{\rho}^2)+1}{D+1}}$$

Questions

1. **Question:** How is this relation used to measure subsystem purity experimentally?

   Answer: This is the basis of randomized measurement protocols. An experimentalist applies random unitaries (from a 2-design) and measures in the computational basis to estimate the collision probability. They then invert this formula to obtain the purity $\mathrm{Tr}(\hat{\rho}^2)$ directly without the exponential cost of full state tomography.

2. **Question:** What is the physical meaning of the term "collision probability" in this context, and what are its boundary limits?

   Answer: The term originates from classical probability theory: for a distribution p, drawing two independent samples yields identical outcomes (a "collision") with probability $\sum_x p_x^2$. In our quantum setting, it is the probability that independent computational basis measurements on two copies of the rotated state $\hat{U}\hat{\rho}\hat{U}^\dagger$ yield matching outcomes. For a pure state, this Haar-averaged probability is maximized at $2/(D+1)$, where quantum projection noise prevents it from reaching 1 because the random rotation spreads the state. For a maximally mixed state, the state is invariant under unitaries, yielding the classical minimum of $1/D$.

Derivation Blueprint: Purity from Randomized Measurements

Board Layout:

- **Left Board Panel:** Write down the definition of the measurement probability $p_U(x)$. State the target purity identity.
- **Middle Panel:** Insert the doubled space trace. Write down the second Haar moment of $\hat{\rho}^{\otimes 2}$ and the corresponding coefficients.
- **Right Panel:** Evaluate the contractions with $\sum_x (|x\rangle \langle x|)^{\otimes 2}$. Show how the D factors cancel out to yield the final clean result.

Common Conceptual Pitfalls: A subtle point in the doubled trace contraction is why both $\mathrm{Tr}[\hat{\mathbb{I}}(|x\rangle \langle x|)^{\otimes 2}]$ and $\mathrm{Tr}[\mathbb{F}(|x\rangle \langle x|)^{\otimes 2}]$ evaluate to D. Since the product operator $(|x\rangle \langle x|)^{\otimes 2}$ is diagonal in the computational basis, contracting it with either the identity $\hat{\mathbb{I}}$ or the swap $\mathbb{F}$ sums the squared diagonal amplitudes, which is identically D. **Physical & Experimental Anchorage:**

- In actual quantum processors, such as Google's Sycamore superconducting device, measuring the purity of a mixed state $\hat{\rho}$ using standard quantum state tomography is experimentally intractable because it scales exponentially as 4^N. By using this randomized measurements protocol, Sycamore can estimate subsystem purity and Renyi entanglement entropy. The device applies random, local single-qubit unitaries, measures in the computational basis, and counts the frequency of matching bitstrings (the collision probability), providing direct access to the state's purity with dramatically reduced sample complexity.

Boundary Limit & Sanity Check: For a pure state ($\mathrm{Tr}(\hat{\rho}^2) = 1$), the average collision probability is exactly $2/(D+1)$, while for a maximally mixed state ($\mathrm{Tr}(\hat{\rho}^2) = 1/D$), it reduces to $1/D$. This matches the expected physical boundary where pure states concentrate and exhibit higher outcome collisions.

Numerical Sandbox: Haar-Randomized Measurement Verification

```
import numpy as np
D = 4  # Hilbert space dimension
# Define a mixed state density matrix
rho = np.diag([0.6, 0.3, 0.1, 0.0])
purity_analytical = np.real(np.trace(rho @ rho))
collision_prob_sum = 0.0
N_samples = 5000
for _ in range(N_samples):
    # Generate Haar-random unitary
    X = np.random.normal(size=(D, D)) + 1j * np.random.normal(size=(D, D))
    Q, R = np.linalg.qr(X)
    U = Q @ np.diag(np.diagonal(R) / np.abs(np.diagonal(R)))
    # Rotate density matrix
    rho_rot = U @ rho @ U.conj().T
    p_U = np.real(np.diagonal(rho_rot))
    collision_prob_sum += np.sum(p_U**2)
collision_prob_sim = collision_prob_sum / N_samples
collision_prob_analytical = (purity_analytical + 1) / (D + 1)
print(f"Purity Tr(rho^2): {purity_analytical:.6f}")
print(f"Simulated Collision Prob:  {collision_prob_sim:.6f}")
print(f"Analytical Collision Prob: {collision_prob_analytical:.6f}")
```

Card 11: Shadow Norm 3^k

Goal: Prove that the shadow norm of a k-local Pauli operator under random single-qubit Clifford measurements scales as 3^k.

Key Trick: Represent the shadow norm as the maximum variance of the classical shadow estimator, and show that it is governed by the probability of measuring in the correct basis.

Step-by-Step Derivation:

1. Under the Pauli classical shadow protocol, each of the N qubits is measured in a uniformly random Pauli basis $b_i \in \{X, Y, Z\}$.

2. The classical shadow estimator for a k-local Pauli string $\hat{P} = \bigotimes_{i=1}^{k} \hat{P}_i$ (acting nontrivially on k active qubits) is given by:
$$\hat{o}_m = \prod_{j=1}^{k} \delta_{b_j,P_j} \cdot 3 \cdot (-1)^{x_j}$$
where $x_j \in \{0, 1\}$ is the measurement outcome of qubit j in basis b_j.

3. The shadow norm is defined as the maximum second moment of the estimator over all possible states $\hat{\rho}$:
$$\|\hat{P}\|_{\text{sh}}^2 = \max_{\hat{\rho}} \mathbb{E}\left[\hat{o}_m^2\right]$$

4. Calculate the second moment of the single-shot estimator:
$$\hat{o}_m^2 = \prod_{j=1}^{k} \left(9\,\delta_{b_j,P_j}\right) = 9^k \prod_{j=1}^{k} \delta_{b_j,P_j}$$
since each matched qubit contributes a factor $3^2 = 9$ and $(-1)^{2x_j} = 1$ regardless of the measurement outcome x_j.

5. The expectation value is taken over the random choice of measurement bases b_j:

$$\mathbb{E}\left[\hat{o}_m^2\right] = \sum_{b_1,\ldots,b_k} P(b_1,\ldots,b_k) \prod_{j=1}^{k} \delta_{b_j,P_j} \cdot 9^k$$

6. Since the basis for each qubit is chosen independently and uniformly from $\{X, Y, Z\}$, the joint probability is:

$$P(b_1,\ldots,b_k) = \left(\frac{1}{3}\right)^k$$

7. The product of Kronecker deltas $\prod_{j=1}^{k} \delta_{b_j,P_j}$ is 1 only when $b_j = P_j$ for all j, which happens for exactly one basis configuration. Thus:

$$\mathbb{E}\left[\hat{o}_m^2\right] = \left(\frac{1}{3}\right)^k \cdot 9^k = 3^k$$

8. Since the second moment is completely independent of the state $\hat{\rho}$ (the measurement outcome signs squared to 1, leaving only the basis match probability), the maximum over all states is trivially:

$$\boxed{\|\hat{P}\|_{\text{sh}}^2 = 3^k}$$

Questions

1. **Question:** Why does the local shadow norm scale exponentially as 3^k for a k-local operator?

   Answer: The exponential scaling 3^k reflects the sample complexity of measuring a k-local observable. Since single-qubit measurement bases are chosen independently at random, the probability that all k measurements match the target Pauli string is $(1/3)^k$. The factor 3^k in the estimator compensates for this to ensure unbiasedness, which projects into a variance scaling as 3^k.

Derivation Blueprint: Shadow Norm 3^k

Board Layout:

- **Left Board Panel:** Write down the classical shadow local estimator formula. Show how the 3 factor is defined for a single qubit.
- **Middle Panel:** Square the estimator. Show why the sign term $(-1)^{2x_j}$ disappears.
- **Right Panel:** Write out the sum over the basis probability configurations. Highlight the $(1/3)^k$ match probability.

Common Conceptual Pitfalls: It is often surprising that the shadow reconstruction norm is completely state-independent. Because the square of the measurement outcomes $(\pm 1)^2 = 1$ is constant, the state-dependent amplitudes vanish entirely from the expectation value of the norm, leaving only the geometric probabilities of the random basis selection. **Boundary Limit & Sanity Check:** Comparing local and global operators reveals the severe sample complexity gap: a local Pauli observable acting on $k \ll N$ qubits has a shadow norm of 3^k, whereas a global Pauli string acting on all N qubits has a shadow norm of 3^N, showing the exponential barrier for global shadow reconstruction.

Active Recall Concept Test: Pauli vs. Global Clifford Shadow Norms

Question: Why does the shadow norm of a k-local Pauli operator scale as 3^k under single-qubit randomized Pauli measurements, whereas it scales as $\mathcal{O}(2^N)$ under global Clifford random measurements?

A) Because global Clifford twirling is a 3-design and averages out third-order moments of unitaries, while local twirling is only a 2-design.

B) Because under global Clifford random twirling, the twirling averages over global unitaries rather than single-qubit rotations, projecting the operator onto a single global depolarizing channel whose shadow norm scales as $D = 2^N$ for the identity and is highly efficient for global operators, whereas local Pauli shadows suffer from the product of independent basis choice probabilities $(1/3)^k$. **[Correct]**

C) Because local Pauli shadows are biased estimators of global quantum states, requiring a larger shadow norm correction factor.

Explanation: In local Pauli shadows, randomized single-qubit measurements are performed independently on each qubit, meaning the choice of measurement basis is local. The probability of choosing the correct basis for all k active qubits of a k-local operator is $(1/3)^k$. To keep the estimator unbiased, this requires scaling by 3^k, which directly carries into the shadow norm. In contrast, global Clifford twirling scrambles the state globally over a 2-design, which depolarizes any operator uniformly over the full 2^N-dimensional space. The resulting global shadow norm scales as $\mathrm{Tr}(\hat{P}^2) + D \approx 2^N$, which is independent of the locality k, making global shadows optimal for global operators but highly inefficient for local ones ($k \ll N$).

Numerical Sandbox: Single-Shot Pauli Shadow Estimator Variance

```
import numpy as np
# Number of qubits k in the operator
k = 2
D = 2**k
# Target Pauli operator (e.g. Z \otimes Z)
Z = np.diag([1.0, -1.0])
ZZ = np.kron(Z, Z)
# State vector (e.g., pure state |00>)
psi = np.zeros(D); psi[0] = 1.0
# Perform single-shot shadow measurements over many samples
N_samples = 100000
estimators = []
for _ in range(N_samples):
    # Randomly choose Pauli bases: 0->X, 1->Y, 2->Z
    bases = np.random.randint(0, 3, size=k)
    # Target operator ZZ has basis (2, 2). If choice matches, compute estimate
    if np.array_equal(bases, [2, 2]):
        # Measure in Z basis: outcome is bitstring from state |00>
        # For |00>, outcomes are 0 and 0 (corresponding to eigenvalue +1 for each)
        estimate = (3**k) * 1.0
    else:
        estimate = 0.0
    estimators.append(estimate)
second_moment = np.mean(np.array(estimators)**2)
print(f"k-local size: {k} | Match probability: (1/3)^{k} = {1/(3**k):.4f}")
print(f"Simulated Second Moment (Shadow Norm): {second_moment:.4f}")
print(f"Analytical Shadow Norm: {3**k:.4f}")
```

Card 15: QFI for Pure States and Metrology Limits

Goal: Prove that the Quantum Fisher Information (QFI) $\mathcal{F}_Q$ for a pure state $|\psi\rangle$ under a generator $\hat{G}$ is four times the variance of the generator:

$$\mathcal{F}_Q = 4\,\mathrm{Var}[\hat{G}]$$

Key Trick: Simplify the Symmetric Logarithmic Derivative (SLD) equation using the rank-1 projection properties of pure states.

Step-by-Step Derivation:

1. The Symmetric Logarithmic Derivative $\hat{L}$ is defined by:

$$\partial_\theta \hat{\rho} = \frac{1}{2}\left(\hat{L}\hat{\rho} + \hat{\rho}\hat{L}\right)$$

2. For a pure state $\hat{\rho} = |\psi\rangle\langle\psi|$, the derivative is:

$$\partial_\theta \hat{\rho} = -i[\hat{G}, \hat{\rho}]$$

3. Propose an ansatz for the Symmetric Logarithmic Derivative of a pure state:

$$\hat{L} = 2\partial_\theta \hat{\rho} = 2\left(|\partial_\theta \psi\rangle\langle\psi| + |\psi\rangle\langle\partial_\theta \psi|\right)$$

We verify this ansatz systematically. Since $\hat{\rho}^2 = \hat{\rho}$ for any pure state, taking the parameter derivative of both sides yields:

$$\partial_\theta \hat{\rho} = \partial_\theta(\hat{\rho}^2) = (\partial_\theta \hat{\rho})\hat{\rho} + \hat{\rho}(\partial_\theta \hat{\rho})$$

This is identical in structure to the SLD definition $\partial_\theta \hat{\rho} = \frac{1}{2}\left(\hat{L}\hat{\rho} + \hat{\rho}\hat{L}\right)$ when $\hat{L} = 2\partial_\theta \hat{\rho}$.

4. Verify that this ansatz satisfies the SLD equation by direct substitution. Using the parameter encoding state $|\psi\rangle = e^{-i\hat{G}\theta}|\psi_0\rangle$, the derivative is:

$$|\partial_\theta \psi\rangle = -i\hat{G}|\psi\rangle \implies \langle \partial_\theta \psi| = i\langle \psi|\hat{G}$$

The normalization $\langle \psi|\psi\rangle = 1$ implies that the derivative of the inner product is:

$$\partial_\theta \langle \psi|\psi\rangle = \langle \partial_\theta \psi|\psi\rangle + \langle \psi|\partial_\theta \psi\rangle = 0 \implies \mathrm{Re}\langle \psi|\partial_\theta \psi\rangle = 0$$

Since $\langle \psi|\partial_\theta \psi\rangle = -i\langle \psi|\hat{G}|\psi\rangle = -i\langle \hat{G}\rangle$, substituting our ansatz into the right-hand side of the SLD definition gives:

$$\frac{1}{2}(\hat{L}\hat{\rho} + \hat{\rho}\hat{L}) = (|\partial_\theta \psi\rangle\langle \psi| + |\psi\rangle\langle \partial_\theta \psi|)|\psi\rangle\langle \psi| + |\psi\rangle\langle \psi|(|\partial_\theta \psi\rangle\langle \psi| + |\psi\rangle\langle \partial_\theta \psi|)$$

Using $\langle \psi|\psi\rangle = 1$ and simplifying the terms:

$$= |\partial_\theta \psi\rangle\langle \psi| + |\psi\rangle\langle \partial_\theta \psi|\langle \psi|\psi\rangle\langle \psi| + |\psi\rangle\langle \psi|\partial_\theta \psi\rangle\langle \psi| + |\psi\rangle\langle \partial_\theta \psi|$$

$$= |\partial_\theta \psi\rangle\langle \psi| + |\psi\rangle\langle \partial_\theta \psi| + |\psi\rangle\left(\langle \psi|\partial_\theta \psi\rangle + \langle \partial_\theta \psi|\psi\rangle\right)\langle \psi|$$

Since the real part of $\langle \psi|\partial_\theta \psi\rangle$ is zero, the sum in the parenthesis vanishes: $\langle \psi|\partial_\theta \psi\rangle + \langle \partial_\theta \psi|\psi\rangle = -i\langle \hat{G}\rangle + i\langle \hat{G}\rangle = 0$. Thus, the equation simplifies exactly to the derivative:

$$\frac{1}{2}(\hat{L}\hat{\rho} + \hat{\rho}\hat{L}) = |\partial_\theta \psi\rangle\langle \psi| + |\psi\rangle\langle \partial_\theta \psi| = \partial_\theta \hat{\rho}$$

This confirms that the ansatz $\hat{L} = 2\partial_\theta \hat{\rho}$ is the unique and exact solution for the SLD of a pure state!

5. The QFI is defined as:

$$\mathcal{F}_Q = \mathrm{Tr}(\hat{\rho}\hat{L}^2) = \langle \psi|\hat{L}^2|\psi\rangle$$

6. Substitute $\hat{L}|\psi\rangle = -2i\left(\hat{G} - \langle \hat{G}\rangle\right)|\psi\rangle$:

$$\mathcal{F}_Q = \left(\langle \psi|\hat{L}\right)\left(\hat{L}|\psi\rangle\right) = 4\langle \psi|\left(\hat{G} - \langle \hat{G}\rangle\right)^2|\psi\rangle = \boxed{4\left(\langle \hat{G}^2\rangle - \langle \hat{G}\rangle^2\right) = 4\,\mathrm{Var}[\hat{G}]}$$

Questions

1. **Question:** What are the fundamental limits of parameter estimation sensitivity in quantum metrology?

   Answer: For an uncorrelated product state of N qubits under a local generator $\hat{G} = \sum_j \hat{g}_j$, the variance scales as $\mathcal{O}(N)$, giving the Standard Quantum Limit (SQL) $\mathcal{F}_Q \sim N$. For a maximally entangled GHZ state, the variance scales as $\mathcal{O}(N^2)$, achieving the Heisenberg Limit $\mathcal{F}_Q \sim N^2$, which represents the ultimate limit of quantum precision allowed by quantum mechanics.

Derivation Blueprint: QFI for Pure States

Board Layout:

- **Left Board Panel:** Define the parameter-encoding operation and the Symmetric Logarithmic Derivative (SLD) equation.
- **Middle Panel:** Write down the pure state SLD ansatz and verify it step-by-step.
- **Right Panel:** Calculate the QFI trace and obtain the final variance formula. List the SQL and Heisenberg limits in a summary box.

Common Conceptual Pitfalls: The relation $\langle\psi|\partial_\theta\psi\rangle = -i\langle G\rangle$ requires careful execution of the parametric derivative. Evaluating the derivative of the parameter-encoded state vector $\partial_\theta(e^{-i\hat{G}\theta}\,|\psi_0\rangle) = -i\hat{G}e^{-i\hat{G}\theta}\,|\psi_0\rangle = -i\hat{G}\,|\psi\rangle$ makes the appearance of the expectation value clear. **Boundary Limit & Sanity Check:** If the probe state $|\psi\rangle$ is an eigenstate of the generator $\hat{G}$ (so $\hat{G}\,|\psi\rangle = g\,|\psi\rangle$), the variance vanishes identically ($\mathrm{Var}[\hat{G}] = 0$), yielding zero metrological sensitivity ($\mathcal{F}_Q = 0$). Parameter shifts under $\hat{G}$ only apply an unobservable global phase.

Active Recall Concept Test: Eigenstate Insensitivity and Parameter Encoding

Question: Why does the Quantum Fisher Information $\mathcal{F}_Q$ vanish identically if the probe state $|\psi\rangle$ is an eigenstate of the metrological generator $\hat{G}$?

A) Because eigenstates are stationary states under time evolution, preventing any phase accumulation.

B) Because the parameterized state is $|\psi(\theta)\rangle = e^{-i\hat{G}\theta}\,|\psi\rangle = e^{-ig\theta}\,|\psi\rangle$. The parameter θ is encoded purely as an overall global phase, which has no observable physical consequences, meaning the state vector itself is static in projective Hilbert space. Thus, the variance $\mathrm{Var}[\hat{G}] = \langle\psi|\,\hat{G}^2\,|\psi\rangle - \langle\psi|\,\hat{G}\,|\psi\rangle^2 = g^2 - g^2 = 0$, giving zero QFI. **[Correct]**

C) Because the Symmetric Logarithmic Derivative is undefined for eigenstates, leading to a singular metrological variance.

Explanation: Quantum metrology operates by measuring the change in a state vector induced by parameter shifts: $|\psi(\theta)\rangle = e^{-i\hat{G}\theta}\,|\psi_0\rangle$. If $|\psi_0\rangle$ is an eigenstate of $\hat{G}$ with eigenvalue g, then $|\psi(\theta)\rangle = e^{-ig\theta}\,|\psi_0\rangle$. The parameter θ appears only as a global phase factor. In quantum mechanics, states that differ by a global phase are identical in projective Hilbert space and yield identical expectation values for any observable, meaning no measurement can estimate θ. Mathematically, this manifests as a vanishing variance $\mathrm{Var}[\hat{G}] = 0 \implies \mathcal{F}_Q = 0$.

Numerical Sandbox: Quantum Fisher Information SLD Variance Verification

```
import numpy as np
# Number of qubits in metrology system
N = 3
D = 2**N
# Local generator G = \sum_i Z_i (total spin projection)
z = np.diag([1.0, -1.0])
G = np.zeros((D, D))
for i in range(N):
    # Construct Z_i acting on qubit i
    op = np.eye(1)
    for j in range(N):
        if j == i:
            op = np.kron(op, z)
        else:
            op = np.kron(op, np.eye(2))
    G += op
# State vector: maximally entangled GHZ state (|000> + |111>)/sqrt(2)
psi = np.zeros(D)
psi[0] = 1.0 / np.sqrt(2)
psi[D-1] = 1.0 / np.sqrt(2)
# Compute variance of generator G
mean_G = np.real(psi.conj().T @ G @ psi)
mean_G2 = np.real(psi.conj().T @ (G @ G) @ psi)
variance_G = mean_G2 - mean_G**2
qfi_analytical = 4 * variance_G
print(f"GHZ State metrology with N={N} qubits")
print(f"Expectation <G>: {mean_G:.4f} | <G^2>: {mean_G2:.4f}")
print(f"Variance Var(G): {variance_G:.4f}")
print(f"QFI 4*Var(G) (reaches Heisenberg limit N^2): {qfi_analytical:.4f}")
```

Card 19: Unbiasedness of Pauli Classical Shadows

Goal: Prove that the local classical shadow reconstruction channel $\mathcal{M}(\hat{\rho}) = \mathbb{E}_U[\hat{U}^\dagger \mathcal{M}_{\text{comp}}(\hat{U}\hat{\rho}\hat{U}^\dagger)\hat{U}]$ averages to $(3\hat{\rho} - \hat{I})/2$ for a single qubit, and show that the inverse channel is $\mathcal{M}^{-1}(\hat{\rho}) = 3\hat{\rho} - \hat{I}$.

Key Trick: Write the computational measurement channel explicitly in terms of projectors, and apply the second Haar moment to evaluate the twirled channel.

Step-by-Step Derivation:

1. The computational measurement channel on a single qubit is:

$$\mathcal{M}_{\text{comp}}(\hat{\sigma}) = \sum_{x\in\{0,1\}} \langle x| \hat{\sigma} |x\rangle |x\rangle \langle x| = \langle 0| \hat{\sigma} |0\rangle |0\rangle \langle 0| + \langle 1| \hat{\sigma} |1\rangle |1\rangle \langle 1|$$

2. The Pauli shadow measurement twirls this channel over the single-qubit Clifford group (or equivalently, Haar unitaries on $D = 2$ since it is a 2-design):

$$\mathcal{M}(\hat{\rho}) = \mathbb{E}_U \left[\hat{U}^\dagger \mathcal{M}_{\text{comp}} \left(\hat{U}\hat{\rho}\hat{U}^\dagger\right) \hat{U}\right] = \sum_{x\in\{0,1\}} \mathbb{E}_U \left[\left(\langle x| \hat{U}\hat{\rho}\hat{U}^\dagger |x\rangle\right) \hat{U}^\dagger |x\rangle \langle x| \hat{U}\right]$$

3. Since the channel is linear, we can decompose $\hat{\rho}$ into its identity and traceless Pauli parts: $\hat{\rho} = \frac{\hat{I}}{2} + \vec{r} \cdot \vec{\sigma}$.

- **Identity twirl:** Since $\mathcal{M}(\hat{I}/2) = \hat{I}/2$ (every step is trace-preserving):

$$\mathcal{M}(\hat{I}) = \hat{I}$$

- **Pauli twirl:** Let $\hat{P}$ be any traceless Pauli operator. By Card 1, the twirl over $\hat{P}$ maps it to a multiple of itself (by symmetry of $SU(2)$ twirling):

$$\mathcal{M}(\hat{P}) = c\,\hat{P}$$

4. To find the coefficient c, evaluate the twirl of $\hat{\sigma}_z$ explicitly: Choosing U from the single-qubit Clifford group twirls $\hat{\sigma}_z$. The Pauli operators are mapped to X, Y, Z with equal probability $1/3$, and computational measurement yields ± 1: The twirled channel acts on any state $\hat{\rho}$ as:

$$\mathcal{M}(\hat{\rho}) = \frac{\hat{\rho} + \hat{I}}{3} \implies 3\mathcal{M}(\hat{\rho}) - \hat{I} = \hat{\rho}$$

Thus, the inverse channel $\mathcal{M}^{-1}(\hat{\sigma})$ is:

$$\boxed{\mathcal{M}^{-1}(\hat{\sigma}) = 3\hat{\sigma} - \hat{I}}$$

Questions

1. **Question:** How does the single-qubit shadow reconstruction extend to N qubits?

   Answer: Because single-qubit Clifford unitaries are applied independently to each qubit, the global reconstruction channel factorizes as a tensor product: $\mathcal{M}^{(N)} = \bigotimes_{j=1}^{N} \mathcal{M}^{(1)}$. The inverse channel is simply the tensor product of the local inverse channels: $\mathcal{M}^{-1} = \bigotimes_{j=1}^{N} \left(3\hat{\sigma}_j - \hat{I}\right)$, making multi-qubit shadow reconstruction highly parallelizable and efficient.

Derivation Blueprint: Unbiasedness of Pauli Classical Shadows

Board Layout:

- **Left Board Panel:** Write down the definition of the twirled channel $\mathcal{M}(\hat{\rho})$. Explain the qubit-by-qubit factorization.
- **Middle Panel:** Decompose the state $\hat{\rho}$ into its identity and traceless Pauli parts. Show how the twirl preserves the identity and scales the Pauli terms by $1/3$.
- **Right Panel:** Combine the parts to solve for the inverse channel. Show the final tensor product formula for N qubits.

Common Conceptual Pitfalls: The scaling factor of $1/3$ for the Pauli terms in local shadow estimation corresponds to the probability of selecting the matching measurement basis. For a single-qubit Pauli operator, a random measurement matches the correct basis in exactly one out of the three possible choices $\{X, Y, Z\}$. **Boundary Limit & Sanity Check:** Measuring the state $|0\rangle$ in the Z basis yields the single-shot classical shadow snapshot $\hat{\rho}^{(\text{sh})} = 3\,|0\rangle\langle 0| - \hat{I}$. While the snapshot itself is non-physical with eigenvalues 2 and -1, taking its statistical expectation value over infinite shots converges exactly to the physical density matrix.

Numerical Sandbox: Pauli Classical Shadow Reconstruction Verification

```
import numpy as np
# Define a pure single-qubit density matrix rho
theta = np.pi/6; phi = np.pi/4
state = np.array([np.cos(theta/2), np.exp(1j*phi)*np.sin(theta/2)])
rho = np.outer(state, np.conj(state))
# Pauli matrices
X = np.array([[0.0, 1.0], [1.0, 0.0]])
Y = np.array([[0.0, -1j], [1j, 0.0]])
Z = np.array([[1.0, 0.0], [0.0, -1.0]])
Paulis = [X, Y, Z]

reconstruction_sum = np.zeros((2, 2), dtype=complex)
N_samples = 100000
for _ in range(N_samples):
    # Random Choice of basis: 0->X, 1->Y, 2->Z
    b = np.random.randint(0, 3)
    P = Paulis[b]
    # Compute basis state transition probabilities
    eigenvals, eigenvecs = np.linalg.eigh(P)
    # Projective measurement outcome
    p_0 = np.real(np.conj(eigenvecs[:, 0]).T @ rho @ eigenvecs[:, 0])
    outcome_idx = 0 if np.random.rand() < p_0 else 1
    measured_state = eigenvecs[:, outcome_idx]
    # Single-shot shadow snapshot: 3 * |s><s| - I
    snapshot = 3.0 * np.outer(measured_state, np.conj(measured_state)) - np.eye(2)
    reconstruction_sum += snapshot
reconstruction_avg = reconstruction_sum / N_samples
print(f"True Density Matrix rho:\n{np.real(rho)}")
print(f"Reconstructed Shadow rho (averaging snapshots converges to true):\n{np.real(reconst
```

7 Quantum Machine Learning & Complexity

Exercise 7.1: Barren Plateau Scaling from the Weingarten Formula

Motivation:

The **barren plateau** phenomenon is the central obstruction to training deep parametrized quantum circuits. When a circuit forms a unitary 2-design, the variance of any cost function gradient decays **exponentially** in the number of qubits N. This result follows directly from the **Weingarten calculus** of Chapter 3 — the same machinery that computes Haar averages of polynomial expressions in U and $U^\dagger$.

This exercise makes the connection explicit: we compute $\mathbb{E}[C^2]$ by applying the $k = 2$ Weingarten formula term by term, showing how the permutation sums, Kronecker deltas, and trace contractions conspire to produce the exponentially small variance $\sim N \cdot 2^{-(N+1)}$.

Preliminaries / Toolbox

Before solving this exercise, the student needs the following machinery from earlier chapters.

1. Unitary 2-design (Definition 5.1): A circuit ensemble $\{U(\boldsymbol{\theta})\}$ is a 2-design if, for any polynomial $P(U, U^\dagger)$ of degree ≤ 2 in each, $\mathbb{E}_{\boldsymbol{\theta}}[P] = \mathbb{E}_{\text{Haar}}[P]$. In practice: sufficiently deep random brickwork circuits form approximate 2-designs.

2. The $k = 2$ Weingarten formula (Theorem 3.2):

$$\mathbb{E}\big[U_{i_1 j_1} U_{i_2 j_2} \overline{U}_{\ell_1 m_1} \overline{U}_{\ell_2 m_2}\big] = \sum_{\pi,\sigma \in S_2} \mathrm{Wg}(\pi^{-1}\sigma,\, D) \prod_{a=1}^{2} \delta_{i_a, \ell_{\pi(a)}}\, \delta_{j_a, m_{\sigma(a)}}.$$

The Weingarten function values at $k = 2$:

$$\mathrm{Wg}(\mathrm{id}, D) = \frac{1}{D^2 - 1}, \qquad \mathrm{Wg}\big((12), D\big) = \frac{-1}{D(D^2 - 1)}.$$

3. Tracelessness and variance: If $\mathrm{Tr}(O) = 0$, then $\mathbb{E}[C] = \mathrm{Tr}(O)/D = 0$ by the first moment formula, so $\mathrm{Var}[C] = \mathbb{E}[C^2]$.

4. Parameter-shift rule: The gradient $\partial C/\partial\theta_i$ for a Pauli rotation gate inherits the variance scaling of C up to a constant prefactor $\sim 1/2$.

5. Pauli trace rules: $\mathrm{Tr}(Z_j) = 0$, $\mathrm{Tr}(Z_j^2) = D$, $\mathrm{Tr}(Z_j Z_k) = 0$ for $j \neq k$.

Notation: $D = 2^N$ (Hilbert space dimension), $|0\rangle = |0^N\rangle$ (computational zero state), O (cost observable).

Problem Statement:

A parametrized circuit on N qubits forms a unitary 2-design. The cost function is

$$C(\boldsymbol{\theta}) = \langle 0 | U^\dagger O\, U | 0 \rangle,$$

with $O = \sum_{j=1}^{N} Z_j$ (traceless, $\mathrm{Tr}(O) = 0$, $\mathrm{Tr}(O^2) = ND$, $D = 2^N$).

(a) Using the $k = 2$ Weingarten formula, compute $\mathrm{Var}[C] = \mathbb{E}[C^2]$ (since $\mathbb{E}[C] = 0$).

(b) Determine the measurement shot cost ν to resolve the gradient.

(c) For per-gate error $p = 10^{-3}$, estimate the critical depth L_{crit}.

Pedantic Step-by-Step Solution:

Part (a): Computing $\mathbb{E}[C^2]$ via Weingarten calculus

Step 1: Write C^2 in matrix elements

$$C = \langle 0 | U^\dagger O U | 0 \rangle = \sum_{a,b} \overline{U}_{a,0}\, O_{ab}\, U_{b,0}.$$

Therefore:

$$C^2 = \sum_{a_1, b_1, a_2, b_2} O_{a_1 b_1}\, O_{a_2 b_2}\, U_{b_1,0}\, U_{b_2,0}\, \overline{U}_{a_1,0}\, \overline{U}_{a_2,0}.$$

This is a $k = 2$ moment with indices: $i_1 = b_1$, $j_1 = 0$, $i_2 = b_2$, $j_2 = 0$, $\ell_1 = a_1$, $m_1 = 0$, $\ell_2 = a_2$, $m_2 = 0$.

Step 2: Apply the Weingarten formula

Substituting into the $k = 2$ formula, the j-deltas all become $\delta_{0,0_{\sigma(a)}} = 1$ for both permutations (since all j's and m's are 0):

$$\mathbb{E}[C^2] = \sum_{\pi,\sigma} \mathrm{Wg}(\pi^{-1}\sigma) \underbrace{\left(\prod_a \delta_{j_a, m_{\sigma(a)}}\right)}_{=1} \sum_{a_1,b_1,a_2,b_2} O_{a_1b_1} O_{a_2b_2} \prod_a \delta_{b_a, a_{\pi(a)}}.$$

Step 3: Evaluate the four (π, σ) contributions

Case $\pi = \mathrm{id}$: $\delta_{b_1,a_1}\,\delta_{b_2,a_2}$.

$$\sum O_{a_1a_1} O_{a_2a_2} = \mathrm{Tr}(O)^2 = 0. \qquad \text{(traceless } O\text{)}$$

Case $\pi = (12)$: $\delta_{b_1,a_2}\,\delta_{b_2,a_1}$.

$$\sum O_{a_1a_2} O_{a_2a_1} = \mathrm{Tr}(O^2) = ND.$$

Step 4: Assemble with Weingarten weights

The $\pi = \mathrm{id}$ contribution vanishes. For $\pi = (12)$, sum over σ:

$$\mathbb{E}[C^2] = \mathrm{Tr}(O^2)\big[\mathrm{Wg}((12)^{-1} \cdot \mathrm{id}) + \mathrm{Wg}((12)^{-1} \cdot (12))\big] = ND\,\big[\mathrm{Wg}((12)) + \mathrm{Wg}(\mathrm{id})\big].$$

$$= ND\left[\frac{-1}{D(D^2-1)} + \frac{1}{D^2-1}\right] = ND \cdot \frac{D-1}{D(D^2-1)} = ND \cdot \frac{1}{D(D+1)} = \frac{N}{D+1}.$$

Step 5: Final result

$$\boxed{\mathbb{E}[C^2] = \frac{N}{D+1} \approx \frac{N}{D} = N \cdot 2^{-N}.}$$

The gradient variance (from the parameter-shift rule, Step 4 in Preliminaries):

$$\mathrm{Var}\left[\frac{\partial C}{\partial \theta_i}\right] \sim \frac{N}{2D} = N \cdot 2^{-(N+1)}.$$

The gradient is exponentially suppressed — this is the barren plateau.

Part (b): Shot cost

Each shot gives a gradient estimate with shot variance $\sim N/(2\nu)$. Resolvability requires:

$$\frac{N}{2\nu} < \frac{N}{2D} \quad \Rightarrow \quad \boxed{\nu > D = 2^N.}$$

Part (c): Critical depth

Fidelity decays as $F \sim e^{-pL}$. Noise dominates when $pL \sim 1$:

$$\boxed{L_{\text{crit}} \sim 1/p = 10^3 \text{ layers.}}$$

Exercise 7.2: Local vs. Global Cost Functions in Deep Circuits

Motivation:

Not all cost functions are equally trainable. A **global** cost (e.g., projecting onto the target state) and a **local** cost (e.g., measuring single-qubit observables) both suffer from barren plateaus in 2-design circuits — but the **rate** of exponential decay is different. The local cost has gradient variance $\sim 1/(N \cdot 2^N)$, while the global cost decays as $\sim 1/2^{2N}$ — an exponential advantage for the local cost. This exercise derives both scalings via the same Weingarten calculus and quantifies the advantage.

Preliminaries / Toolbox

Before diving into the solution, recall the following concepts:

1. Cost Functions in VQAs: The objective is $C(\boldsymbol{\theta}) = \langle 0|U^\dagger(\boldsymbol{\theta}) O U(\boldsymbol{\theta})|0\rangle$.

2. Global vs. Local Observables: A global cost relies on an observable that acts non-trivially on all qubits, e.g., $O_G = |0\rangle\langle 0|^{\otimes N}$. A local cost is a sum of local terms, e.g., $O_L = \frac{1}{N}\sum_j |0\rangle\langle 0|_j$.

3. Barren Plateaus (2-design): If $U(\boldsymbol{\theta})$ forms a 2-design, the variance of the cost function is $\mathrm{Var}[C] = \frac{\mathrm{Tr}(O^2)}{D(D+1)}$ where $D = 2^N$.

4. Trace properties: $\mathrm{Tr}(A \otimes B) = \mathrm{Tr}(A)\,\mathrm{Tr}(B)$. The dimension is $D = 2^N$.

Problem Statement:

For a 2-design circuit on N qubits ($D = 2^N$), compare two traceless cost observables:

$$O_G = |0\rangle\langle 0|^{\otimes N} - \frac{I}{D} \qquad \text{(global projector, traceless)},$$

$$O_L = \frac{1}{N}\sum_{j=1}^{N} Z_j \qquad \text{(normalized local observable, traceless)}.$$

(a) Compute $\mathrm{Tr}(O_G^2)$ and $\mathrm{Tr}(O_L^2)$.
(b) Using the Weingarten formula, determine the gradient variance scaling for both costs.
(c) Compute the ratio $\mathrm{Var}_L/\mathrm{Var}_G$ and interpret.

Pedantic Step-by-Step Solution:

Part (a): Computing $\mathrm{Tr}(O^2)$ for each cost

Global cost O_G:

$$O_G^2 = \left(|0\rangle\langle 0|^{\otimes N} - \frac{I}{D}\right)^2 = |0\rangle\langle 0|^{\otimes N} - \frac{2}{D}|0\rangle\langle 0|^{\otimes N} + \frac{I}{D^2}.$$

$$\mathrm{Tr}(O_G^2) = 1 - \frac{2}{D} + \frac{1}{D} = 1 - \frac{1}{D} = \frac{D-1}{D}.$$

For large D: $\mathrm{Tr}(O_G^2) \approx 1$.

Local cost O_L:

$$O_L^2 = \frac{1}{N^2}\sum_{j,k} Z_j Z_k.$$

- **Diagonal** ($j = k$): $\mathrm{Tr}(Z_j^2) = D$. There are N such terms.
- **Off-diagonal** ($j \neq k$): $Z_j Z_k$ is a tensor product with two Z factors and $N-2$ identities. Since $\mathrm{Tr}(Z) = 0$, $\mathrm{Tr}(Z_j Z_k) = 0$.

$$\mathrm{Tr}(O_L^2) = \frac{1}{N^2}\cdot N \cdot D = \frac{D}{N}.$$

Part (b): Gradient variance via Weingarten calculus

From Exercise 7.1, we derived the exact Weingarten result for any traceless observable O:

$$\mathbb{E}[C^2] = \frac{\mathrm{Tr}(O^2)}{D(D+1)}.$$

(This followed from the $k = 2$ Weingarten formula: the $\pi = \mathrm{id}$ contribution vanished because $\mathrm{Tr}(O) = 0$, and the $\pi = (12)$ contribution gave $\mathrm{Tr}(O^2) \times [\mathrm{Wg}(\mathrm{id}) + \mathrm{Wg}((12))] = \mathrm{Tr}(O^2)/(D(D+1))$.)
Applying this to both costs:
Global:

$$\mathrm{Var}_G = \frac{(D-1)/D}{D(D+1)} = \frac{D-1}{D^2(D+1)} \approx \frac{1}{D^2} = 2^{-2N}.$$

Local:

$$\mathrm{Var}_L = \frac{D/N}{D(D+1)} = \frac{1}{N(D+1)} \approx \frac{1}{ND} = \frac{1}{N \cdot 2^N}.$$

Part (c): Ratio and interpretation

$$\frac{\mathrm{Var}_L}{\mathrm{Var}_G} = \frac{D^2(D+1)}{D \cdot N \cdot (D+1)} \cdot \frac{D}{D-1} \approx \frac{D}{N} = \frac{2^N}{N}.$$

$$\boxed{\frac{\mathrm{Var}_L}{\mathrm{Var}_G} \approx \frac{2^N}{N} \gg 1.}$$

The local cost has an **exponentially larger** gradient than the global cost.
Physical interpretation:

- The global cost $O_G = |0\rangle\langle 0|^{\otimes N}$ is a rank-1 projector — it asks "are you *exactly* in the target state?" Since $\mathrm{Tr}(O_G^2) \approx 1$ is $O(1)$, the Weingarten formula gives $\mathrm{Var}_G \sim 1/D^2$: the cost landscape is exponentially flat.
- The local cost $O_L = \frac{1}{N}\sum Z_j$ sums single-qubit measurements. Since $\mathrm{Tr}(O_L^2) \sim D/N$, the variance gets a factor D/N boost. Each local term "sees" only part of the system, so the information is not diluted over the full D-dimensional Hilbert space.
- Both costs still have barren plateaus (exponential decay), but the exponent is halved for the local cost: 2^{-N} vs 2^{-2N}. In practice, this can mean the difference between trainable ($N \sim 20$) and untrainable.

Exercise 7.3: Barren Plateaus through the Quantum Fisher Information

1. The Geometric Picture

Consider a parameterized circuit $U(\boldsymbol{\theta})$ preparing the state $|\psi(\boldsymbol{\theta})\rangle = U(\boldsymbol{\theta})|0\rangle$. The cost function is $C(\boldsymbol{\theta}) = \langle\psi(\boldsymbol{\theta})|O|\psi(\boldsymbol{\theta})\rangle$ for some observable O.
The **Quantum Fisher Information** (QFI) with respect to the parameter θ_k measures how sensitively the quantum state responds to infinitesimal changes in θ_k:

$$F_Q(\theta_k) = 4\left(\langle\partial_k\psi|\partial_k\psi\rangle - |\langle\psi|\partial_k\psi\rangle|^2\right).$$

This is precisely **four times the Fubini-Study metric** g_{kk}^{FS} on the projective Hilbert space $\mathbb{CP}^{D-1}$:

$$F_Q(\theta_k) = 4\, g_{kk}^{\mathrm{FS}}.$$

When $F_Q(\theta_k) \to 0$, the state $|\psi\rangle$ becomes **insensitive** to θ_k: an infinitesimal change in θ_k produces a negligible change in the quantum state. If the state doesn't change, the cost function doesn't change, and the gradient vanishes.
This provides a purely geometric, observable-independent explanation of barren plateaus: the parameter landscape becomes **metrically flat** in the Fubini-Study sense.

2. Mathematical Toolbox

Tool 1 — QFI for pure states: For $|\psi(\theta)\rangle = e^{-i\theta G}|\psi_0\rangle$ with generator G,

$$F_Q(\theta) = 4\,\mathrm{Var}_{|\psi_0\rangle}(G) = 4\left(\langle G^2\rangle - \langle G\rangle^2\right).$$

Tool 2 — Cramér-Rao bound on gradients: The variance of $\partial C/\partial\theta_k$ over the parameter distribution is bounded by:

$$\mathrm{Var}\left[\frac{\partial C}{\partial \theta_k}\right] \leq \|O\|^2 \cdot F_Q(\theta_k).$$

If $F_Q \to 0$, the gradient is forced to vanish regardless of the observable O.

Tool 3 — Haar first moment (Weingarten $k = 1$):

$$\mathbb{E}_U[U^\dagger A\, U] = \frac{\mathrm{Tr}(A)}{D}\, I.$$

Tool 4 — Haar second moment (Weingarten $k = 2$):

$$\mathbb{E}_U[U^\dagger A\, U \otimes U^\dagger B\, U] = \alpha(A,B)\, I \otimes I + \beta(A,B)\,\mathrm{SWAP},$$

where $\alpha = \frac{D\ \mathrm{Tr}(A)\,\mathrm{Tr}(B) - \mathrm{Tr}(AB)}{D(D^2-1)}$ and $\beta = \frac{D\ \mathrm{Tr}(AB) - \mathrm{Tr}(A)\,\mathrm{Tr}(B)}{D(D^2-1)}$.

3. Exercise Statement

Consider a circuit $U(\theta) = V_2\, e^{-i\theta G}\, V_1$ where V_1, V_2 are Haar-random and G is a fixed Hermitian generator (e.g. a single Pauli).

(a) Show that the QFI with respect to θ is $F_Q(\theta) = 4\,\mathrm{Var}_{|\psi_1\rangle}(G)$ where $|\psi_1\rangle = V_1|0\rangle$.

(b) Show that for Haar-random V_1, the expected QFI is:

$$\mathbb{E}_{V_1}[F_Q] = \frac{4}{D+1}\left(\mathrm{Tr}(G^2) - \frac{(\mathrm{Tr}\ G)^2}{D}\right).$$

(c) For a traceless Pauli generator G with $G^2 = I$ and $\mathrm{Tr}(G) = 0$, show that $\mathbb{E}[F_Q] = \frac{4D}{D+1} \xrightarrow{D\to\infty} 4$. Compare this with the gradient variance $\sim N/D$.

(d) Using the pointwise Cauchy–Schwarz bound $|\partial_\theta C|^2 \leq \mathrm{Var}_\psi[O]\, F_Q$ and the Haar average $\mathbb{E}_\psi[\mathrm{Var}_\psi[O]] = \mathrm{Tr}(O^2)/D$, show that the gradient variance obeys $\mathrm{Var}[\partial_\theta C] \lesssim \mathrm{Tr}(O^2)/D$. This is looser than the exact second-moment result $\mathrm{Var}[\partial_\theta C] \approx \mathrm{Tr}(O^2)/(2D^2)$ of the main text, but it gives exponential suppression by the same observable-projection mechanism.

4. Step-by-Step Derivation

Part (a): QFI from the variance of the generator

At $\theta = 0$, the state is $|\psi_1\rangle = V_1|0\rangle$ and the parameter acts via $e^{-i\theta G}$. The QFI formula for pure states gives:

$$F_Q = 4\left(\langle\psi_1|G^2|\psi_1\rangle - \langle\psi_1|G|\psi_1\rangle^2\right) = 4\,\mathrm{Var}_{|\psi_1\rangle}(G). \quad \checkmark$$

Note: the post-rotation V_2 does not affect F_Q because the Fubini-Study metric is invariant under global unitaries. The QFI captures the intrinsic sensitivity of the state manifold, not the measurement basis.

Part (b): Haar-averaged QFI

We need $\mathbb{E}_{V_1}[\langle\psi_1|G^2|\psi_1\rangle]$ and $\mathbb{E}_{V_1}[\langle\psi_1|G|\psi_1\rangle^2]$.

First term (using Tool 3, $k = 1$):

$$\mathbb{E}[\langle G^2\rangle] = \mathrm{Tr}\left(\frac{I}{D}G^2\right) = \frac{\mathrm{Tr}(G^2)}{D}.$$

Second term (using Tool 4, $k=2$): Since $\langle G\rangle^2 = \langle\psi_1|G|\psi_1\rangle^2$, this is a second-moment integral:

$$\mathbb{E}[\langle G\rangle^2] = \frac{\mathrm{Tr}(G)^2 + \mathrm{Tr}(G^2)}{D(D+1)}.$$

Combining:

$$\mathbb{E}[F_Q] = 4\left(\frac{\mathrm{Tr}(G^2)}{D} - \frac{\mathrm{Tr}(G)^2 + \mathrm{Tr}(G^2)}{D(D+1)}\right) = \frac{4}{D+1}\left(\mathrm{Tr}(G^2) - \frac{\mathrm{Tr}(G)^2}{D}\right). \quad \checkmark$$

Part (c): Pauli generator and the trainability paradox

For a traceless Pauli G ($G^2 = I$, $\mathrm{Tr}(G) = 0$, $\mathrm{Tr}(G^2) = D$):

$$\mathbb{E}[F_Q] = \frac{4D}{D+1} \xrightarrow{D\to\infty} 4.$$

This is an $\mathcal{O}(1)$ constant! The QFI does **not** vanish exponentially. The quantum state *is* sensitive to the parameter θ.

But recall that the gradient variance $\mathrm{Var}[\partial C/\partial\theta] \sim N/D \to 0$ exponentially. This seems contradictory: the state moves in Hilbert space, but the cost doesn't change?

Resolution: The state $|\psi(\theta)\rangle$ moves along a geodesic in $\mathbb{CP}^{D-1}$, but the observable O projects onto a tiny subspace. The motion is "perpendicular" to the cost landscape. The QFI tells us the state moves; the barren plateau tells us it moves in a direction that doesn't matter for the observable.

This is the **curse of expressibility**: the circuit explores exponentially many directions, but only $\mathcal{O}(1)$ of them align with a given cost function.

Part (d): The observable-weighted QFI bound

The gradient is $\partial_\theta C = 2\,\mathrm{Im}\langle\psi|O|\partial_\theta\psi\rangle$. Cauchy–Schwarz gives the *pointwise* bound on the squared gradient:

$$|\partial_\theta C|^2 \le \mathrm{Var}_\psi[O]\cdot F_Q,$$

so $\mathrm{Var}_\theta[\partial_\theta C] = \mathbb{E}_\theta[|\partial_\theta C|^2] \le \mathbb{E}[\mathrm{Var}_\psi[O]]\,F_Q$ (the mean gradient vanishes). For Haar-random states, $\mathbb{E}[\mathrm{Var}_\psi[O]] = \mathrm{Tr}(O^2)/D$. Combined with $F_Q \sim \mathcal{O}(1)$:

$$\mathrm{Var}[\partial_\theta C] \lesssim \frac{\mathrm{Tr}(O^2)}{D}\cdot\frac{4D}{D+1} \sim \frac{4\,\mathrm{Tr}(O^2)}{D}.$$

For a global observable with $\mathrm{Tr}(O^2) = \mathcal{O}(1)$, this gives exponential suppression $\sim 1/D$. (This Cauchy–Schwarz bound is looser than the exact $\mathrm{Tr}(O^2)/(2D^2)$ by a factor D; both vanish exponentially.)

5. Physical Interpretation: Geometry vs Trainability

The QFI analysis separates the barren plateau into two distinct effects:

— Quantity — Scaling — Meaning — —:—:—:— — $F_Q(\theta)$ — $\mathcal{O}(1)$ — The state *does* move in Hilbert space — — $\mathrm{Var}_{|\psi\rangle}(O)$ — $\mathcal{O}(1/D)$ — The observable *cannot distinguish* nearby states — — $\mathrm{Var}[\partial C/\partial\theta]$ — $\mathcal{O}(1/D)$ — The cost landscape is **flat** —

The barren plateau is not a failure of the parameterization to move the state—it is a failure of the observable to *detect* the motion. The Fubini-Study geodesic ball explored by the parameter θ has an exponentially large volume, but the "shadow" of this ball onto the cost function is exponentially thin.

This motivates two strategies:

1. **Local observables:** If O acts on k qubits, $\mathrm{Tr}(O^2) \sim 2^k$ and $\mathrm{Var}[\partial C] \sim 2^k/D$, which is polynomial for $k = \mathcal{O}(\log N)$.

2. **Structured ansätze:** Restrict the circuit to a low-dimensional submanifold of $\mathbb{CP}^{D-1}$ where F_Q and the cost function are aligned.

8. Visual Proof: QFI vs Gradient Scaling

Exercise 7.4: Born Machine Expressivity and Concentration

Motivation:

A **Born machine** is a quantum generative model where samples are drawn from the Born probability distribution $p(x) = |\langle x|U(\boldsymbol{\theta})|0\rangle|^2$. Generative modeling aims to learn complex, highly correlated probability distributions.
This exercise explores the expressivity limits of Born machines. First, product states (no entanglement) can only generate factorized distributions with limited inverse participation ratios (IPR). Second, deep chaotic circuits (unitaries forming a 2-design) generate **Porter-Thomas distributions**, which exhibit an "entropy deficit" relative to the uniform distribution. Finally, we prove that classical correlations in the generated distribution are strictly bounded by the quantum entanglement entropy of the underlying state.

Preliminaries / Toolbox

1. Inverse Participation Ratio (IPR): $C_2 = \sum_x p(x)^2$. It measures the sparsity/concentration of a distribution. $C_2 = 1$ for a delta function (completely deterministic), and $C_2 = 1/D$ for the uniform distribution (maximally spread).
2. Porter-Thomas Distribution: For a Haar-random pure state, each Born probability $p(x)$ is drawn from a Beta distribution $p_x \sim \text{Beta}(1, D-1)$, which approximates an exponential distribution $\Pr(p) \approx De^{-Dp}$ for large D.
3. Beta Integral Identity: If $x \sim \text{Beta}(\alpha, \beta)$, then $\mathbb{E}[x \ln x] = \frac{\alpha}{\alpha+\beta}[\psi(\alpha+1) - \psi(\alpha+\beta+1)]$, where $\psi(z)$ is the digamma function.
4. Data Processing Inequality: The mutual information between classical bitstrings X_A and X_B obtained by local measurements on subsystems A and B cannot exceed the quantum mutual information $I_Q(A:B)$ of the pre-measurement state.

Problem Statement:

(a) Show that for a product state $|\psi\rangle = \bigotimes_{j=1}^N |\psi_j\rangle$ with local Z-magnetizations $m_j = \langle Z_j \rangle$, the IPR is $C_2 = 2^{-N}\prod_j(1+m_j^2)$. Evaluate this for (i) all $m_j = 0$ and (ii) all $m_j = \pm 1$.
(b) For Haar-random states, show that the average Shannon entropy is $\mathbb{E}[H] = \psi(D+1) - \psi(2)$, and compute the entropy deficit $\Delta_H = \ln D - \mathbb{E}[H]$.
(c) Let X_A and X_B be classical bitstrings from measuring disjoint bipartite systems A and B. Prove $I(X_A : X_B) \leq 2S(\rho_A)$.

Pedantic Step-by-Step Solution:

Part (a): IPR of a Product State

For a product state, the Born probability factorizes: $p(x) = \prod_{j=1}^N p_j(x_j)$, where $p_j(0) = |\alpha_j|^2$ and $p_j(1) = |\beta_j|^2$. The magnetization is $m_j = p_j(0) - p_j(1)$.
Since $p_j(0) + p_j(1) = 1$, we have $p_j(0) = \frac{1+m_j}{2}$ and $p_j(1) = \frac{1-m_j}{2}$.
The IPR also factorizes:

$$C_2 = \sum_x p(x)^2 = \prod_{j=1}^N (p_j(0)^2 + p_j(1)^2) = \prod_{j=1}^N \left[\left(\frac{1+m_j}{2}\right)^2 + \left(\frac{1-m_j}{2}\right)^2\right].$$

$$= \prod_{j=1}^N \frac{1+2m_j+m_j^2+1-2m_j+m_j^2}{4} = \prod_{j=1}^N \frac{2+2m_j^2}{4} = 2^{-N}\prod_{j=1}^N (1+m_j^2). \quad \checkmark$$

Limits:

- (i) $m_j = 0$ (equatorial state, e.g., $|+\rangle^{\otimes N}$): $C_2 = 2^{-N} \prod 1 = 1/D$. (Uniform distribution)
- (ii) $m_j = \pm 1$ (computational basis state): $C_2 = 2^{-N} \prod 2 = 1$. (Delta function)

Part (b): Porter-Thomas Entropy Deficit

The Shannon entropy is $H = -\sum_x p(x) \ln p(x)$. By linearity of expectation, $\mathbb{E}[H] = -D\,\mathbb{E}[p_x \ln p_x]$.
Using $p_x \sim \text{Beta}(1, D-1)$, apply the beta integral identity with $\alpha = 1, \beta = D-1$:

$$\mathbb{E}[p_x \ln p_x] = \frac{1}{D}\big[\psi(2) - \psi(D+1)\big].$$

$$\mathbb{E}[H] = -D \cdot \frac{1}{D}\big[\psi(2) - \psi(D+1)\big] = \psi(D+1) - \psi(2). \quad \checkmark$$

Using the asymptotic expansion of the digamma function $\psi(z) \approx \ln z - \frac{1}{2z}$ for large z, and $\psi(2) = 1 - \gamma_E$ (where $\gamma_E \approx 0.577$ is the Euler-Mascheroni constant):

$$\mathbb{E}[H] \approx \ln(D+1) - (1-\gamma_E) \approx \ln D - (1-\gamma_E).$$

The entropy deficit relative to the uniform distribution ($H_{\text{unif}} = \ln D$) is:

$$\Delta_H = \ln D - \mathbb{E}[H] = 1 - \gamma_E \approx 0.4228 \text{ nats}.$$

This $O(1)$ gap shows that Haar-random Born machines do *not* generate the uniform distribution; they have "speckle" fluctuations that reduce entropy.

Part (c): Entanglement bounds classical correlations

The process of drawing classical samples is conceptually a local quantum measurement map $\mathcal{M} = \mathcal{M}_A \otimes \mathcal{M}_B$ on the pure state $|\psi\rangle\langle\psi|$. The post-measurement classically-correlated state is $\sigma_{AB} = \sum_{x_A, x_B} P(x_A, x_B) |x_A, x_B\rangle\langle x_A, x_B|$.
The classical mutual information is $I(X_A : X_B) = I_Q(\sigma_{AB})$.
Because local measurements are CPTP maps, the **Data Processing Inequality** implies:

$$I(X_A : X_B) = I_Q(\mathcal{M}(|\psi\rangle\langle\psi|)) \leq I_Q(|\psi\rangle\langle\psi|).$$

For a pure bipartite state $|\psi\rangle$, the quantum mutual information is:

$$I_Q(A:B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB}) = S(\rho_A) + S(\rho_A) - 0 = 2S(\rho_A).$$

Thus, $I(X_A : X_B) \leq 2S(\rho_A)$. $\checkmark$
Consequence: A Born machine cannot learn probability distributions with long-range variable correlations (high classical mutual information) unless the underlying quantum circuit generates sufficient long-range quantum entanglement.

Exercise 7.5: QELM Feature Concentration under Haar Scrambling

1. The Physical Setup

Consider a quantum device with two registers:

- An **input register** of N_{in} qubits, encoding classical data x via a state $\rho_{\text{in}}(x)$.
- A **reservoir register** of N_{res} qubits, initialized in the fiducial state $|0\rangle^{\otimes N_{\text{res}}}$.

The total Hilbert space dimension is $D = 2^{N_{\text{in}} + N_{\text{res}}}$. A fixed (but random) unitary U acts on the joint system, and features are extracted by measuring in the computational basis:

$$f_b(x) = \mathrm{Tr}\big(|b\rangle\langle b|\, U\, \sigma(x)\, U^\dagger\big), \qquad \sigma(x) = \rho_{\text{in}}(x) \otimes |0\rangle\langle 0|_{\text{res}}.$$

The D real numbers $\{f_b(x)\}_{b=0}^{D-1}$ form the feature vector that a classical linear model is trained on. The central question is: *how much information about x survives in these features when U is a deep, chaotic circuit?*

2. The POVM Perspective: Naimark Dilation in Reverse

One way to understand the QELM is to recognize that the entire pipeline—tensoring with the reservoir, applying U, and projecting onto $|b\rangle$—defines an **effective measurement** on the input register alone.

Tracing over the reservoir degrees of freedom, each computational basis outcome b corresponds to a **POVM element** acting on the input space $\mathcal{H}_{\text{in}}$:

$$M_b \;=\; \langle 0|_{\text{res}}\, U^\dagger\, |b\rangle\langle b|\, U\, |0\rangle_{\text{res}} \;\in\; \mathcal{L}(\mathcal{H}_{\text{in}}).$$

These satisfy the POVM axioms:

1. **Positivity:** $M_b \geq 0$ (since $M_b = A_b^\dagger A_b$ where $A_b = \langle b|U|0\rangle_{\text{res}}$).
2. **Completeness:** $\sum_{b=0}^{D-1} M_b = I_{\text{in}}$ (since $\{|b\rangle\}$ is a complete basis and U is unitary).
3. **Informativeness:** Each M_b has rank at most $D_{\text{res}} = 2^{N_{\text{res}}}$, so the POVM is **informationally complete** when $D_{\text{res}} \geq D_{\text{in}}$.

The feature is simply the Born probability under this POVM: $f_b(x) = \mathrm{Tr}(M_b\, \rho_{\text{in}}(x))$.

This is precisely the **Naimark dilation theorem** run in reverse: a projective measurement on a larger space realizes a generalized POVM on the subsystem.

3. Mathematical Toolbox

The following results from Haar integration and Weingarten calculus are needed:

Tool 1 — First Moment (Weingarten $k = 1$): For any operator A on $\mathcal{H}$ with $\dim \mathcal{H} = D$,

$$\mathbb{E}_{U\sim\text{Haar}}\big[U^\dagger A\, U\big] = \frac{\mathrm{Tr}(A)}{D}\, I_D.$$

This follows from Schur's lemma: the integral must commute with all unitaries, hence must be proportional to the identity. The coefficient is fixed by taking the trace of both sides.

Tool 2 — Second Moment (Weingarten $k = 2$): The collision probability uses the $k = 2$ Weingarten formula. Physically, this is the quantum analog of classical collision probability $\sum_x p_x^2$ (the probability that two independent draws from a distribution yield the same outcome). Here, it represents the probability that independent computational-basis measurements of two copies of the rotated state $U|\psi\rangle$ yield matching outcomes. For a pure state $|\psi\rangle$,

$$\mathbb{E}_U\left[\sum_{b=0}^{D-1} |\langle b|U|\psi\rangle|^4\right] = \frac{\mathrm{Tr}(\rho^2) + (\mathrm{Tr}\,\rho)^2}{D+1} = \frac{2}{D+1},$$

where we used $\mathrm{Tr}(\rho^2) = 1$ and $\mathrm{Tr}(\rho) = 1$ for a pure state. At this pure limit, quantum projection noise restricts the average collision probability to $2/(D+1)$ rather than 1 because the random rotation spreads the wave function. The Weingarten functions for S_2 are $\mathrm{Wg}(\mathrm{id}, D) = 1/(D^2-1)$ and $\mathrm{Wg}((12), D) = -1/(D(D^2-1))$.

Tool 3 — Haar Symmetry: The Haar measure is invariant under basis permutations: $\mathbb{E}[f_b^2]$ is independent of b, so $\mathbb{E}[f_b^2] = \frac{1}{D}\mathbb{E}[\sum_b f_b^2]$.

Tool 4 — Shot Noise Statistics: Estimating a probability p from ν Bernoulli trials gives a standard error $\sigma_{\text{shot}} = \sqrt{p(1-p)/\nu}$.

4. Exercise Statement

Assume U is drawn from the Haar measure on $U(D)$ and the input state is pure.

(a) Signal Death: Show that the Haar-averaged POVM element is $\mathbb{E}_U[M_b] = \frac{1}{D} I_{\text{in}}$, and consequently that $\mathbb{E}_U[f_b(x)] = 1/D$ for all inputs x.

(b) Speckle Variance: Show that the feature variance is $\mathrm{Var}_U[f_b] = \frac{D-1}{D^2(D+1)}$.
(c) Exponential Shot Cost: Show that resolving the input-dependent signal from measurement shot noise requires $\nu > D = 2^N$ shots.

5. Step-by-Step Derivation

Part (a): The Haar average trivializes the POVM

We compute the average of each POVM element over the Haar measure. Using Tool 1 (Weingarten $k = 1$):

$$\mathbb{E}_U[M_b] = \langle 0|_{\mathrm{res}}\, \mathbb{E}_U\big[U^\dagger |b\rangle\langle b| U\big]\, |0\rangle_{\mathrm{res}} = \langle 0|_{\mathrm{res}}\, \frac{I_D}{D}\, |0\rangle_{\mathrm{res}} = \frac{1}{D}\, I_{\mathrm{in}}.$$

On average, every POVM element M_b is proportional to the identity on the input space. It carries **zero directional information** about the input state. Every input produces the same Born probability:

$$\mathbb{E}_U[f_b(x)] = \mathrm{Tr}\left(\frac{I_{\mathrm{in}}}{D}\, \rho_{\mathrm{in}}(x)\right) = \frac{1}{D}.$$

Geometrically, in the space of POVM elements, the Haar average projects every M_b onto the unique $\mathrm{SU}(D)$-invariant point: the normalized identity. This is the POVM-theoretic manifestation of *information scrambling*.

Part (b): Speckle fluctuations carry the signal

The information about the input x lives entirely in the fluctuations of M_b around its mean I_{in}/D. Using Tool 2 (Weingarten $k = 2$), the collision probability for a pure input is:

$$\mathbb{E}_U\left[\sum_b f_b^2\right] = \frac{2}{D+1}.$$

By Tool 3 (Haar symmetry), $\mathbb{E}_U[f_b^2] = \frac{2}{D(D+1)}$.
The variance of each feature is:

$$\mathrm{Var}_U[f_b] = \frac{2}{D(D+1)} - \frac{1}{D^2} = \frac{2D-(D+1)}{D^2(D+1)} = \frac{D-1}{D^2(D+1)}.$$

For large D, $\mathrm{Var}[f_b] \sim 1/D^2$. The useful features are exponentially small perturbations on top of the uniform background $1/D$.

Part (c): Exponential measurement cost

By Tool 4, the shot noise from ν measurements is $\sigma_{\mathrm{shot}} \approx 1/\sqrt{\nu D}$. For the signal ($\sim 1/D$) to dominate:

$$\frac{1}{D} > \frac{1}{\sqrt{\nu D}} \implies \nu > D = 2^{N_{\mathrm{in}}+N_{\mathrm{res}}}.$$

An exponential shot budget is required to **detect** the presence of an input-dependent signal.

6. Physical Interpretation: The Information-Extraction Hierarchy

— Scrambling Regime — POVM Quality — Feature Variance — Shot Cost — ML Utility — —:– —:—–:—–:—–:—– — None (identity U) — Projective on input — $\mathcal{O}(1)$ — $\mathcal{O}(1)$ — Input-limited — — Partial (finite depth) — Structured, rank-deficient — $\mathcal{O}(1/\mathrm{poly}(D))$ — $\mathrm{poly}(D)$ — Useful — — Full (Haar random) — Trivial ($\propto I$) — $\mathcal{O}(1/D^2)$ — $\mathcal{O}(D)$ — Useless —

The sweet spot for quantum machine learning lies at the **edge of scrambling**: deep enough that the POVM elements explore a rich subspace, but shallow enough that they retain directional sensitivity to the input.

9. Visual Proof: Porter-Thomas Speckle Distribution

Exercise 7.6: Injection Contractivity and Restarting Overhead in QRC

Motivation:

In Quantum Reservoir Computing (QRC), temporal data is processed by sequentially injecting inputs into a small subsystem of a quantum reservoir, followed by unitary scrambling. A functional reservoir must exhibit the **fading memory property**: it must eventually forget inputs from the distant past.

This memory loss is driven by the physical act of data injection, which involves discarding (tracing out) the injection qubit to reset it. This exercise proves that the injection map is strictly contractive under the trace distance, providing the mathematical mechanism for fading memory. We also analyze the severe experimental overhead of reading out the reservoir using standard projective measurements, motivating the need for weak continuous measurements.

Preliminaries / Toolbox

1. The Injection Map: To inject classical data x_n into qubit 1, the QRC protocol traces out the current state of qubit 1 and prepares the pure state $|\psi(x_n)\rangle$. The map on the full reservoir state ρ is:

$$\mathcal{E}_x(\rho) = \mathrm{Tr}_1(\rho) \otimes |\psi(x)\rangle\langle\psi(x)|_1.$$

2. Kraus Decomposition: Any completely positive, trace-preserving (CPTP) map can be written as $\mathcal{E}(\rho) = \sum_a K_a \rho K_a^\dagger$, where $\sum_a K_a^\dagger K_a = I$.

3. Trace Distance: $D_{\mathrm{tr}}(\rho, \sigma) = \frac{1}{2}\|\rho - \sigma\|_1$. It is a metric for the distinguishability of two quantum states.

4. Contractivity: All CPTP maps are contractive under the trace distance: $D_{\mathrm{tr}}(\mathcal{E}(\rho), \mathcal{E}(\sigma)) \leq D_{\mathrm{tr}}(\rho, \sigma)$.

5. Tensor Product Norm: For a tensor product state $A \otimes B$, the trace norm factorizes: $\|A \otimes B\|_1 = \|A\|_1 \cdot \|B\|_1$.

Problem Statement:

(a) Write an explicit Kraus decomposition for the injection map $\mathcal{E}_x$ and verify completeness $\sum K_a^\dagger K_a = I$.

(b) Show algebraically that $\mathcal{E}_x$ is contractive: $\|\mathcal{E}_x(\rho) - \mathcal{E}_x(\sigma)\|_1 \leq \|\rho - \sigma\|_1$.

(c) In the standard restarting protocol, measuring the reservoir at time step n destroys the state, requiring the sequence to be replayed from $t = 1$. For a time series of length T and shot count ν, calculate the total number of unitary sequence runs.

Analytical Derivation

Part (a): Kraus decomposition

The partial trace over qubit 1 projects it onto the computational basis, $\sum_{a=0,1}\langle a|\cdot|a\rangle$, while leaving the remaining qubits (the bulk, denoted $\bar{1}$) unchanged. We then tensor in $|\psi(x)\rangle$.

The Kraus operators are:

$$K_a = (I_{\bar{1}} \otimes |\psi(x)\rangle_1)(I_{\bar{1}} \otimes \langle a|_1), \quad a \in \{0, 1\}.$$

Action on state:

$$\sum_a K_a \rho K_a^\dagger = \sum_a (I_{\bar{1}} \otimes |\psi(x)\rangle_1)\langle a|_1 \rho |a\rangle_1 (I_{\bar{1}} \otimes \langle\psi(x)|_1)$$

$$= \left(\sum_a \langle a|_1 \rho |a\rangle_1\right) \otimes |\psi(x)\rangle\langle\psi(x)|_1 = \mathrm{Tr}_1(\rho) \otimes |\psi(x)\rangle\langle\psi(x)|_1. \quad \checkmark$$

Completeness: Since $\langle\psi(x)|\psi(x)\rangle = 1$:

$$\sum_a K_a^\dagger K_a = \sum_a I_{\bar{1}} \otimes |a\rangle\langle a|_1 = I_{\bar{1}} \otimes I_1 = I. \quad \checkmark$$

Part (b): Contractivity and Fading Memory

$$\|\mathcal{E}_x(\rho) - \mathcal{E}_x(\sigma)\|_1 = \|\operatorname{Tr}_1(\rho - \sigma) \otimes |\psi\rangle\langle\psi|\|_1.$$

Because the trace norm of a tensor product factorizes, and $\||\psi\rangle\langle\psi|\|_1 = 1$:

$$= \|\operatorname{Tr}_1(\rho - \sigma)\|_1.$$

By the data processing inequality (monotonicity of trace distance under the partial trace channel):

$$\|\operatorname{Tr}_1(\Delta)\|_1 \leq \|\Delta\|_1.$$

Thus, $\|\mathcal{E}_x(\rho) - \mathcal{E}_x(\sigma)\|_1 \leq \|\rho - \sigma\|_1$. ✓

Physical interpretation: This strict reduction (the inequality is generally strict unless Δ has no support on qubit 1) demonstrates that the injection physically destroys information about the reservoir's past trajectory, enforcing fading memory.

Part (c): Restarting Readout Overhead

To extract a feature at step n, we require ν shots. Because measurement is projective and destroys the quantum state, we cannot continue to step $n+1$. We must replay the sequence from $t = 1$ up to $t = n$ for every shot.

$$\text{Total runs} = \nu \sum_{n=1}^{T} n = \nu \frac{T(T+1)}{2}.$$

This gives an overhead of $\mathcal{O}(\nu T^2)$. For typical sequences ($T \sim 100$) and shot budgets ($\nu \sim 1000$), this runs into the millions, limiting the scalability of projective QRC and motivating weak-measurement paradigms.

Exercise 7.7: Hardware Noise and Quantum Kernel Concentration

1. The Physical Setup

A **quantum kernel** is a similarity measure between two classical data points x and x', defined by the overlap of their quantum feature states:

$$\kappa(x, x') = |\langle\phi(x')|\phi(x)\rangle|^2 = \operatorname{Tr}\big(\rho(x)\,\rho(x')\big),$$

where $|\phi(x)\rangle = U(x)|0\rangle$ is the quantum feature state and $\rho(x) = |\phi(x)\rangle\langle\phi(x)|$.

This kernel is a valid positive semi-definite kernel (by construction), and can be plugged into any classical kernel machine (SVM, kernel ridge regression, Gaussian processes). The power of the approach lies in the exponentially large feature space $\mathcal{H} = (\mathbb{C}^2)^{\otimes N}$ that the quantum circuit explores.

But there is a catch: if the circuit $U(x)$ is too expressive (too deep, too random), the feature states $|\phi(x)\rangle$ become essentially Haar-random, and the kernel concentrates around zero for $x \neq x'$.

2. Connection to Haar Measure Theory

The key insight is that the quantum kernel $\kappa(x, x')$ is nothing but the **fidelity** between two quantum states. When the encoding circuit $U(x)$ forms a unitary 2-design (or is Haar-random), we can compute its statistics exactly.

Consider the relative circuit $V = U(x')^\dagger U(x)$. If the encoding is sufficiently expressive, V behaves as a Haar-random unitary for generic pairs (x, x') with $x \neq x'$. Then:

$$\kappa(x, x') = |\langle 0|V|0\rangle|^2,$$

which is the overlap of a Haar-random state with a fixed reference.

3. Preliminaries / Toolbox

1. Fidelity Statistics: For a Haar-random unitary $V \in U(D)$,

$$\mathbb{E}[|\langle 0|V|0\rangle|^2] = \frac{1}{D}, \qquad \mathrm{Var}[|\langle 0|V|0\rangle|^2] = \frac{D-1}{D^2(D+1)}.$$

2. Kernel Matrix Properties: A kernel matrix $K_{ij} = \kappa(x_i, x_j)$ must be positive semi-definite. Its spectrum determines the effective dimensionality of the feature space. A flat kernel ($K_{ij} \approx \delta_{ij}/D$) has rank ≈ 1 and is useless for classification.

3. Expressibility (from Ch. 5): The expressibility of a parameterized circuit $U(\theta)$ measures how close its output distribution is to the Haar measure. It is quantified by the frame potential:

$$\mathcal{F}^{(2)} = \int |\langle \phi(\theta)|\phi(\theta')\rangle|^4 \, d\theta \, d\theta'.$$

For the Haar measure, $\mathcal{F}^{(2)}_{\text{Haar}} = 2/(D(D+1))$.

4. Geometric Measure of Distinguishability: The kernel value $\kappa(x, x')$ is the cosine-squared of the angle between $|\phi(x)\rangle$ and $|\phi(x')\rangle$ on the complex projective space $\mathbb{CP}^{D-1}$. When $\kappa \to 0$, the states are orthogonal (maximally distinguishable); when $\kappa = 1$, they are identical.

4. Exercise Statement

(a) Show that for a Haar-random encoding, the expected off-diagonal kernel value is $\mathbb{E}[\kappa(x, x')] = 1/D$.

(b) Show that the variance of the kernel is $\mathrm{Var}[\kappa] = \frac{D-1}{D^2(D+1)} \sim 1/D^2$, so the kernel matrix concentrates around $K \approx I/D + (1 - 1/D)\,\mathrm{diag}$.

(c) Using SymPy, verify the exact formula for the kernel mean and variance symbolically.

(d) Numerically construct the kernel matrix for M Haar-random feature states at $N = 2, 3, 4$ qubits, and visualize the concentration phenomenon.

5. Analytical Derivation

Part (a): Average Kernel Value

For two distinct inputs $x \neq x'$, the relative circuit $V = U(x')^\dagger U(x)$ is Haar-random. The kernel is:

$$\kappa(x, x') = |\langle 0|V|0\rangle|^2.$$

By the first moment of the Haar measure:

$$\mathbb{E}_V\big[V|0\rangle\langle 0|V^\dagger\big] = \frac{I}{D}.$$

Taking the expectation value in the $|0\rangle$ state:

$$\mathbb{E}[\kappa] = \langle 0|\frac{I}{D}|0\rangle = \frac{1}{D}. \quad \checkmark$$

For $N = 10$ qubits, this gives $\mathbb{E}[\kappa] = 1/1024 \approx 0.001$. The kernel is essentially zero for all distinct input pairs.

Part (b): Kernel Variance and Concentration

We need $\mathbb{E}[\kappa^2] = \mathbb{E}[|\langle 0|V|0\rangle|^4]$. This is the second moment of the fidelity, computed via the collision probability formula:

$$\mathbb{E}[|\langle 0|V|0\rangle|^4] = \frac{2}{D(D+1)}.$$

Therefore:

$$\mathrm{Var}[\kappa] = \frac{2}{D(D+1)} - \frac{1}{D^2} = \frac{2D - (D+1)}{D^2(D+1)} = \frac{D-1}{D^2(D+1)}. \quad \checkmark$$

The standard deviation is $\sigma_\kappa \sim 1/D$, which is of the same order as the mean. This means the kernel matrix takes the form:

$$K_{ij} \approx \delta_{ij} + \frac{1}{D}(1-\delta_{ij}) + \mathcal{O}(1/D),$$

which is approximately a rescaled identity matrix. Such a kernel assigns equal similarity to all pairs and is **trivially non-informative** for any classification task.

6. Physical Interpretation

The concentration of the quantum kernel is a direct consequence of the **curse of dimensionality** in exponentially large Hilbert spaces. Two random points on the unit sphere in $\mathbb{C}^D$ are nearly orthogonal with overwhelming probability when D is large.

This has implications for quantum advantage in kernel methods:

- **Too little expressibility** $\Rightarrow$ the feature states lie in a low-dimensional manifold, and the quantum kernel reduces to a classical one.
- **Too much expressibility** $\Rightarrow$ the feature states fill the Hilbert space uniformly (Haar-like), and the kernel concentrates around zero.
- **The sweet spot** $\Rightarrow$ the circuit maps data into a structured, problem-dependent subspace with geometric properties that a classical kernel cannot efficiently reproduce.

This mirrors the barren plateau phenomenon (Exercise 7.1) and the QELM concentration (Exercise 7.4): scrambling destroys the gradients, the features, *and* the kernels.

Flashcards, Active Recall & Derivations

Card 12: Barren Plateau (Full Proof)

Goal: Prove that the variance of the gradient $\mathrm{Var}\left[\frac{\partial C}{\partial \theta_i}\right]$ for a deep random parameterized quantum circuit scales exponentially as 2^{-N}.

Key Trick: Express the gradient using the parameter-shift rule as a commutator, double the space to write its variance as a trace, and apply the second Haar moment to the unitary blocks.

Step-by-Step Derivation:

1. Let the cost function be $C(\vec{\theta}) = \mathrm{Tr}\left(\hat{U}(\vec{\theta})\hat{\rho}_0\hat{U}^\dagger(\vec{\theta})\hat{O}\right)$.
2. Assume the circuit is structured as $\hat{U}(\vec{\theta}) = \hat{V}e^{-i\theta_i\hat{P}_i/2}\hat{W}$, where $\hat{V}$ and $\hat{W}$ are independent Haar-random unitary blocks.
3. The derivative of the cost function with respect to θ_i is:
$$\frac{\partial C}{\partial \theta_i} = \frac{i}{2}\mathrm{Tr}\left([\hat{P}_i, \hat{O}_V]\hat{\rho}_W\right)$$
where $\hat{O}_V = \hat{V}^\dagger\hat{O}\hat{V}$ and $\hat{\rho}_W = \hat{W}\hat{\rho}_0\hat{W}^\dagger$.
4. Write the variance of the gradient (since the mean gradient is zero):
$$\mathrm{Var}\left[\frac{\partial C}{\partial \theta_i}\right] = \mathbb{E}\left[\left(\frac{\partial C}{\partial \theta_i}\right)^2\right] = -\frac{1}{4}\mathbb{E}_{V,W}\left[\mathrm{Tr}\left([\hat{P}_i, \hat{O}_V]\hat{\rho}_W\right)^2\right]$$
5. Double the space to rewrite the squared trace:
$$\mathrm{Tr}\left([\hat{P}_i, \hat{O}_V]\hat{\rho}_W\right)^2 = \mathrm{Tr}\left[\left([\hat{P}_i, \hat{O}_V] \otimes [\hat{P}_i, \hat{O}_V]\right)(\hat{\rho}_W \otimes \hat{\rho}_W)\right]$$

6. First average over $\hat{V}$ using the second Haar moment: Since $\hat{O}_V$ contains $\hat{V}$, we evaluate $\mathbb{E}_V[\hat{O}_V \otimes \hat{O}_V] = c_I \,\hat{\mathbb{I}} + c_{\mathbb{F}}\,\mathbb{F}$. Assuming $\hat{O}$ is traceless ($\mathrm{Tr}(\hat{O}) = 0$), the coefficients simplify:

$$c_I = -\frac{\mathrm{Tr}(\hat{O}^2)}{D(D^2-1)}, \qquad c_{\mathbb{F}} = \frac{D\,\mathrm{Tr}(\hat{O}^2)}{D(D^2-1)} = \frac{\mathrm{Tr}(\hat{O}^2)}{D^2-1}$$

Thus:

$$\mathbb{E}_V[\hat{O}_V \otimes \hat{O}_V] = \frac{\mathrm{Tr}(\hat{O}^2)}{D^2-1}\left(\mathbb{F} - \frac{\hat{\mathbb{I}}}{D}\right)$$

7. Substitute this back into the commutator terms. First, expand the commutator product on the doubled space:

$$[\hat{P}_i, \hat{O}_V] \otimes [\hat{P}_i, \hat{O}_V] = \left(\hat{P}_i\hat{O}_V - \hat{O}_V\hat{P}_i\right) \otimes \left(\hat{P}_i\hat{O}_V - \hat{O}_V\hat{P}_i\right)$$

Distribute the terms using $(\hat{A}\hat{B}) \otimes (\hat{C}\hat{D}) = (\hat{A} \otimes \hat{C})(\hat{B} \otimes \hat{D})$:

$$\begin{aligned}[\hat{P}_i, \hat{O}_V] \otimes [\hat{P}_i, \hat{O}_V] = {} & (\hat{P}_i \otimes \hat{P}_i)(\hat{O}_V \otimes \hat{O}_V) - (\hat{P}_i \otimes \hat{I})(\hat{O}_V \otimes \hat{O}_V)(\hat{I} \otimes \hat{P}_i) \\ & - (\hat{I} \otimes \hat{P}_i)(\hat{O}_V \otimes \hat{O}_V)(\hat{P}_i \otimes \hat{I}) + (\hat{O}_V \otimes \hat{O}_V)(\hat{P}_i \otimes \hat{P}_i)\end{aligned}$$

Now take the expectation value over $\hat{V}$. Substitute $\mathbb{E}_V[\hat{O}_V \otimes \hat{O}_V] = c_{\mathbb{F}}\left(\mathbb{F} - \frac{\hat{\mathbb{I}}}{D}\right)$ where $c_{\mathbb{F}} = \frac{\mathrm{Tr}(\hat{O}^2)}{D^2-1}$. Since the identity operator $\hat{\mathbb{I}}$ commutes with all operators, its contribution to each of the four distributed terms evaluates to:

$$(\hat{P}_i\otimes\hat{P}_i)\hat{\mathbb{I}} - (\hat{P}_i\otimes\hat{I})\hat{\mathbb{I}}(\hat{I}\otimes\hat{P}_i) - (\hat{I}\otimes\hat{P}_i)\hat{\mathbb{I}}(\hat{P}_i\otimes\hat{I}) + \hat{\mathbb{I}}(\hat{P}_i\otimes\hat{P}_i) = \hat{P}_i\otimes\hat{P}_i - \hat{P}_i\otimes\hat{P}_i - \hat{P}_i\otimes\hat{P}_i + \hat{P}_i\otimes\hat{P}_i = 0$$

Thus, the identity term $\hat{\mathbb{I}}/D$ vanishes completely. We are left only with the contractions of the SWAP operator $\mathbb{F}$:

$$\mathbb{E}_V\left[[\hat{P}_i, \hat{O}_V] \otimes [\hat{P}_i, \hat{O}_V]\right] = c_{\mathbb{F}}\left[(\hat{P}_i \otimes \hat{P}_i)\mathbb{F} - (\hat{P}_i \otimes \hat{I})\mathbb{F}(\hat{I} \otimes \hat{P}_i) - (\hat{I} \otimes \hat{P}_i)\mathbb{F}(\hat{P}_i \otimes \hat{I}) + \mathbb{F}(\hat{P}_i \otimes \hat{P}_i)\right]$$

Using the permutation property $(\hat{A} \otimes \hat{B})\mathbb{F} = \mathbb{F}(\hat{B} \otimes \hat{A})$ and $\hat{P}_i^2 = \hat{I}$, we simplify each term:

- $(\hat{P}_i \otimes \hat{P}_i)\mathbb{F} = \mathbb{F}(\hat{P}_i \otimes \hat{P}_i)$
- $(\hat{P}_i \otimes \hat{I})\mathbb{F}(\hat{I} \otimes \hat{P}_i) = \mathbb{F}(\hat{I} \otimes \hat{P}_i)(\hat{I} \otimes \hat{P}_i) = \mathbb{F}(\hat{I} \otimes \hat{P}_i^2) = \mathbb{F}$
- $(\hat{I} \otimes \hat{P}_i)\mathbb{F}(\hat{P}_i \otimes \hat{I}) = \mathbb{F}(\hat{P}_i \otimes \hat{I})(\hat{P}_i \otimes \hat{I}) = \mathbb{F}(\hat{P}_i^2 \otimes \hat{I}) = \mathbb{F}$
- $\mathbb{F}(\hat{P}_i \otimes \hat{P}_i)$

Combine these back to yield:

$$\mathbb{E}_V\left[[\hat{P}_i, \hat{O}_V] \otimes [\hat{P}_i, \hat{O}_V]\right] = \frac{2\,\mathrm{Tr}(\hat{O}^2)}{D^2-1}\left(\mathbb{F}(\hat{P}_i \otimes \hat{P}_i) - \mathbb{F}\right)$$

(Here $\mathbb{F}(\hat{P}_i \otimes \hat{P}_i) - \mathbb{F}$ is kept as is—it does *not* simplify to $\hat{\mathbb{I}} - \mathbb{F}$.)

8. Average over $\hat{W}$. Since $\hat{\rho}_0 = |0\rangle\langle 0|$ is pure, so is $\hat{\rho}_W$, and its second moment is

$$\mathbb{E}_W[\hat{\rho}_W \otimes \hat{\rho}_W] = \frac{\hat{\mathbb{I}} + \mathbb{F}}{D(D+1)}.$$

9. Contract the two averaged operators. Using the swap-trick identities $\mathrm{Tr}[\mathbb{F}(\hat{A} \otimes \hat{B})] = \mathrm{Tr}(\hat{A}\hat{B})$, $\mathbb{F}^2 = \hat{\mathbb{I}}$, and $\mathrm{Tr}(\hat{P}_i) = 0$,

$$\mathrm{Tr}\Big[\big(\mathbb{F}(\hat{P}_i \otimes \hat{P}_i) - \mathbb{F}\big)\big(\hat{\mathbb{I}} + \mathbb{F}\big)\Big] = D + 0 - D - D^2 = -D^2.$$

Folding in the prefactor $2\,\mathrm{Tr}(\hat{O}^2)/(D^2-1)$, the $1/(D(D+1))$ from the $\hat{W}$-average, the overall $1/4$ from the parameter-shift rule, and the $i^2=-1$ that makes the squared gradient positive, gives

$$\mathrm{Var}\left[\frac{\partial C}{\partial \theta_i}\right] = \frac{D\ \mathrm{Tr}(\hat{O}^2)}{2(D^2-1)(D+1)} \xrightarrow{D\gg 1} \boxed{\frac{\mathrm{Tr}(\hat{O}^2)}{2D^2}}.$$

For a 1-local observable $\hat{O}=\hat{Z}_1$, $\mathrm{Tr}(\hat{O}^2)=D$, so $\mathrm{Var} \approx \dfrac{1}{2D} = 2^{-(N+1)}$.

Questions

1. **Question:** What architectural strategies are used to avoid barren plateaus?

   Answer: Barren plateaus are mathematically inevitable if the variational ansatz is a unitary 2-design and we optimize global cost functions. To mitigate this, one must use local cost functions (where the gradient variance decays only polynomially as $1/\mathrm{poly}(N)$) or initialize parameters close to a good solution using identity-blocked ansätze or pre-training.

Derivation Blueprint: Barren Plateau (Full Proof)

Board Layout:

- **Left Board Panel:** Write out the VQA setup, the parameterized circuit, and the gradient expression as a commutator.
- **Middle Panel:** Show the doubled space expansion. Perform the expectation value over $\hat{V}$ using the second Haar moment, explaining why the identity term drops out.
- **Right Panel:** Perform the expectation value over $\hat{W}$ and complete the contraction. Highlight the final 2^{-N} exponential scaling.

Common Conceptual Pitfalls: Expanding and contracting the commutator terms with the swap operator $\mathbb{F}$ involves a highly complex tensor structure. One should write out all four terms of the product explicitly to track how the swap operator links the inputs and outputs of the various subsystems. **Physical & Experimental Anchorage:**

- In Variational Quantum Algorithms (VQAs) implemented on NISQ processors (like IBM Quantum or superconducting chips), the barren plateau phenomenon is a severe roadblock. If we initialize the ansatz parameters randomly, the gradient of the cost function concentrates exponentially around zero, making it impossible for standard classical optimizers (like gradient descent or SPSA) to find a descent direction. To mitigate this, practitioners use **local cost functions** (where gradients decay only polynomially) or initialize the ansatz with **identity-blocked parameters** to restrict the early optimization path to a shallow, searchable subspace.

Boundary Limit & Sanity Check: A local cost function $\hat{O}=\hat{Z}_1$ has $\mathrm{Tr}(\hat{O}^2)=D$, giving a gradient variance that scales as $1/(2D)=2^{-(N+1)}$, whereas a global cost function $\hat{O}=|0\rangle\langle 0|^{\otimes N}$ has $\mathrm{Tr}(\hat{O}^2)=1$, which severely suppresses the variance to $1/(2D^2)=2^{-(2N+1)}$.

Numerical Sandbox: Barren Plateau Gradient Variance Verification

```
import numpy as np
# Compare N=2 and N=3 qubit gradient variance
for N in [2, 3]:
    D = 2**N
    Z = np.array([[1, 0], [0, -1]], dtype=complex)
    Y = np.array([[0, -1j], [1j, 0]], dtype=complex)
    I = np.eye(2, dtype=complex)
    def embed_first(op, N):
        # op on qubit 1, identity elsewhere: op (x) I (x) ... (x) I
        M = op
        for _ in range(N - 1):
            M = np.kron(M, I)
        return M
    # 1-local cost operator O = Z_1 = Z (x) I^{(N-1)} (already traceless)
    O = embed_first(Z, N)
    # Generator P = Y_1 = Y (x) I^{(N-1)}
    P = embed_first(Y, N)

    gradients = []
    N_samples = 2000
    for _ in range(N_samples):
        # Generate random Haar unitary V
        X = np.random.normal(size=(D, D)) + 1j * np.random.normal(size=(D, D))
        Q, R = np.linalg.qr(X)
        V = Q @ np.diag(np.diagonal(R) / np.abs(np.diagonal(R)))
        # Rotated operator
        O_V = V.conj().T @ O @ V
        # Random Haar state vector psi (simulates W |0>)
        x = np.random.normal(size=D) + 1j * np.random.normal(size=D)
        psi = x / np.linalg.norm(x)
        # Compute gradient trace element
        grad = -0.5 * np.imag(psi.conj().T @ (P @ O_V - O_V @ P) @ psi)
        gradients.append(grad)
    variance_sim = np.var(gradients)
    trO2 = np.trace(O @ O).real
    var_exact = D * trO2 / (2 * (D**2 - 1) * (D + 1))  # exact (Chapter 7)
    var_leading = trO2 / (2 * D**2)                    # large-D leading order
    print(f"N={N} | Sim: {variance_sim:.6f} | "
          f"Exact: {var_exact:.6f} | Leading 1/(2D): {var_leading:.6f}")
```

Card 13: Kernel Concentration

Goal: Prove that the off-diagonal entries of the quantum kernel matrix $K_{ij} = |\langle\phi(x_i)|\phi(x_j)\rangle|^2$ concentrate exponentially to $1/D$ as $D \to \infty$.

Key Trick: Represent the overlap squared as a trace of the swap operator on the doubled space, and apply the first moment of Haar unitaries to show that the off-diagonal entries act as random projections.

Step-by-Step Derivation:

1. Let $|\phi(x_i)\rangle = \hat{U}(x_i)\,|0\rangle$ be the quantum feature map.

2. For $i \neq j$, the kernel entry is the transition probability:

$$K_{ij} = |\langle\phi(x_i)|\phi(x_j)\rangle|^2 = \mathrm{Tr}\left(|\phi(x_i)\rangle\langle\phi(x_i)|\,|\phi(x_j)\rangle\langle\phi(x_j)|\right)$$

3. If the feature map $\hat{U}(x)$ is highly expressive (modeled as a Haar-random unitary), then $|\phi(x_i)\rangle$ and $|\phi(x_j)\rangle$ act as independent Haar-random states $|\psi_i\rangle$ and $|\psi_j\rangle$.

4. Compute the expectation value over the Haar measure:

$$\mathbb{E}[K_{ij}] = \mathbb{E}_{U_i,U_j}\left[\langle\psi_i|\,|\psi_j\rangle\langle\psi_j|\,|\psi_i\rangle\right]$$

5. Use the first Haar moment (Card 1) to average over $|\psi_j\rangle$:

$$\mathbb{E}_{U_j}[|\psi_j\rangle\langle\psi_j|] = \frac{\hat{I}}{D}$$

6. Substitute this back:

$$\mathbb{E}[K_{ij}] = \mathbb{E}_{U_i}\left[\langle\psi_i|\,\frac{\hat{I}}{D}\,|\psi_i\rangle\right] = \frac{1}{D}$$

7. For qubits, $D = 2^N$. Thus:

$$\boxed{\mathbb{E}[K_{ij}] = 2^{-N} \quad (i \neq j)}$$

Questions

1. **Question:** Why is quantum kernel concentration considered a fundamental trainability limit?

   Answer: If the quantum kernel concentrates exponentially to the identity matrix, the classifier treats every data point as orthogonal to every other data point. It loses all geometric information about the data distribution, preventing generalization to unseen data—a phenomenon known as a quantum kernel barren plateau.

Derivation Blueprint: Kernel Concentration

Board Layout:

- **Left Board Panel:** Write down the definition of the quantum kernel matrix K_{ij} and explain the feature map mapping.
- **Middle Panel:** Write down the first Haar moment twirl and explain why the off-diagonal states act as independent random projections.
- **Right Panel:** Show the final result $K_{ij} = \delta_{ij} + \mathcal{O}(2^{-N})$. Discuss the classification consequences.

Common Conceptual Pitfalls: A common question is why the diagonal elements K_{ii} of the quantum kernel do not exhibit exponential measure concentration. This is because for $i = j$, the diagonal element represents the overlap of a state vector with itself, which is identically normalized to unity: $K_{ii} = |\langle\phi(x_i)|\phi(x_i)\rangle|^2 = 1$. **Boundary Limit & Sanity Check:** As $N \to \infty$, the off-diagonal kernel entries decay exponentially as 2^{-N}, causing the quantum kernel matrix to concentrate to the identity matrix. Under this asymptotic limit, the classifier treats all data points as mutually orthogonal, breaking generalization and causing severe overfitting.